Fundamentals of Backend Communications and Protocols

Understand what’s behind the backend

Introduction

Introduction

- Welcome
- Who this course is for?
- Course Outline

Backend Communication Design Patterns

There are handful of ways backends communicate

Request - Response

Classic, Simple and Everywhere

Request Response Model

- Client sends a Request
- Server parses the Request
- Server processes the Request
- Server sends a Response
- Client parses the Response and consume

Where it is used?

- Web, HTTP, DNS, SSH
- RPC (remote procedure call)
- SQL and Database Protocols
- APIs (REST/SOAP/GraphQL)
- Implemented in variations

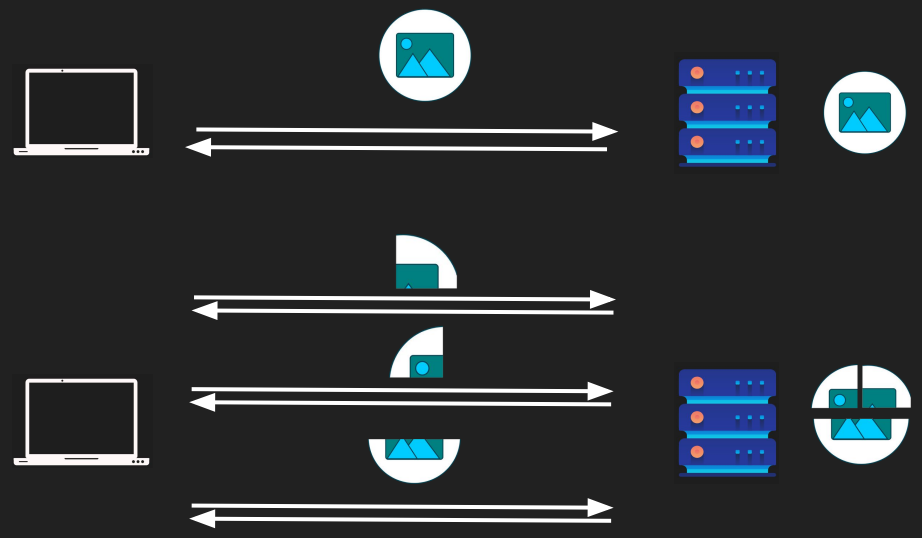
Anatomy of a Request / Response

- A request structure is defined by both client and server
- Request has a boundary
- Defined by a protocol and message format
- Same for the response
- E.g. HTTP Request

```
GET / HTTP/1.1
Headers
<CRLF>
BODY
```

Building an upload image service with request response

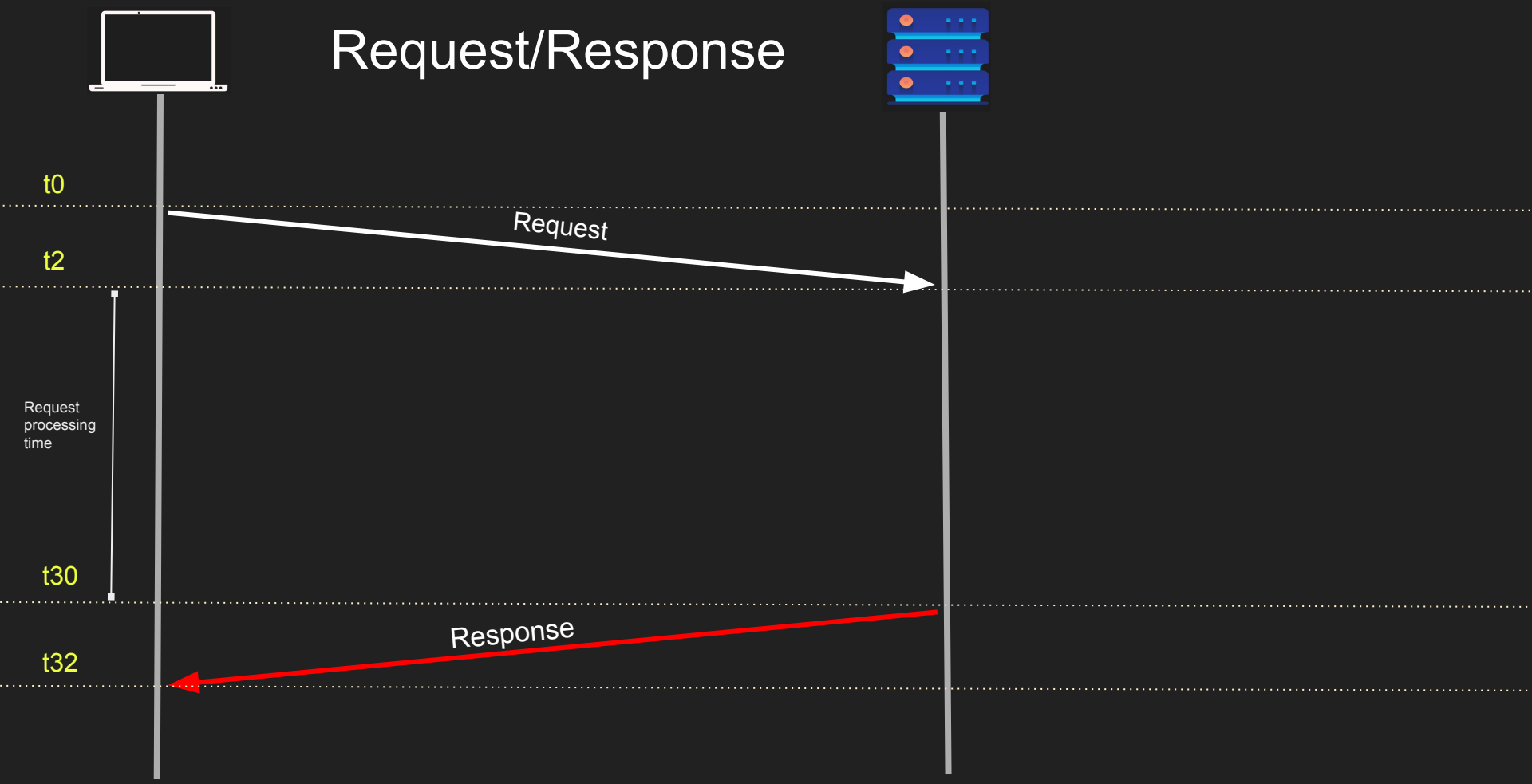
- Send large request with the image (simple)
- Chunk image and send a request per chunk (resumable)



Doesn’t work everywhere

- Notification service
- Chatting application
- Very Long requests
- What if client disconnects?
- We will talk about alternatives

Request/Response



Demo Curl

–trace

Synchronous vs Asynchronous

Can I do work while waiting?

Synchronous I/O

- Caller sends a request and blocks
- Caller cannot execute any code meanwhile
- Receiver responds, Caller unblocks
- Caller and Receiver are in “sync”

Example of an OS synchronous I/O

- Program asks OS to read from disk
- Program main thread is taken off of the CPU
- Read completes, program can resume execution

```
//Program starts
//Program uses CPU to execute stuff
doWork();
//Program reads from disk
//Program can't do anything until file loads
readFile("largefile.dat");
//program resumes
doWork2();
```

Asynchronous I/O

- Caller sends a request
- Caller can work until it gets a response
- Caller either:
  - Checks if the response is ready (epoll)
  - Receiver calls back when it's done (io_uring)
  - Spins up a new thread that blocks
- Caller and receiver are not necessary in sync

Example of an OS asynchronous call (NodeJS)

- Program spins up a secondary thread
- Secondary thread reads from disk, OS blocks it
- Main program still running and executing code
- Thread finish reading and calls back main thread

```
//Program starts
//Program uses CPU to execute stuff
doWork();
//Program requests read from disk
//Program asks to callback when done
//Program moves on to doWork2
readFile("largefile.dat", onReadFinished(theFile));
//file is probably not read yet
//Program happy doing stuff
doWork2();
//someone just called onReadFinished
//processing it.
---->onReadFinished(theFile)
```

Synchronous vs Asynchronous in Request Response

- Synchronicity is a client property
- Most modern client libraries are asynchronous
- E.g. Clients send an HTTP request and do work

Synchronous vs Asynchronous in real life

- Just in case it is still confusing
- In Synchronous communication the caller waits for a response from receiver
  - E.g. Asking someone a question in a meeting
- Asynchronous communication the response can come whenever. Caller and receiver can do anything meanwhile
  - email

Asynchronous workload is everywhere

- Asynchronous Programming (promises/futures)
- Asynchronous backend processing
- Asynchronous commits in postgres
- Asynchronous IO in Linux (epoll, io_uring)
- Asynchronous replication
- Asynchronous OS fsync (fs cache)

Demo NodeJS

Blocking call (loop) and fetch

Push

I want it as soon as possible

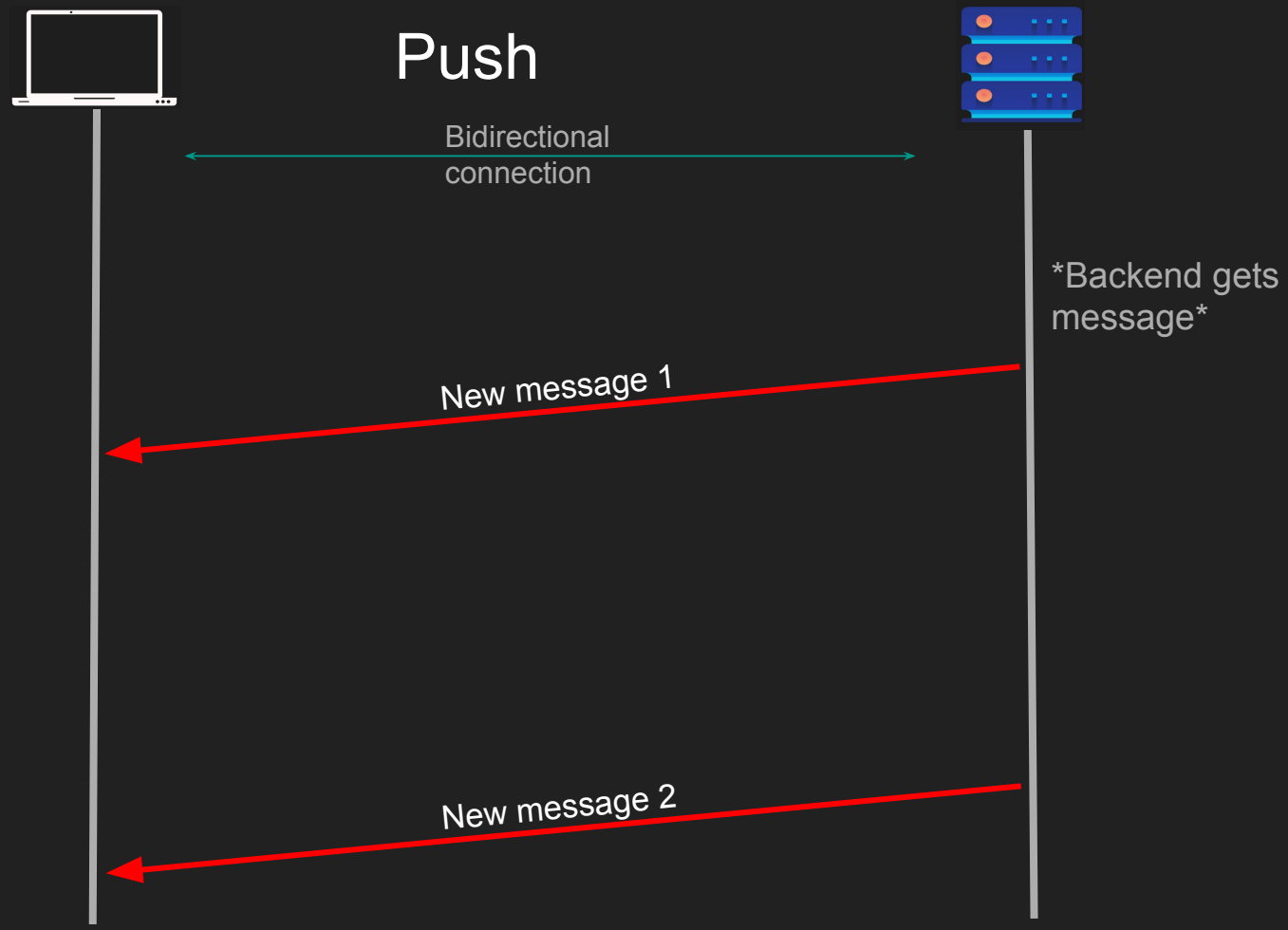
Request/response isn’t always ideal

- Client wants real time notification from backend
  - A user just logged in
  - A message is just received
- Push model is good for certain cases

What is Push?

- Client connects to a server
- Server sends data to the client
- Client doesn’t have to request anything
- Protocol must be bidirectional
- Used by RabbitMQ

Push


Bidirectional connection
Backend gets message
New message 1
New message 2

Push Pros and Cons

- Pros
  - Real time
- Cons
  - Clients must be online
  - Clients might not be able to handle
  - Requires a bidirectional protocol
  - Polling is preferred for light clients

Demo WebSockets

Push notification (users login)

Short Polling

Request is taking a while, I'll check with you later

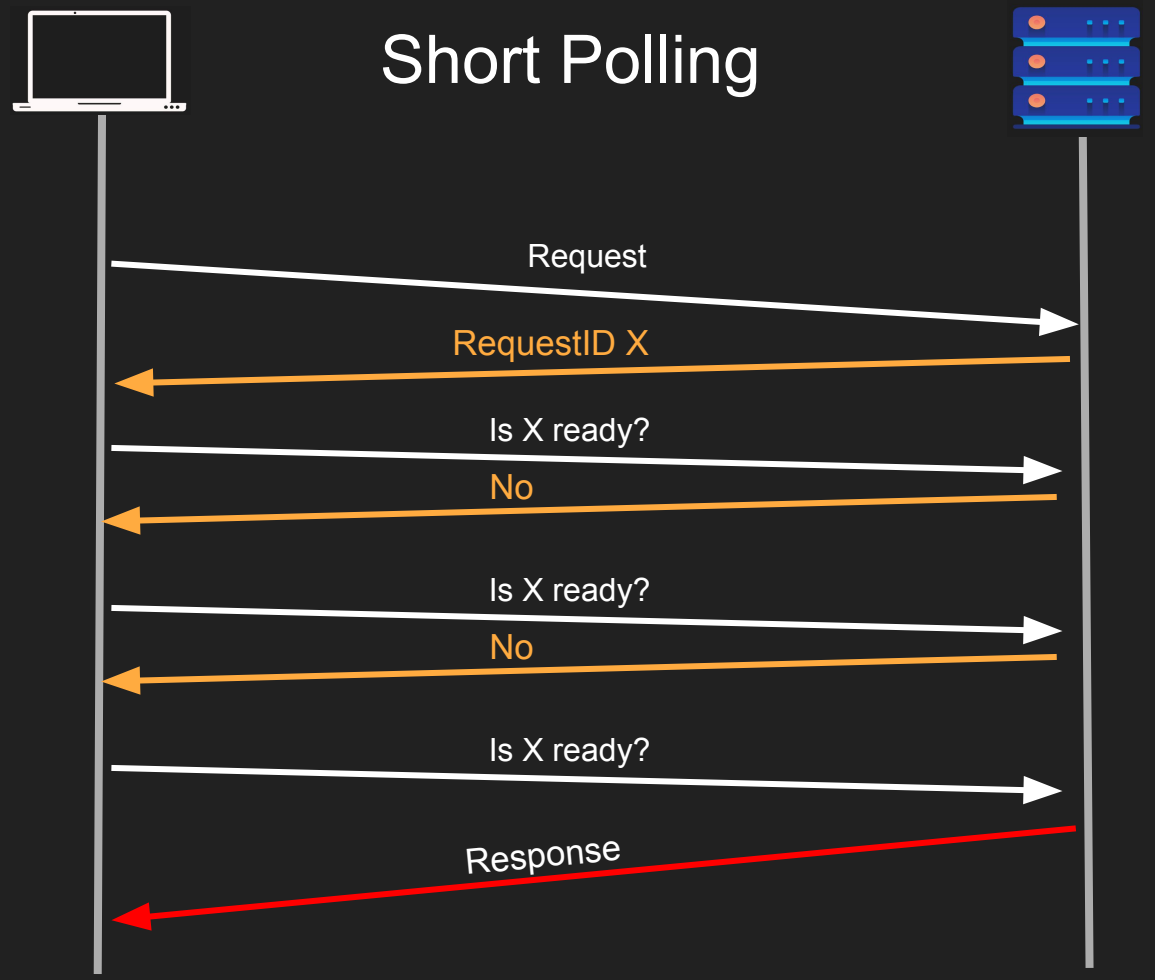
Where request/response isn’t ideal

- A request takes long time to process
  - Upload a youtube video
- The backend want to sends notification
  - A user just logged in
- Polling is a good communication style

What is Short Polling?

- Client sends a request
- Server responds immediately with a handle
- Server continues to process the request
- Client uses that handle to check for status
- Multiple “short” request response as polls

Short Polling



Short Polling Pros and Cons

- Pros
  - Simple
  - Good for long running requests
  - Client can disconnect
- Cons
  - Too chatty
  - Network bandwidth
  - Wasted Backend resources

Demo Job Status

Check if a job is done with progress

Long Polling

Request is taking long, I'll check with you later
But talk to me only when it's ready

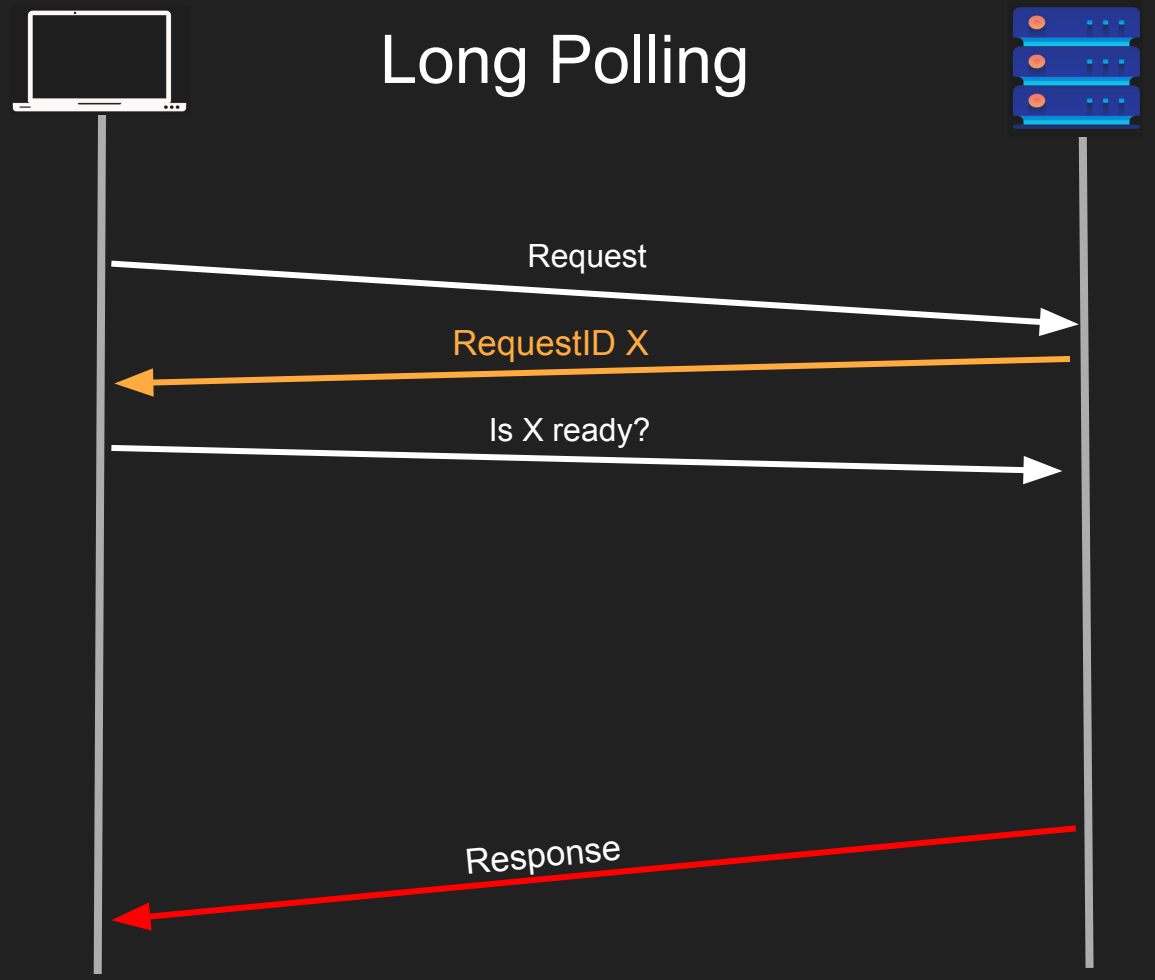
Where request/response & polling isn’t ideal

- A request takes long time to process
  - Upload a youtube video
- The backend want to sends notification
  - A user just logged in
- Short Polling is a good but chatty
- Meet Long polling (Kafka uses it)

What is Long Polling?

- Client sends a request
- Server responds immediately with a handle
- Server continues to process the request
- Client uses that handle to check for status
- Server DOES not reply until it has the response
- So we got a handle, we can disconnect and we are less chatty
- Some variation has timeouts too

Long Polling



Long Polling Pros and Cons

- Pros
  - Less chatty and backend friendly
  - Client can still disconnect
- Cons
  - Not real time

Demo Job Status

Check if a job is done

Server Sent Events

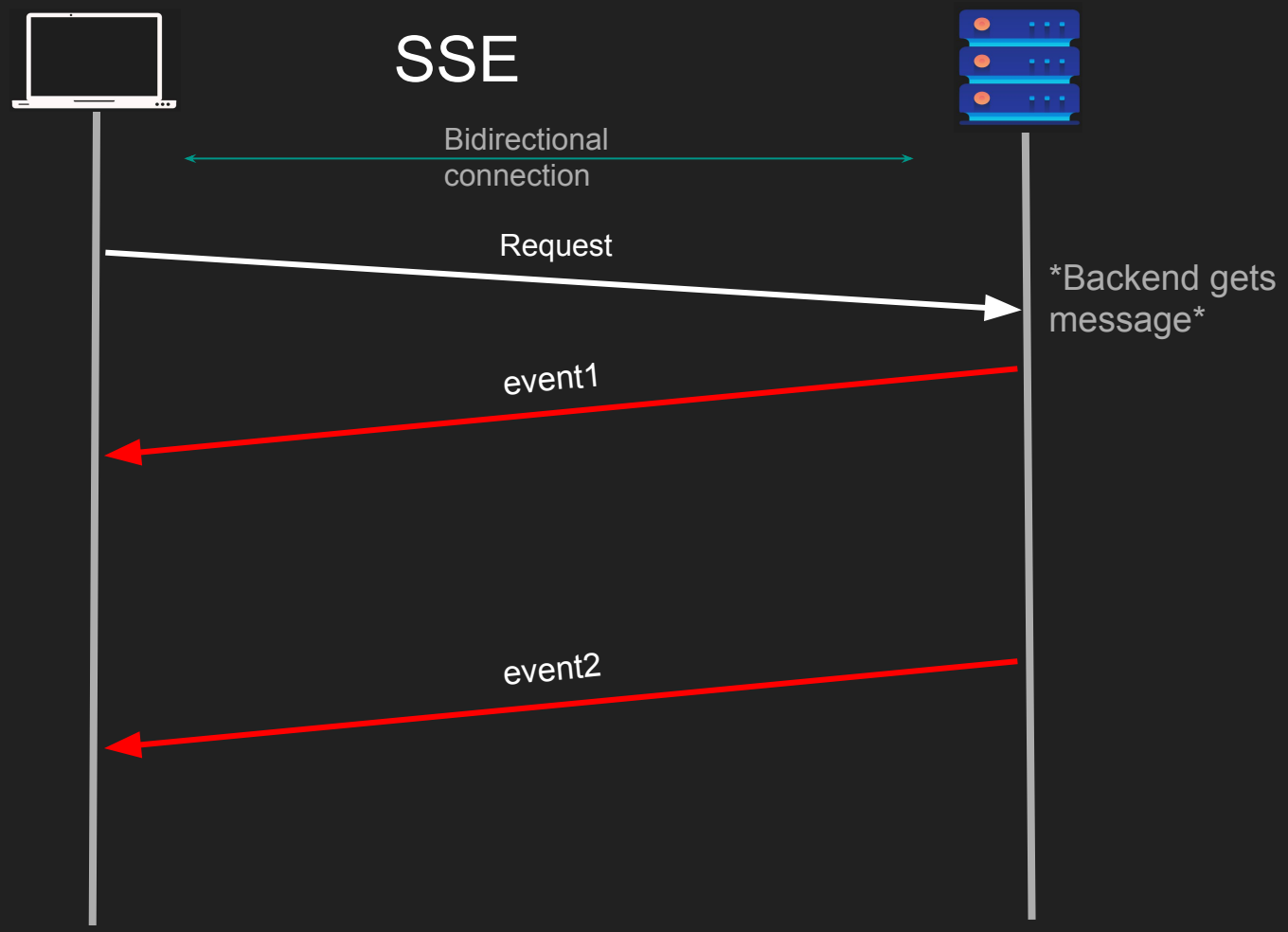
One Request, a very very long response

Limitations of Request/Response

- Vanilla Request/response isn’t ideal for notification backend
- Client wants real time notification from backend
  - A user just logged in
  - A message is just received
- Push works but restrictive
- Server Sent Events work with Request/response
- Designed for HTTP

What is Server Sent Events?

- A response has start and end
- Client sends a request
- Server sends logical events as part of response
- Server never writes the end of the response
- It is still a request but an unending response
- Client parses the streams data looking for events
- Works with request/response (HTTP)


SSE
Bidirectional connection
Request
Backend gets message
event1
event2

Server Sent Events Pros and Cons

- Pros
  - Real time
  - Compatible with Request/response
- Cons
  - Clients must be online
  - Clients might not be able to handle
  - Polling is preferred for light clients
  - HTTP/1.1 problem (6 connections)

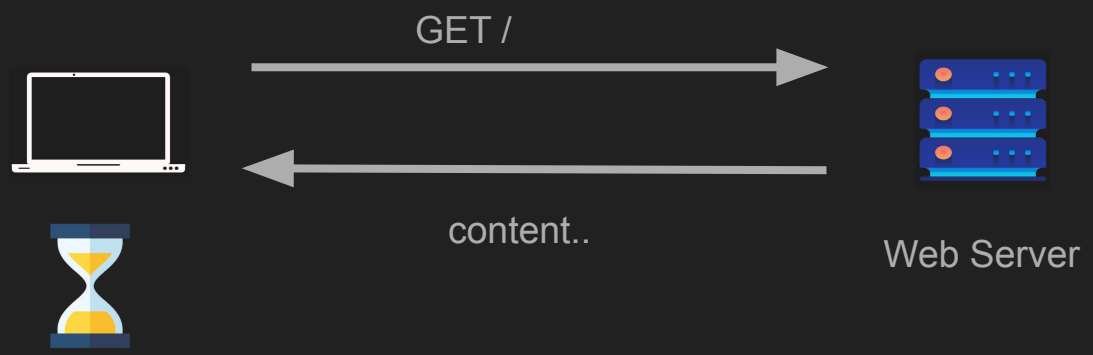
Demo Job Status

Check if a job is done with progress using SSE

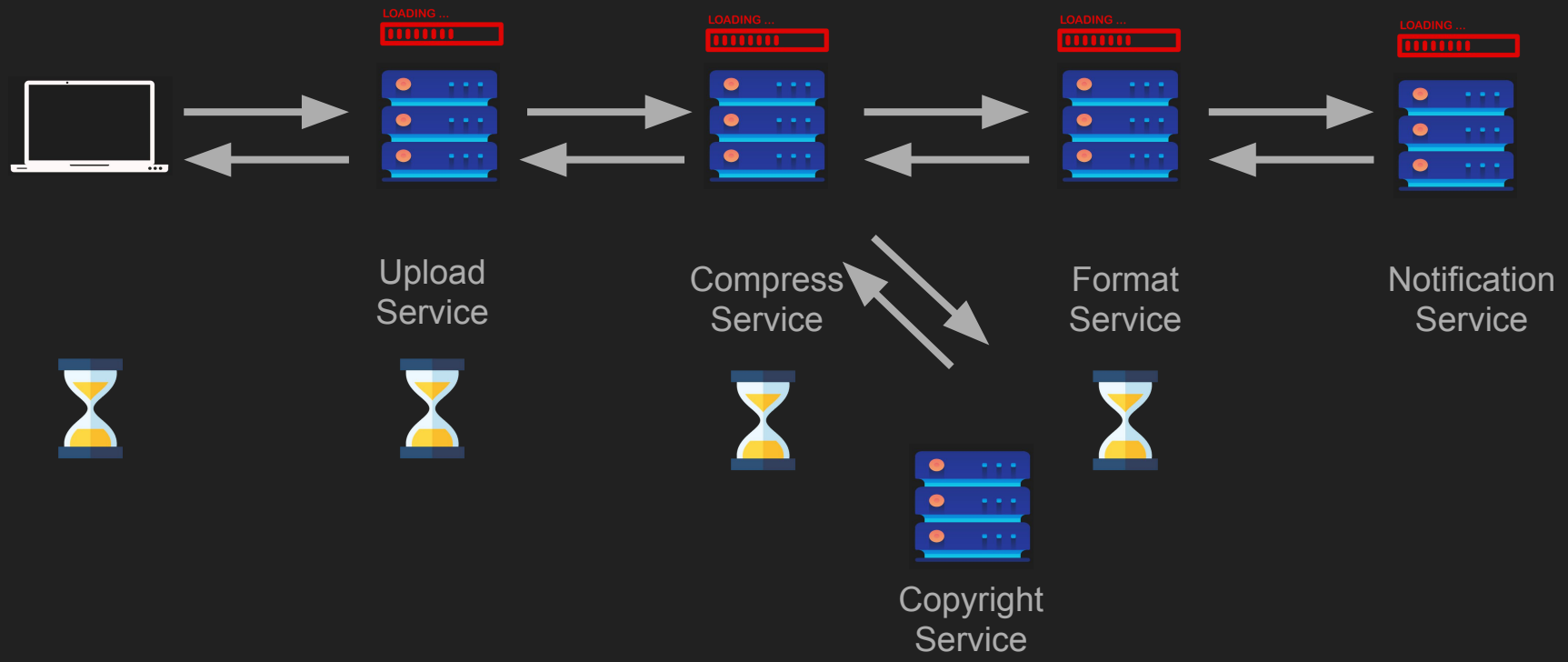
Publish Subscribe

One publisher many readers

Request Response



Where it breaks



Request/Response pros and cons

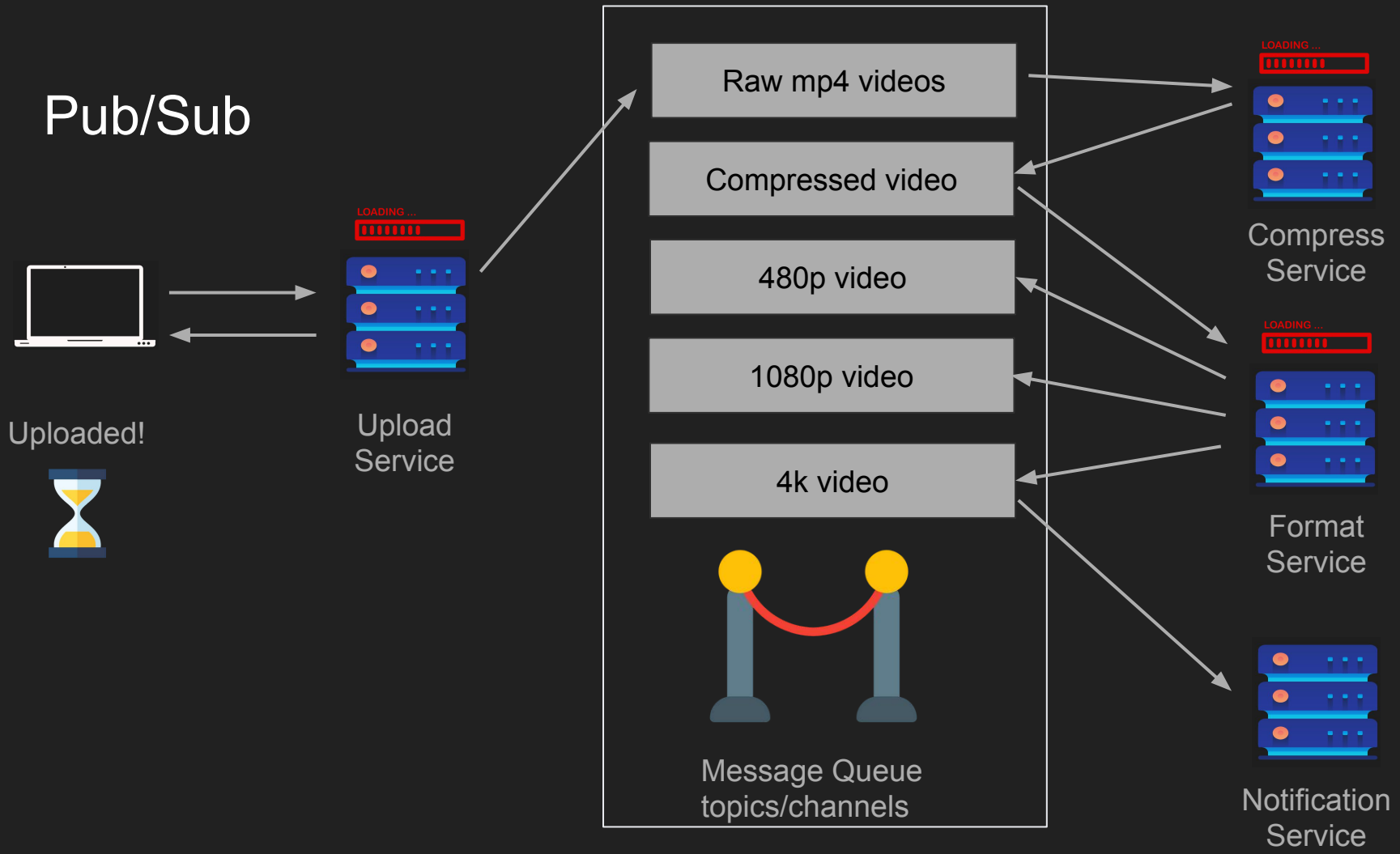
Pros

- Elegant and Simple
- Scalable

Cons

- Bad for multiple receivers
- High Coupling
- Client/Server have to be running
- Chaining, circuit breaking

Pub/Sub


Uploaded!
Upload Service
Raw mp4 videos
Compressed video
480p video
1080p video
4k video
Message Queue topics/channels
Compress Service
Format Service
Notification Service

Pub/Sub pros and cons

Pros

- Scales w/ multiple receivers
- Great for microservices
- Loose Coupling
- Works while clients not running

Cons

- Message delivery issues (Two generals problem)
- Complexity
- Network saturation

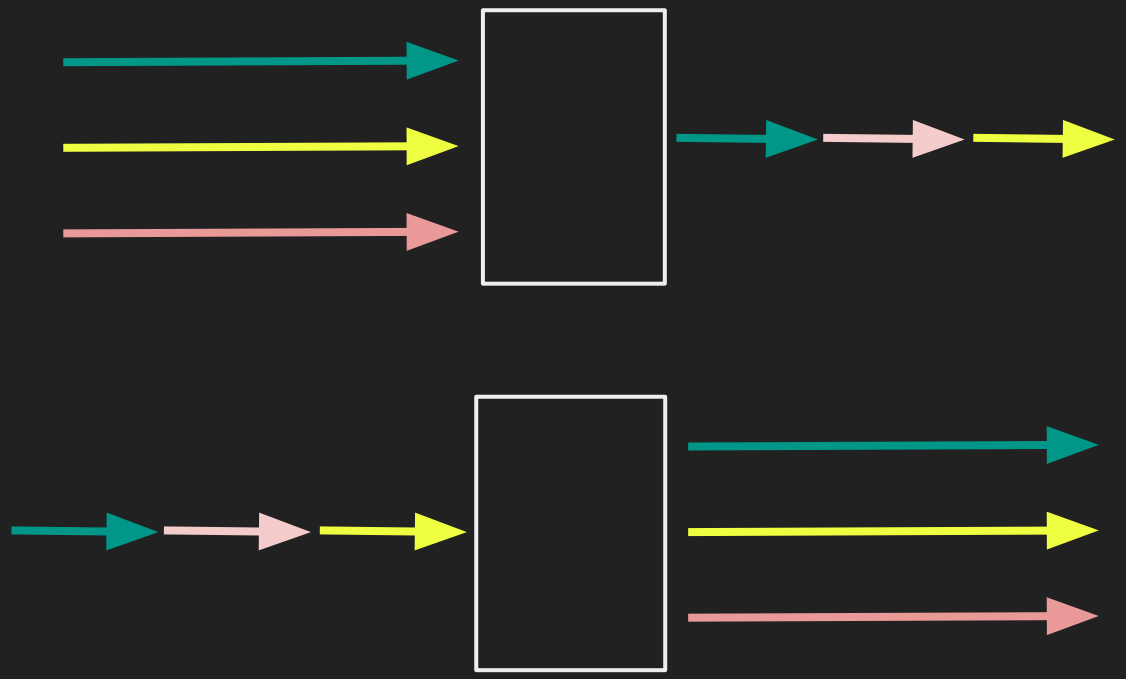
RabbitMQ Demo

Publish a job and have worker pick them up

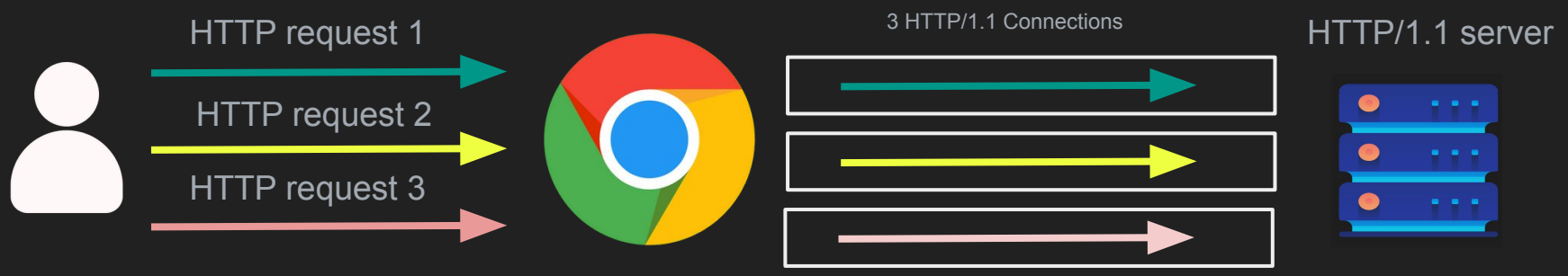
Multiplexing vs Demultiplexing

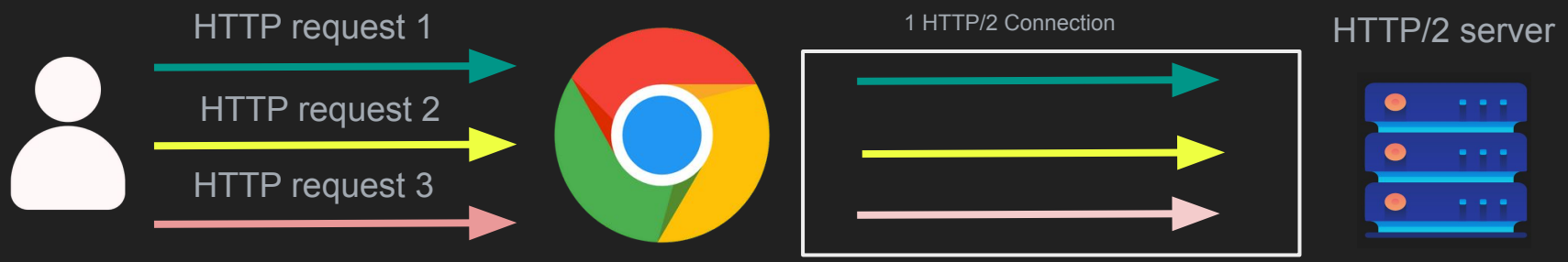
HTTP/2, QUIC,
Connection Pool, MPTCP

Multiplexing vs Demultiplexing

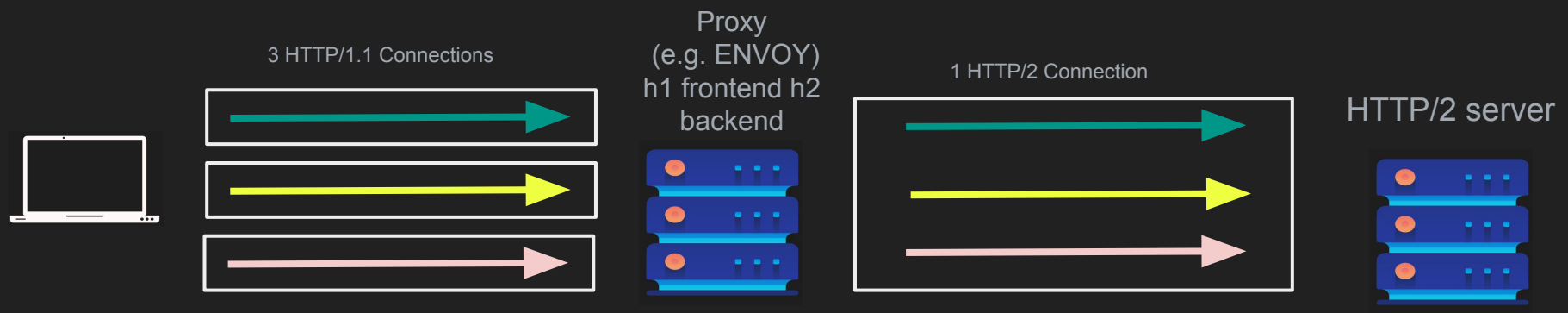


Multiplexing example HTTP/2



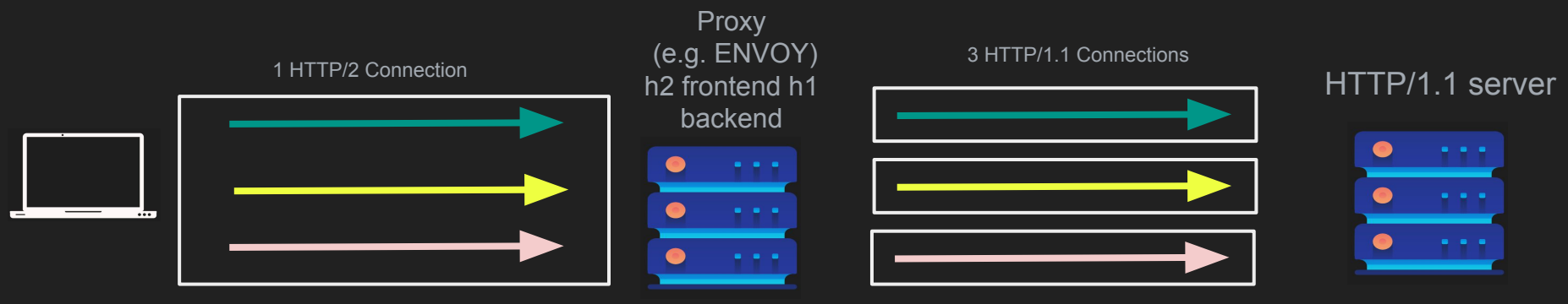


Multiplexing HTTP/2 on the Backend



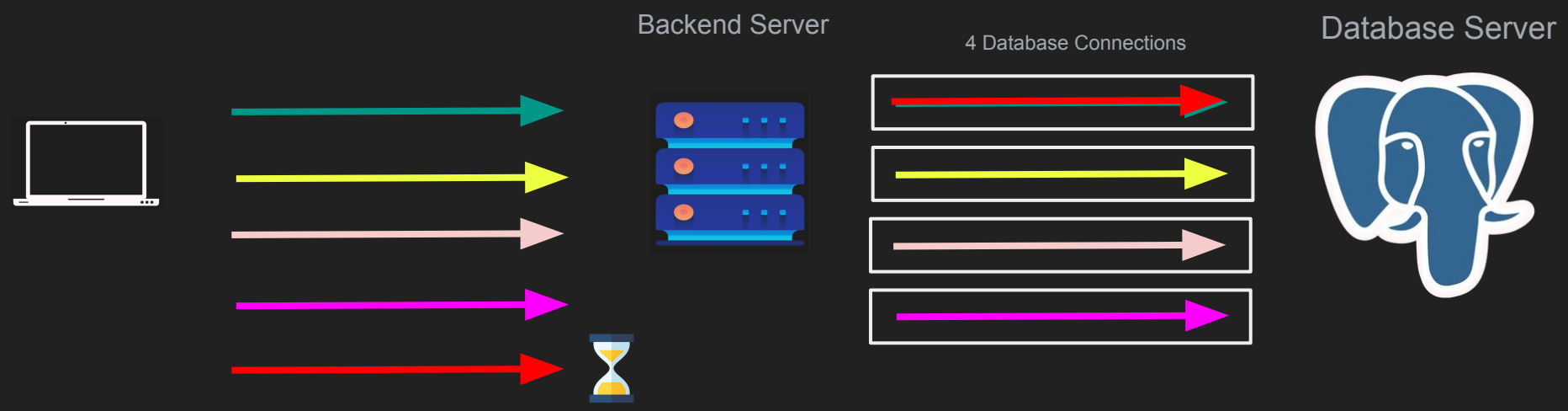
More throughput
High backend resources (CPU for h2)

Demultiplexing HTTP/1.1 on the Backend

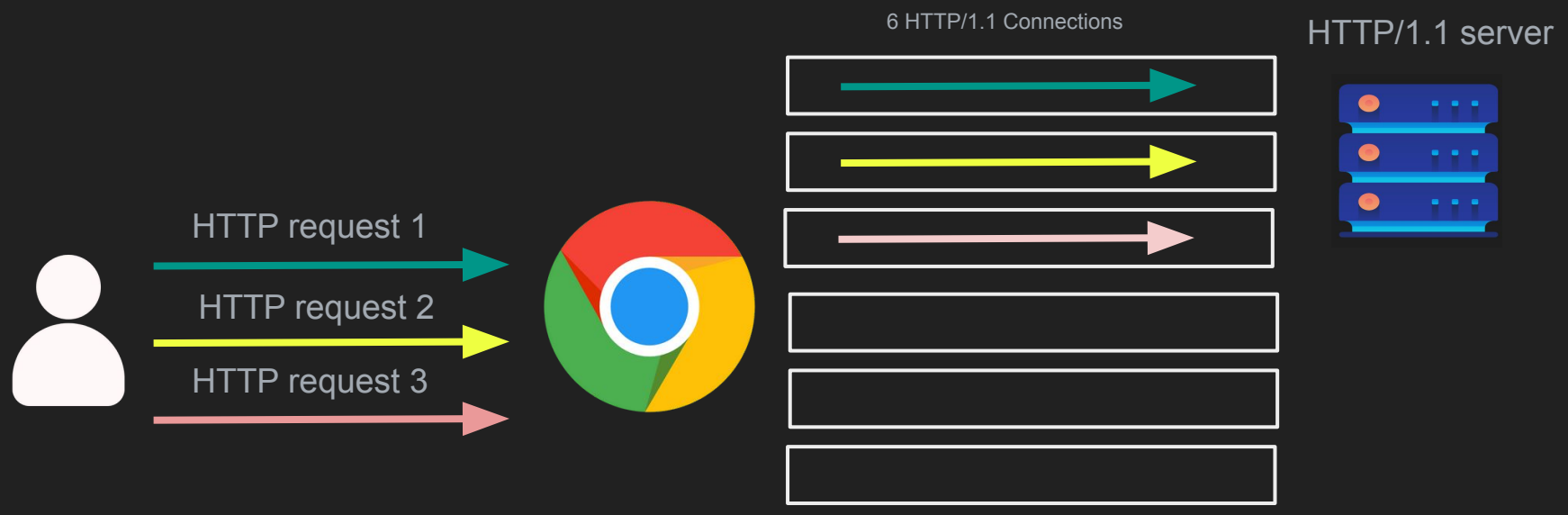


less throughput
low backend resources (simple h1)

Connection Pooling



Browser demultiplexing in HTTP/1.1



Note: Chromes allows up to 6 connections per domain, user's requests are demultiplexed in the 6 connections

Browser connection pool Demo

Up to 6 connections in HTTP/1.1
After that we are blocked

Stateless vs Stateful

Is state stored in the backend?

Stateful vs Stateless backend

- Stateful
  - Stores state about clients in its memory
  - Depends on the information being there
- Stateless
  - Client is responsible to “transfer the state” with every request
  - May store but can safely lose it

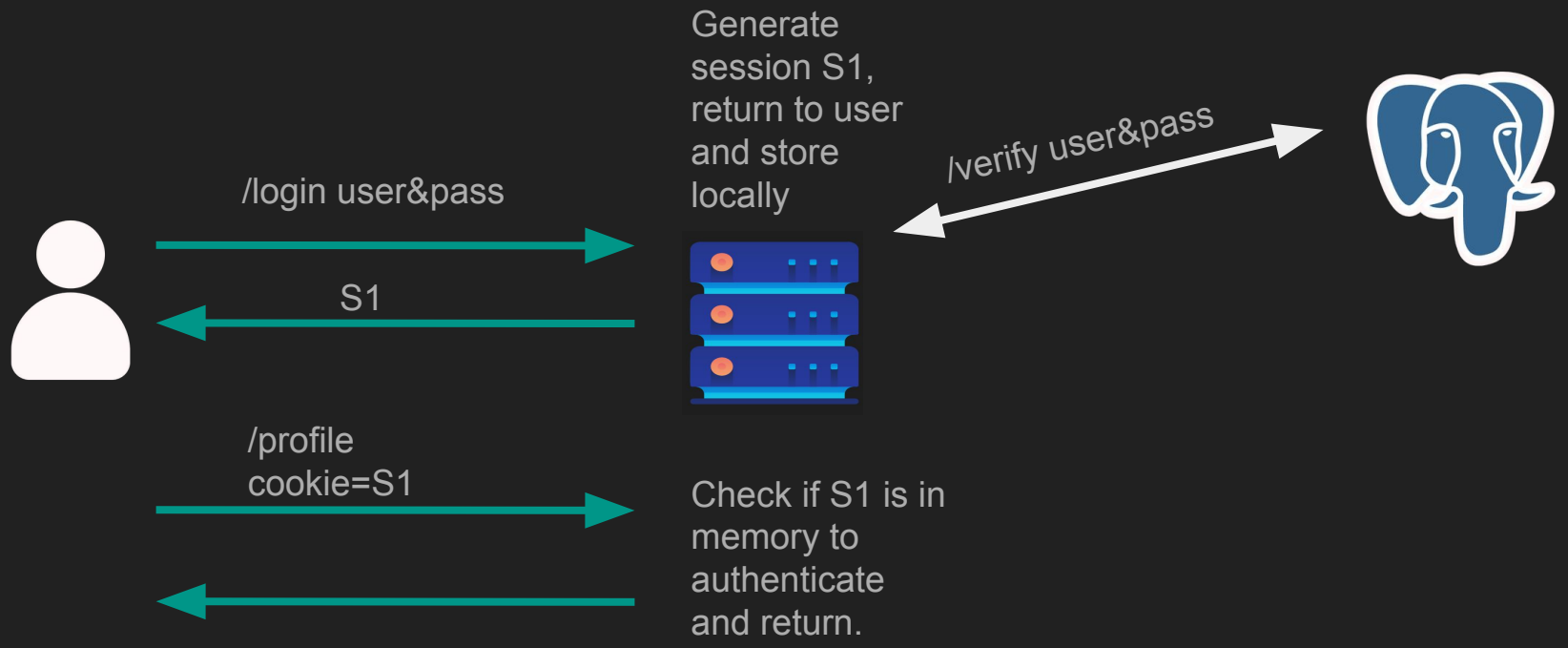
Stateless Backends

- Stateless backends can still store data somewhere else
- Can you restart the backend during idle time while the client workflow continues to work?

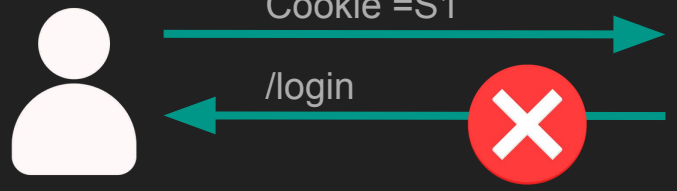
What makes a backend stateless?

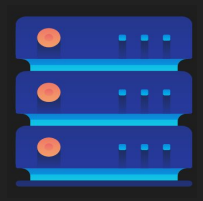
- Stateless backends can store state somewhere else (database)
- The backend remain stateless but the system is stateful
- Can you restart the backend during idle time and the client workflow continue to work?

Stateful backend

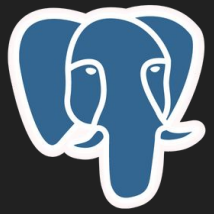


Stateful backend where it breaks

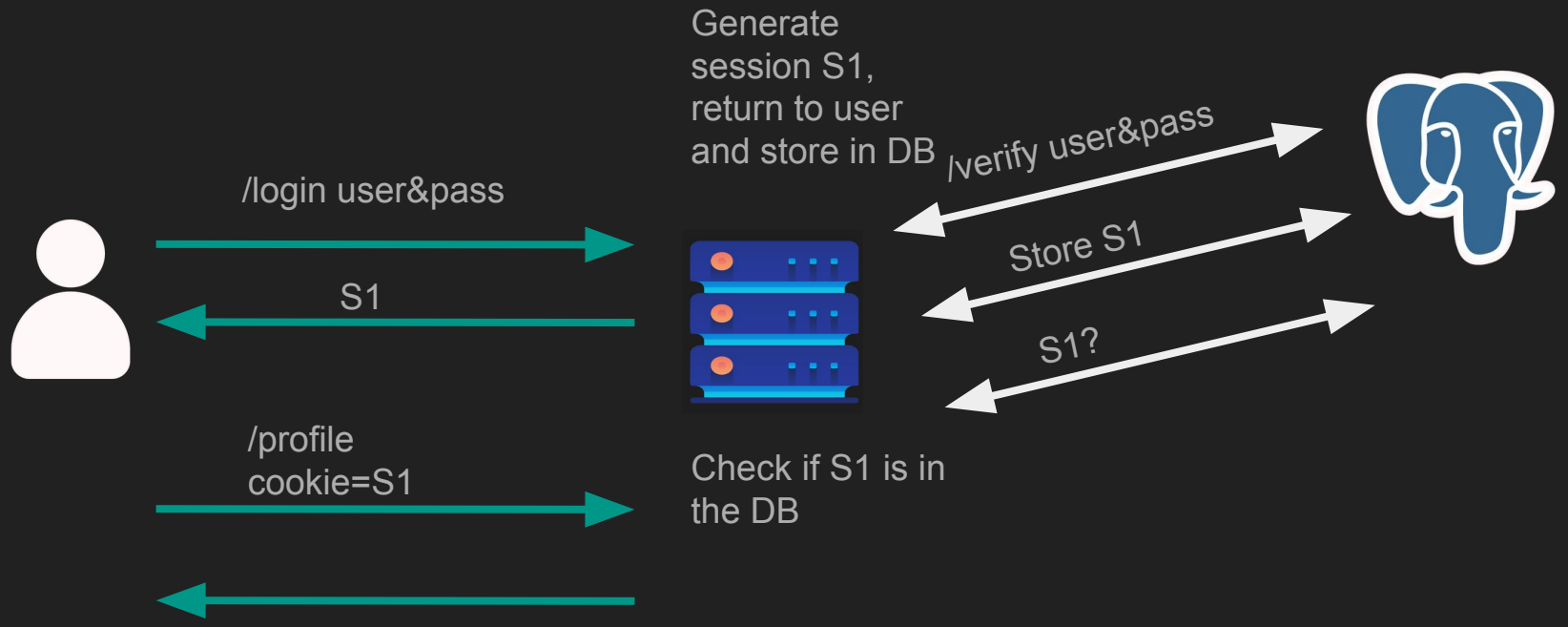




Backend restarted
Sessions are empty



Stateless backend



Backend App is stateless but entire system still stateful (database dies we lose everything)

Stateless vs Stateful protocols

- The Protocols can be designed to store state
- TCP is stateful
  - Sequences, Connection file descriptor
- UDP is stateless
  - DNS send queryID in UDP to identify queries
  - QUIC sends connectionID to identify connection

Stateless vs Stateful protocols

- You can build a stateless protocol on top of a stateful one and vise versa
- HTTP on top of TCP
- If TCP breaks, HTTP blindly create another one
- QUIC on top UDP

Complete Stateless System

- Stateless Systems are rare
- State is carried with every request
- A backend service that relies completely on the input
  - Check if input param is a prime number
- JWT (JSON Web Token)
- Definitions go nowhere

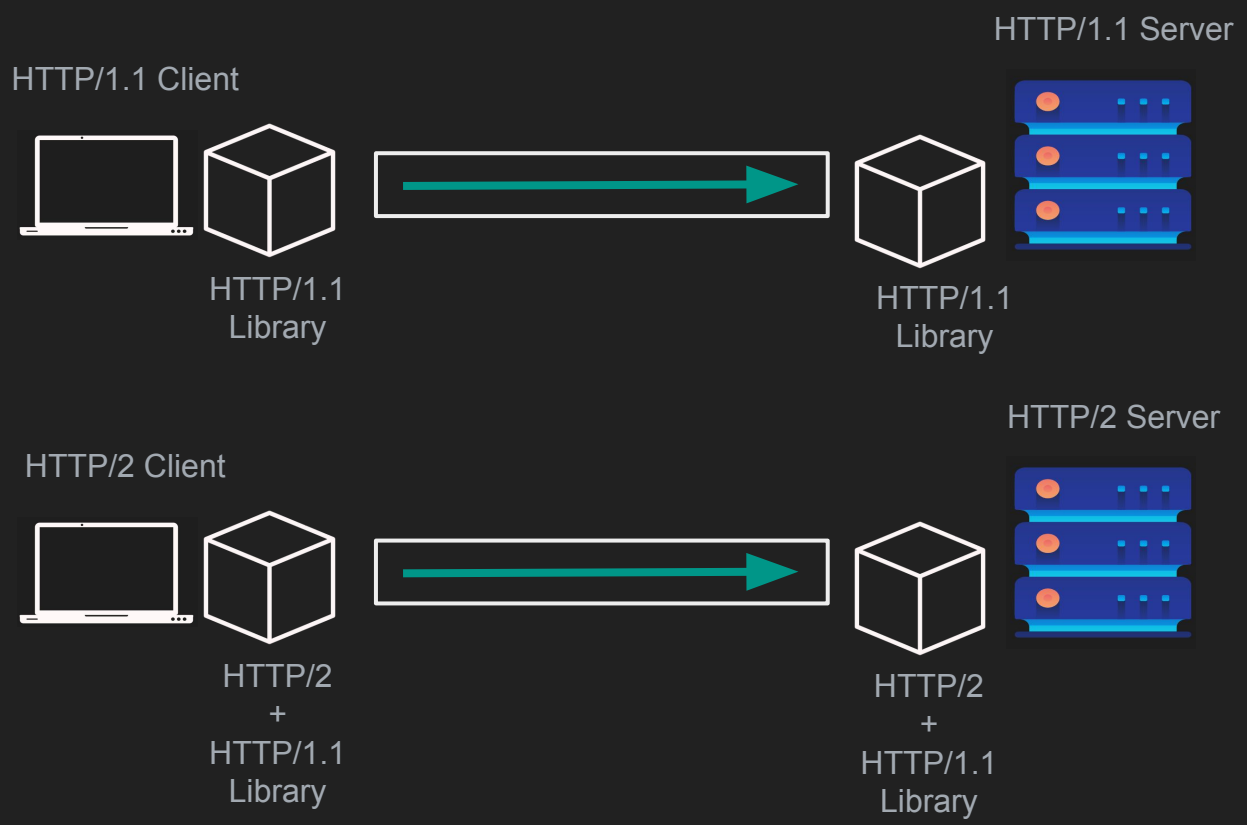
HTTP vs TCP Demo

Restart a TCP server vs Restart an HTTP Server

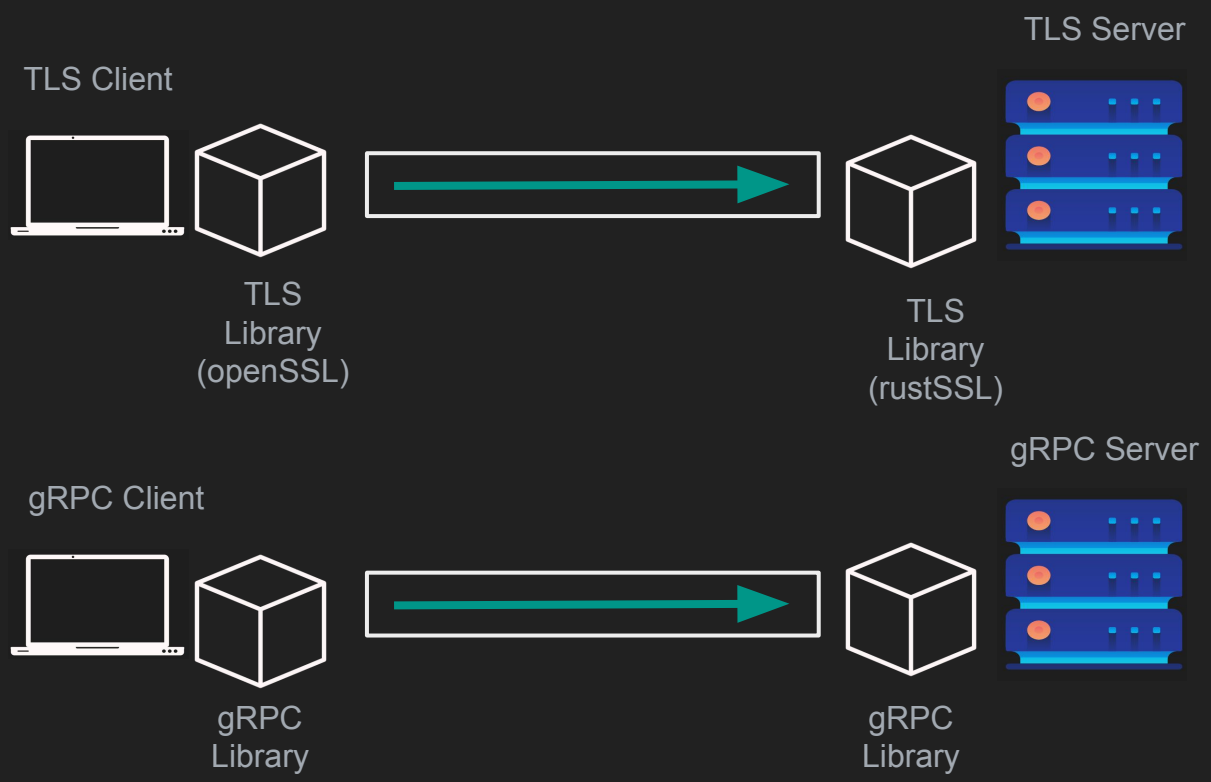
Sidecar Pattern

Thick clients, Thicker backends

Every protocol requires a library



Every protocol requires a library

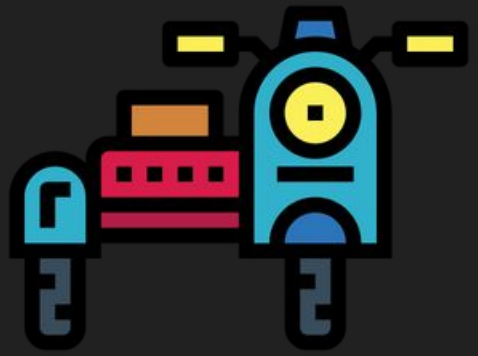


Changing the library is hard

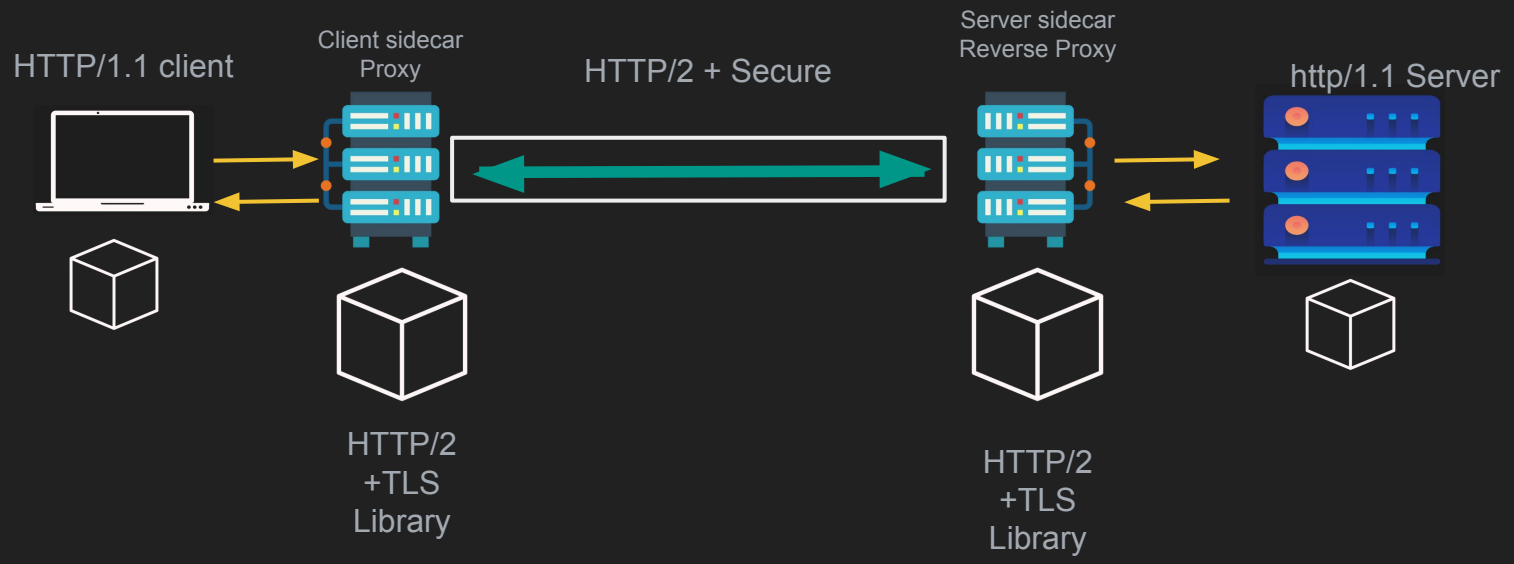
- Once you use the library your app is entrenched
- App & Library “should” be same language
- Changing the library require retesting
- Breaking changes Backward compatibility
- Adding features to the library is hard
- Microservices suffer

When if we delegate communication?

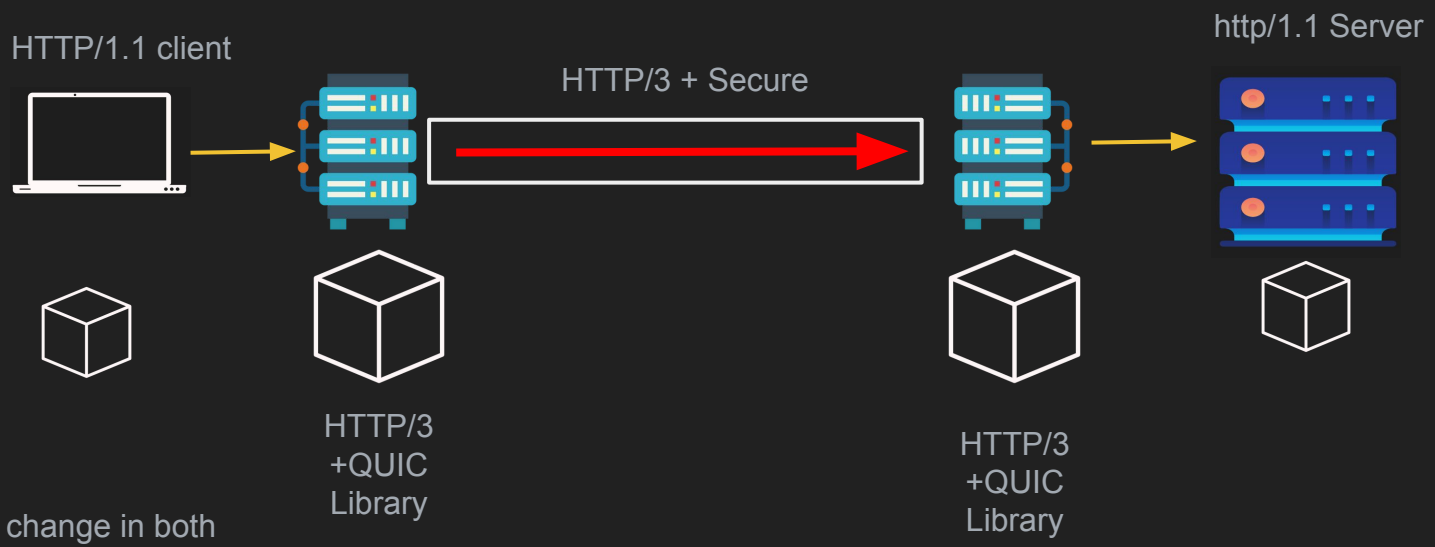
- Proxy communicate instead
- Proxy has the rich library
- Client has thin library (e.g. h1)
- Meet Sidecar pattern
- Each client must have a sidecar proxy



Sidecar Design Pattern



Sidecar Design Pattern - Upgrade to HTTP/3!



No change in both client and server code, just the proxies
But introduces an extra hop

Sidecar Examples

- Service Mesh Proxies
  - Linkerd, Istio, Envoy
- Sidecar Proxy Container
- Must be Layer 7 Proxy

Pros & Cons of Sidecar proxy

Pros

- Language agnostic (polyglot)
- Protocol upgrade
- Security
- Tracing and Monitoring
- Service Discovery
- Caching

Cons

- Complexity
- Latency

Protocols

How the client and backend communicate

Protocol Properties

What to take into account when designing a protocol?

What is a protocol?

- A system that allows two parties to communicate
- A protocol is designed with a set of properties
- Depending on the purpose of the protocol
- TCP, UDP, HTTP, gRPC, FTP

Protocol properties

- Data format
  - Text based (plain text, JSON, XML)
  - Binary (protobuf, RESP, h2, h3)
- Transfer mode
  - Message based (UDP, HTTP)
  - Stream (TCP, WebRTC)
- Addressing system
  - DNS name, IP, MAC
- Directionality
  - Bidirectional (TCP)
  - Unidirectional (HTTP)
  - Full/Half duplex

Protocol properties

- State
  - Stateful (TCP, gRPC, apache thrift)
  - Stateless (UDP, HTTP)
- Routing
  - Proxies, Gateways
- Flow & Congestion control
  - TCP (Flow & Congestion)
  - UDP (No control)
- Error management
  - Error code
  - Retries and timeouts

OSI Model

Open Systems Interconnection model

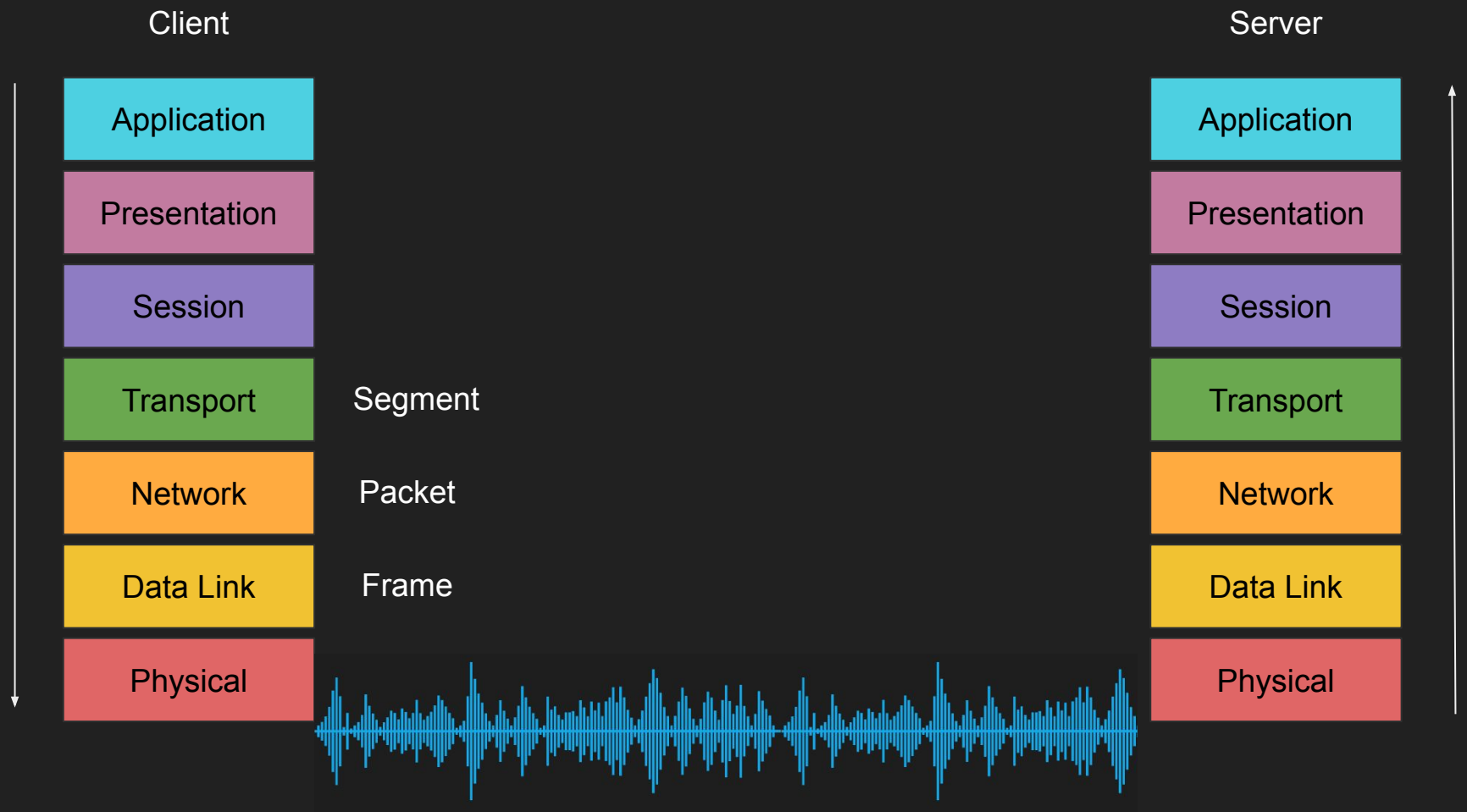
Why do we need a communication model?

- Agnostic applications
  - App doesn’t need to to know network medium
  - Otherwise we need an App for WIFI, ethernet vs LTE vs fiber
- Network Equipment Management
  - Without a standard model, upgrading network equipments becomes difficult
- Decoupled Innovation
  - Innovations can be done in each layer separately without affecting the rest of the models

What is the OSI Model?

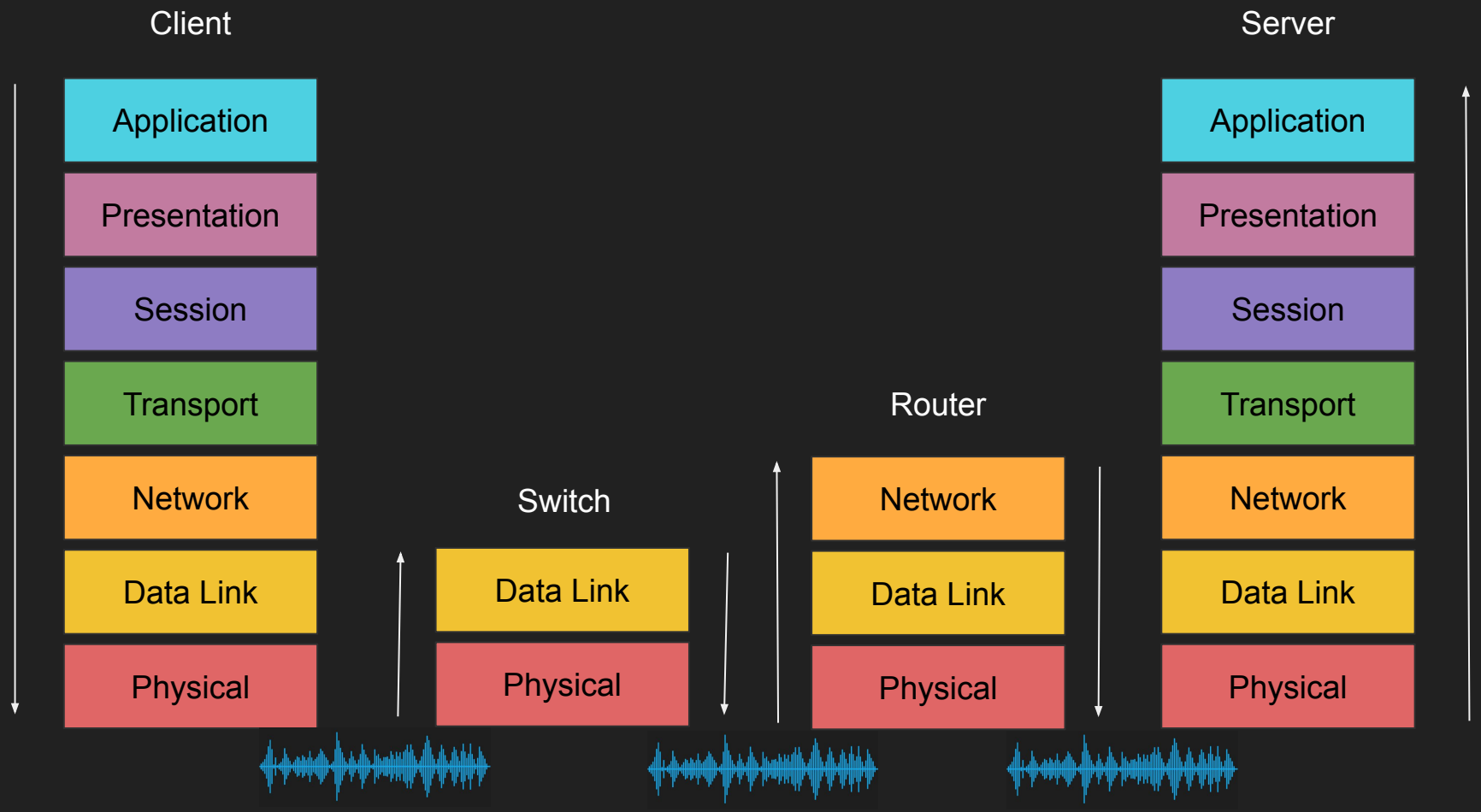
- 7 Layers each describe a specific networking component
- Layer 7 - Application - HTTP/FTP/gRPC
- Layer 6 - Presentation - Encoding, Serialization
- Layer 5 - Session - Connection establishment, TLS
- Layer 4 - Transport - UDP/TCP
- Layer 3 - Network - IP
- Layer 2 - Data link - Frames, Mac address Ethernet
- Layer 1 - Physical - Electric signals, fiber or radio waves

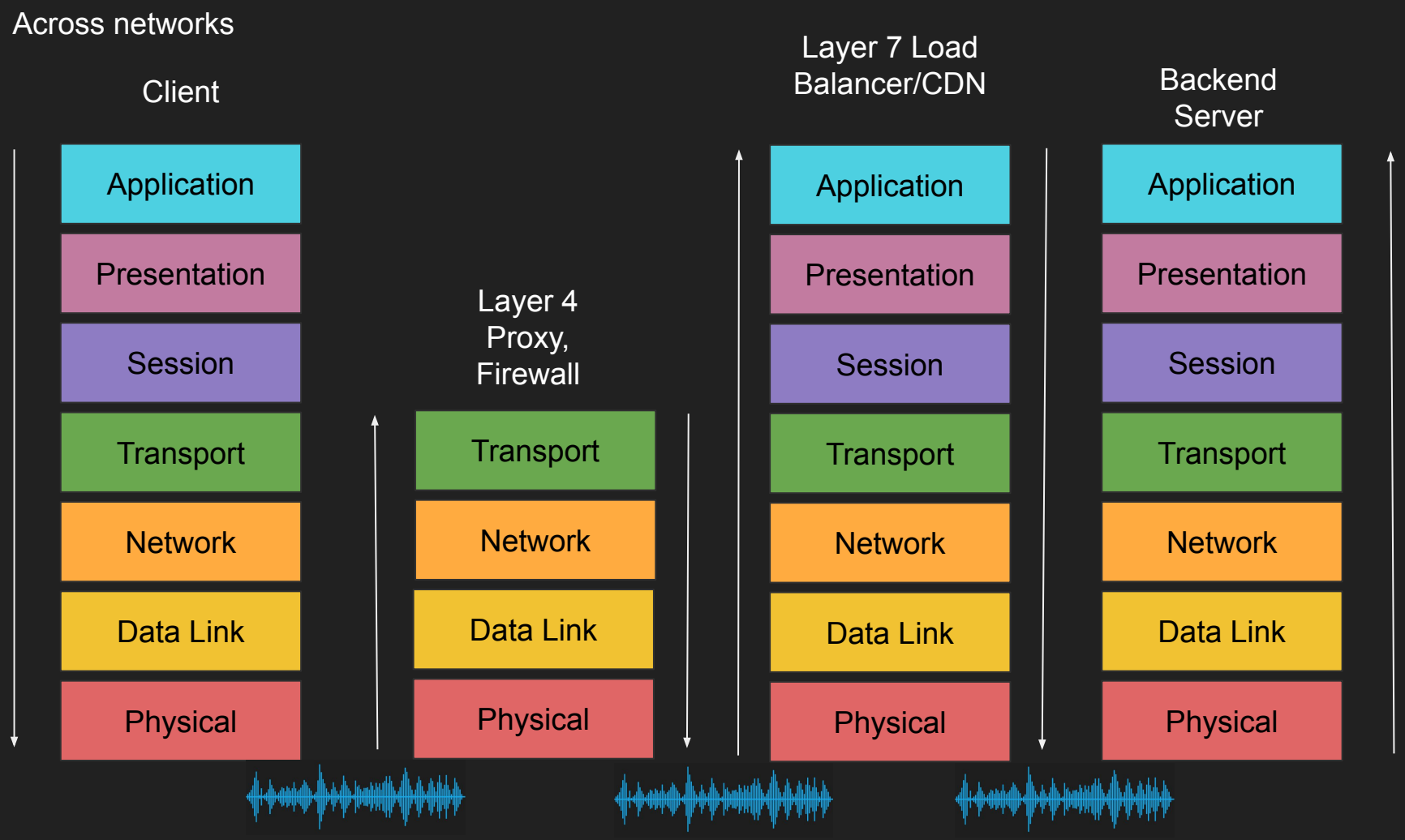
Client sends an HTTPS POST request



Client
Server
SPORT
DPORT
SIP
DIP
SMAC
DMAC
SPORT
DPORT
SIP
DIP
SMAC
DMAC

Across networks




Across networks
Client
Application
Presentation
Session
Transport
Network
Data Link
Physical
Layer 4
Proxy,
Firewall
Transport
Network
Data Link
Physical
Layer 7 Load
Balancer/CDN
Application
Presentation
Session
Transport
Network
Data Link
Physical
Backend
Server
Application
Presentation
Session
Transport
Network
Data Link
Physical

The shortcomings of the OSI Model

- OSI Model has too many layers which can be hard to comprehend
- Hard to argue about which layer does what
- Simpler to deal with Layers 5-6-7 as just one layer, application
- TCP/IP Model does just that

TCP/IP Model

- Much simpler than OSI just 4 layers
- Application (Layer 5, 6 and 7)
- Transport (Layer 4)
- Internet (Layer 3)
- Data link (Layer 2)
- Physical layer is not officially covered in the model

OSI Model Summary

- Why do we need a communication model?
- What is the OSI Model?
- Example
- Each device in the network doesn’t have to map the entire 7 layers
- TCP/IP is simpler model

UDP

User Datagram Protocol

UDP

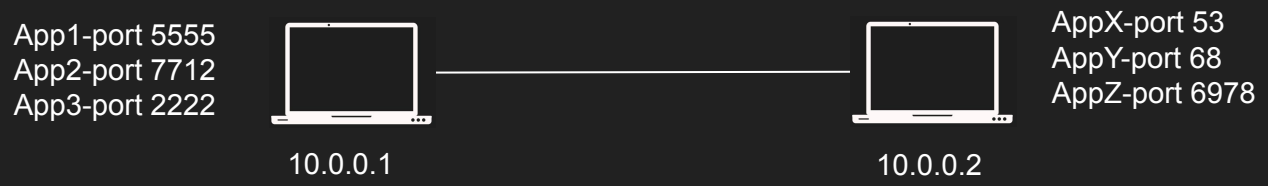
- Stands for User Datagram Protocol
- Message Based Layer 4 protocol
- Ability to address processes in a host using ports
- Simple protocol to send and receive messages
- Prior communication not required (double edge sword)
- Stateless no knowledge is stored on the host
- 8 byte header Datagram

UDP Use cases

- Video streaming
- VPN
- DNS
- WebRTC

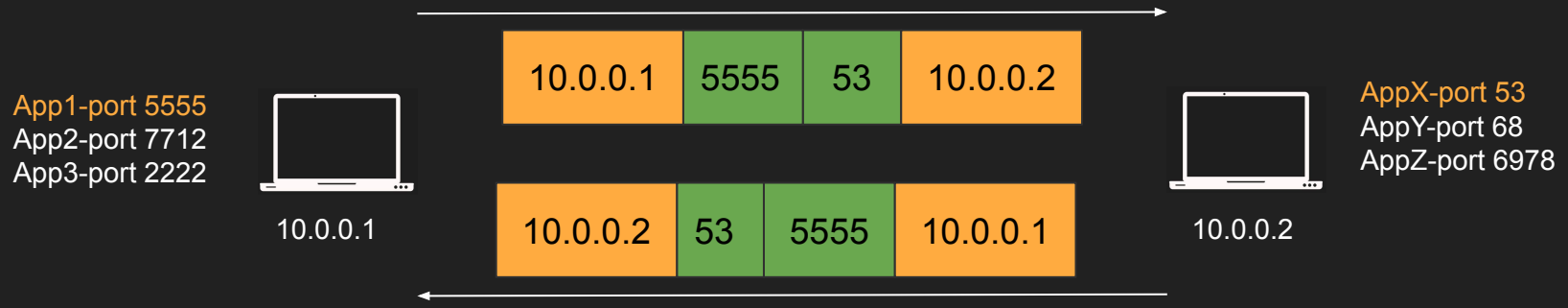
Multiplexing and demultiplexing

- IP target hosts only
- Hosts run many apps each with different requirements
- Ports now identify the “app” or “process”
- Sender multiplexes all its apps into UDP
- Receiver demultiplex UDP datagrams to each app



Source and Destination Port

- App1 on 10.0.0.1 sends data to AppX on 10.0.0.2
- Destination Port = 53
- AppX responds back to App1
- We need Source Port so we know how to send back data
- Source Port = 5555



UDP Pros

- Simple protocol
- Header size is small so datagrams are small
- Uses less bandwidth
- Stateless
- Consumes less memory (no state stored in the server/client)
- Low latency - no handshake , order, retransmission or guaranteed delivery

UDP Cons

- No acknowledgement
- No guarantee delivery
- Connection-less - anyone can send data without prior knowledge
- No flow control
- No congestion control
- No ordered packets
- Security - can be easily spoofed

Summary

- UDP is a simple message based layer 4 protocol
- Uses ports to address processes
- Stateless
- Pros & Cons

TCP

Transmission Control Protocol

TCP

- Stands for Transmission Control Protocol
- Stream based Layer 4 protocol
- Ability to address processes in a host using ports
- “Controls” the transmission unlike UDP which is a firehose
- Connection
- Requires handshake
- 20 bytes headers Segment (can go to 60)
- Stateful

TCP Use cases

- Reliable communication
- Remote shell
- Database connections
- Web communications
- Any bidirectional communication

TCP Connection

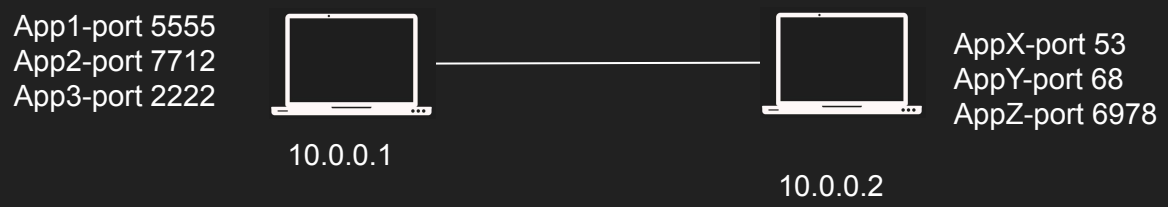
- Connection is a Layer 5 (session)
- Connection is an agreement between client and server
- Must create a connection to send data
- Connection is identified by 4 properties
  - SourceIP-SourcePort
  - DestinationIP-DestinationPort

TCP Connection

- Can’t send data outside of a connection
- Sometimes called socket or file descriptor
- Requires a 3-way TCP handshake
- Segments are sequenced and ordered
- Segments are acknowledged
- Lost segments are retransmitted

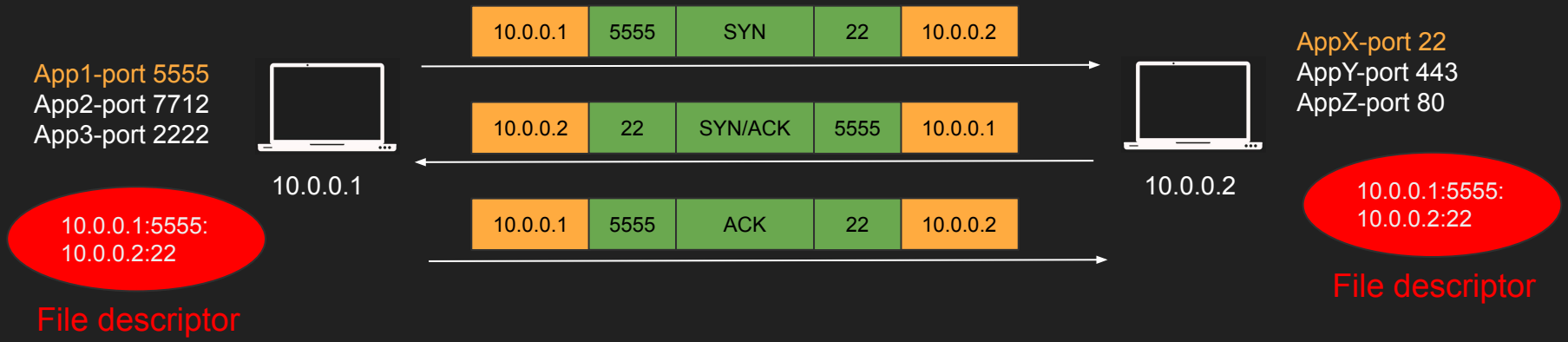
Multiplexing and demultiplexing

- IP target hosts only
- Hosts run many apps each with different requirements
- Ports now identify the “app” or “process”
- Sender multiplexes all its apps into TCP connections
- Receiver demultiplex TCP segments to each app based on connection pairs



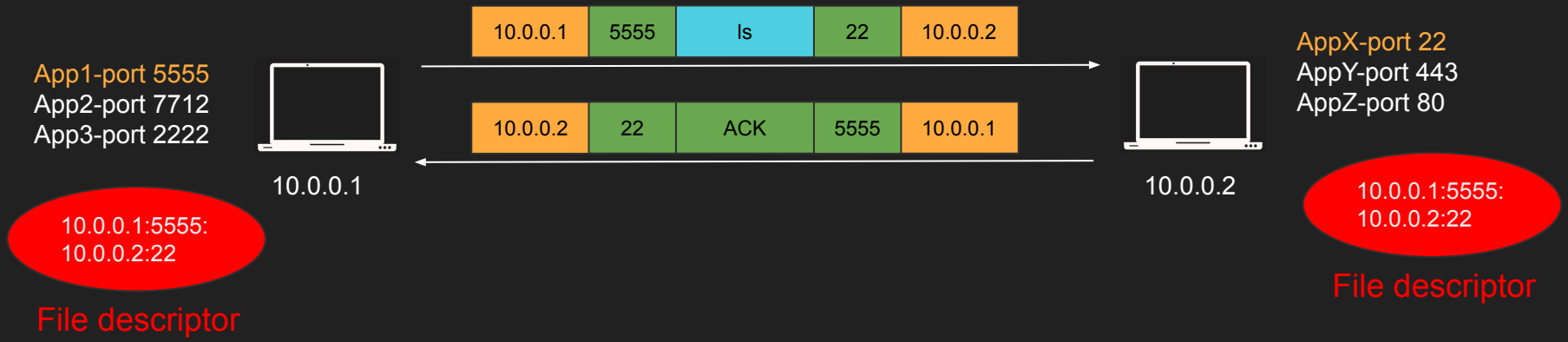
Connection Establishment

- App1 on 10.0.0.1 want to send data to AppX on 10.0.0.2
- App1 sends SYN to AppX to synchronous sequence numbers
- AppX sends SYN/ACK to synchronous its sequence number
- App1 ACKs AppX SYN.
- Three way handshake



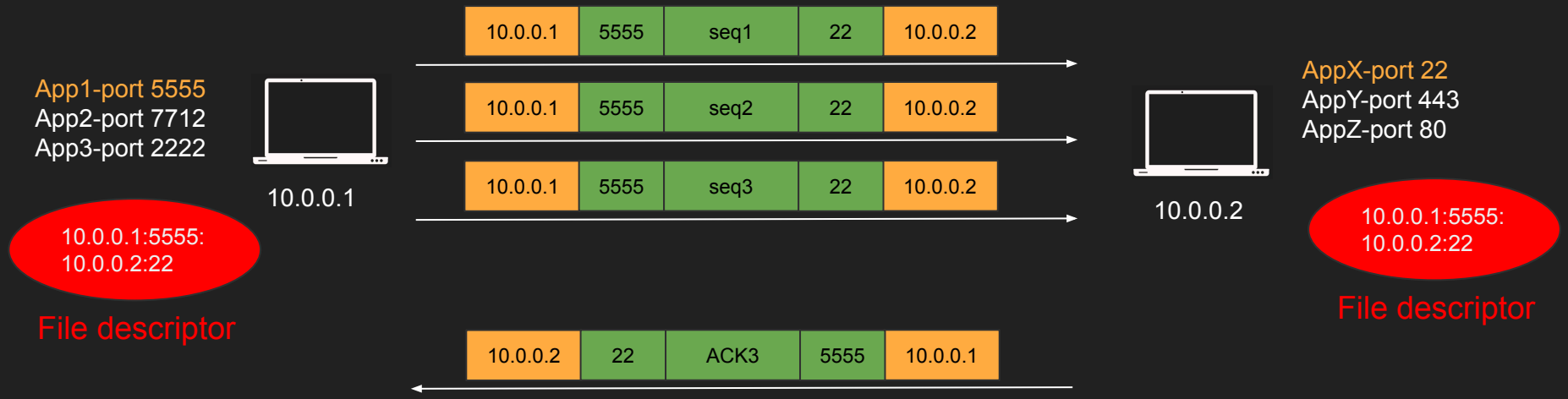
Sending data

- App1 sends data to AppX
- App1 encapsulate the data in a segment and send it
- AppX acknowledges the segment
- Hint: Can App1 send new segment before ack of old segment arrives?



Acknowledgment

- App1 sends segment 1,2 and 3 to AppX
- AppX acknowledge all of them with a single ACK 3

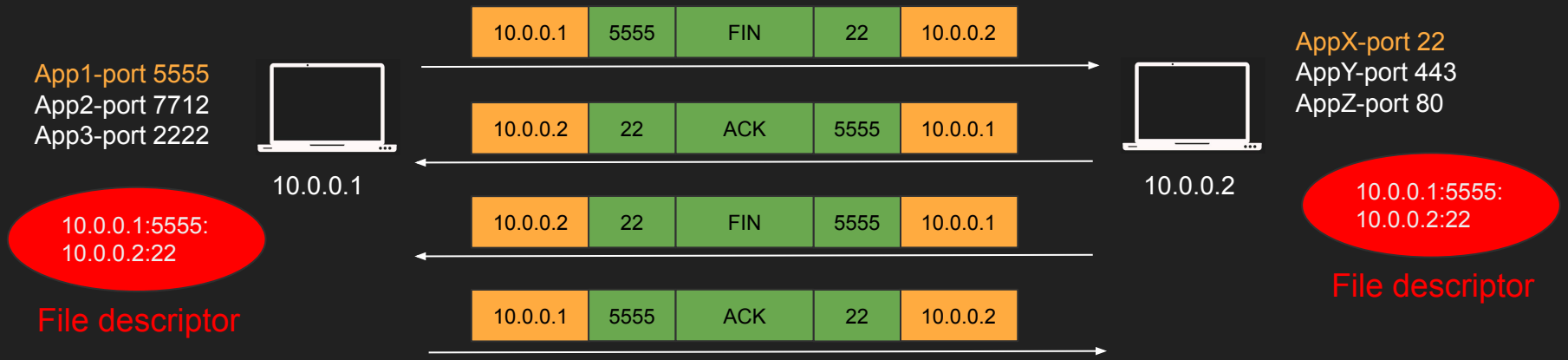


Lost data

- App1 sends segment 1,2 and 3 to AppX
- Seg 3 is lost, AppX acknowledge 3
- App1 resend Seg 3

Closing Connection

- App1 wants to close the connection
- App1 sends FIN, AppX ACK
- AppX sends FIN, App1 ACK
- Four way handshake



TCP Pros

- Guarantee delivery
- No one can send data without prior knowledge
- Flow Control and Congestion Control
- Ordered Packets no corruption or app level work
- Secure and can’t be easily spoofed

TCP Cons

- Large header overhead compared to UDP
- More bandwidth
- Stateful - consumes memory on server and client
- Considered high latency for certain workloads (Slow start/ congestion/ acks)
- Does too much at a low level (hence QUIC)
  - Single connection to send multiple streams of data (HTTP requests)
  - Stream 1 has nothing to do with Stream 2
  - Both Stream 1 and Stream 2 packets must arrive
- TCP Meltdown
  - Not a good candidate for VPN

Summary

- Stands for Transmission Control Protocol
- Layer 4 protocol
- “Controls” the transmission unlike UDP which is a firehose
- Introduces Connection concept
- Retransmission, acknowledgement, guaranteed delivery
- Stateful, connection has a state
- Pros & Cons

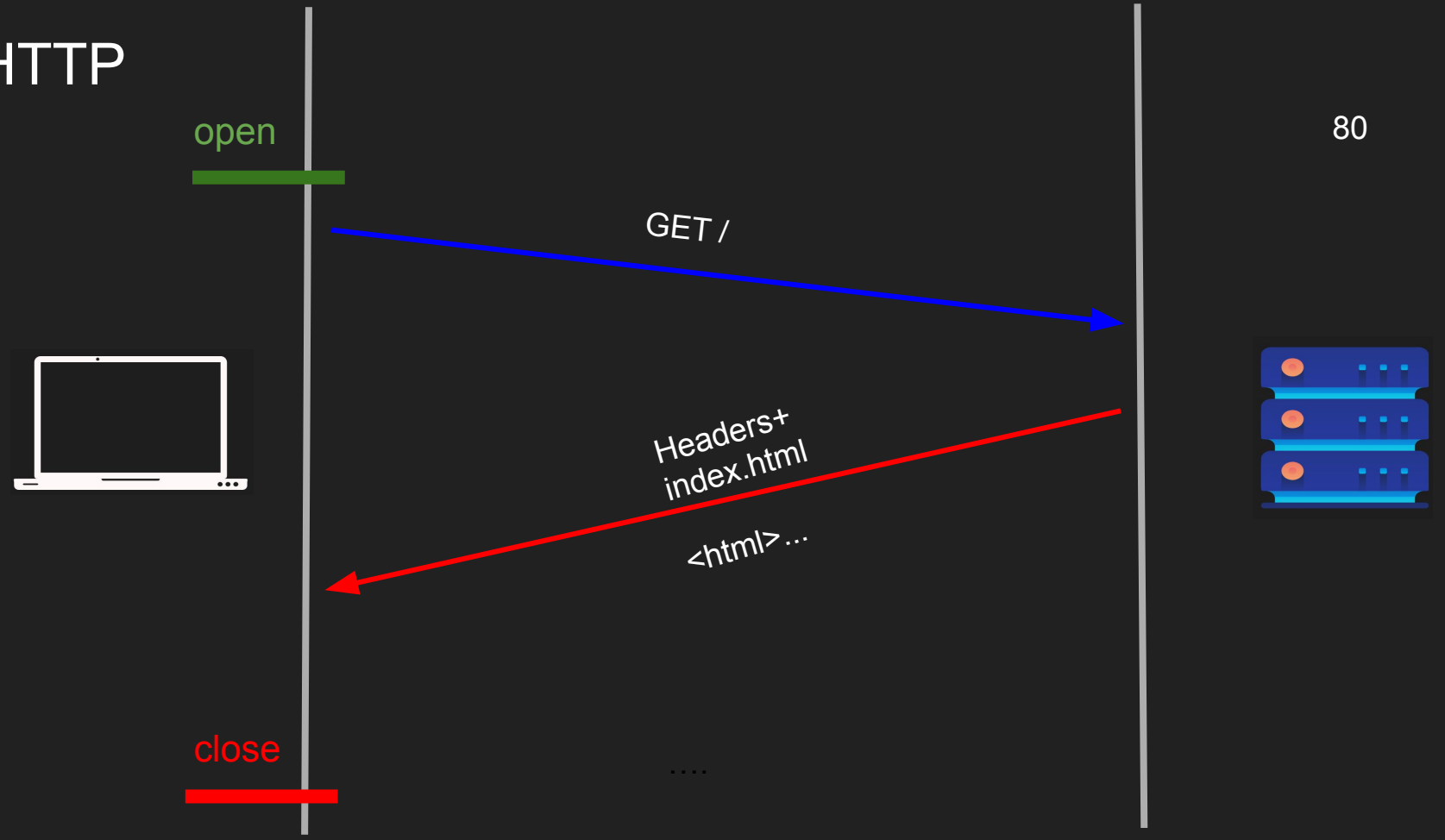
TLS

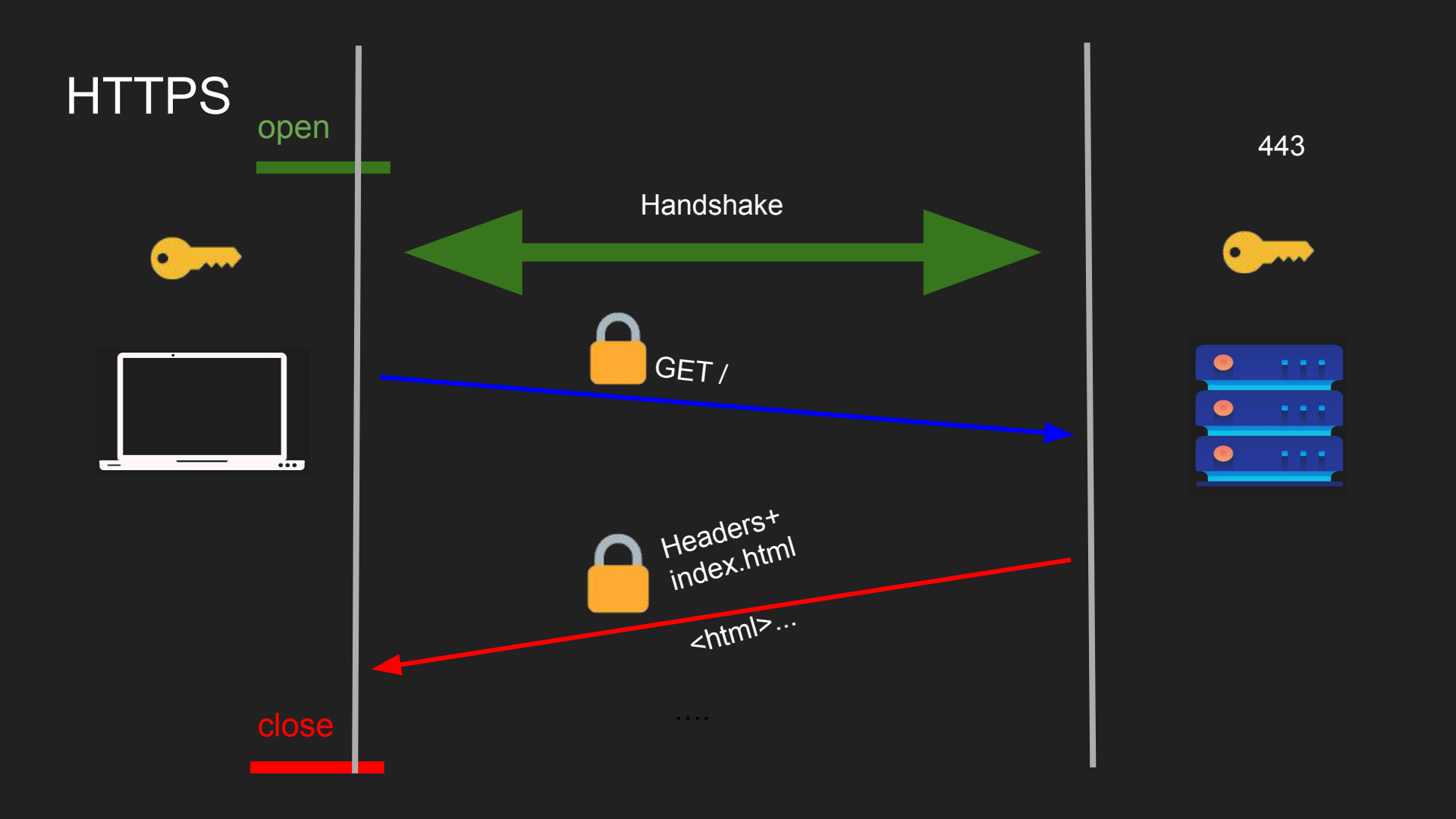
Transport Layer Security

TLS

- Vanilla HTTP
- HTTPS
- TLS 1.2 Handshake
- Diffie Hellman
- TLS 1.3 Improvements

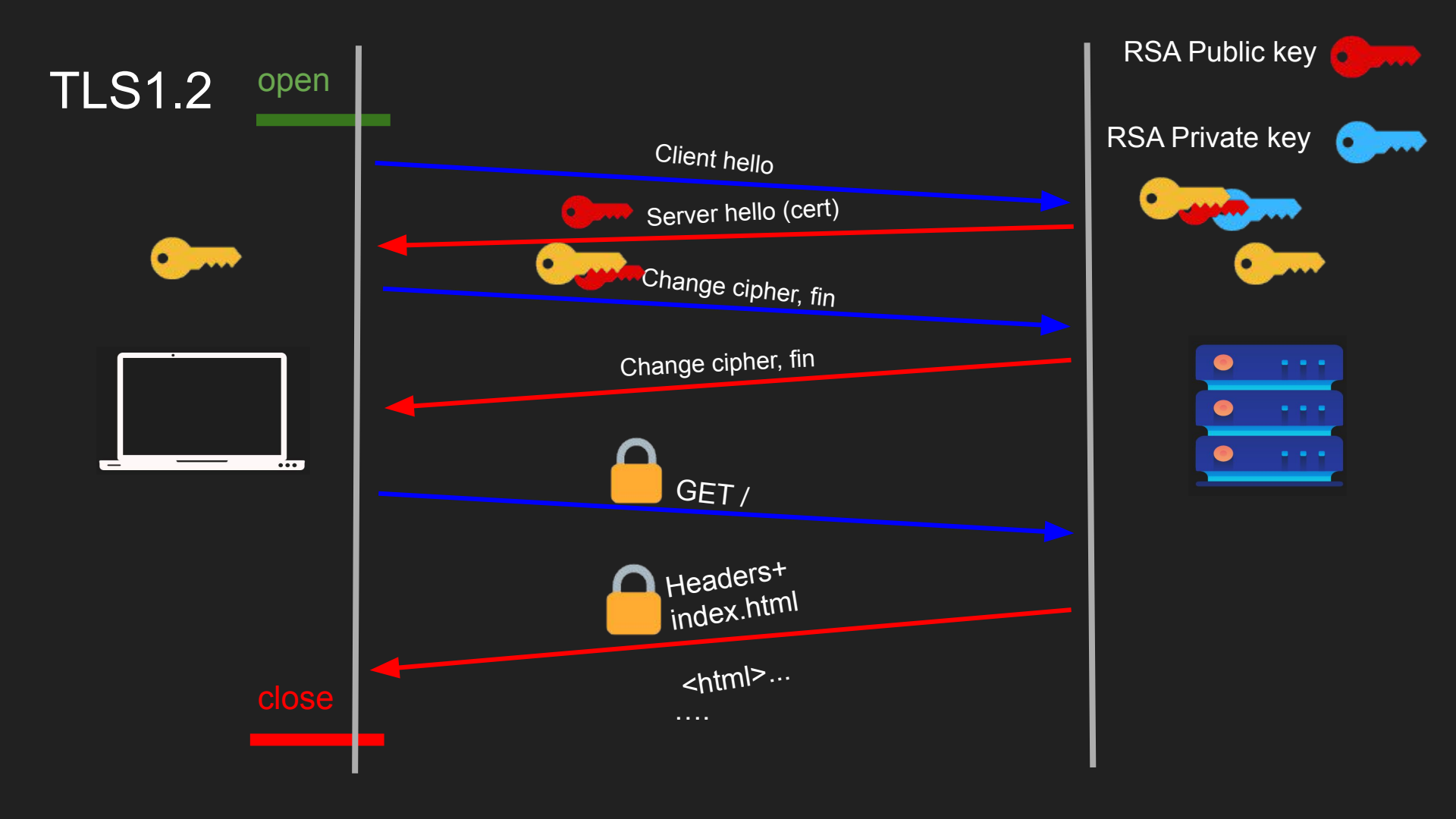
HTTP




HTTPS
open
443
Handshake
GET /
Headers+
index.html
<html>...
....
close

Why TLS

- We encrypt with symmetric key algorithms
- We need to exchange the symmetric key
- Key exchange uses asymmetric key (PKI)
- Authenticate the server
- Extensions (SNI, preshared, 0RTT)


TLS1.2
open
RSA Public key
RSA Private key
Client hello
Server hello (cert)
Change cipher, fin
Change cipher, fin
GET /
Headers+
index.html
<html>...
....
close

Diffie Hellman

Private x

+

Public g,n

+

Private y

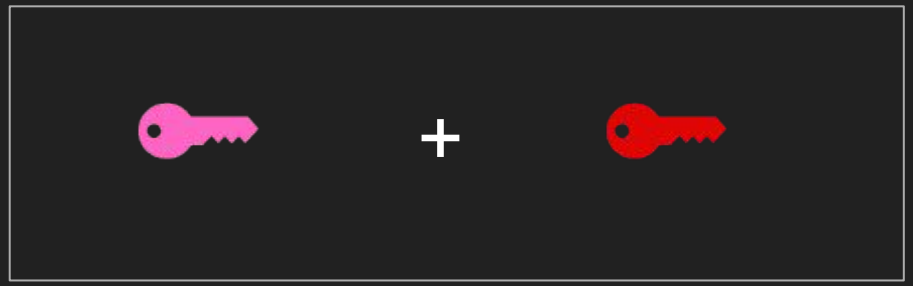
=

Symmetric key

Diffie Hellman

Public/
Unbreakable
/can be shared
g^x % n

Public/
Unbreakable
/can be shared
g^y % n

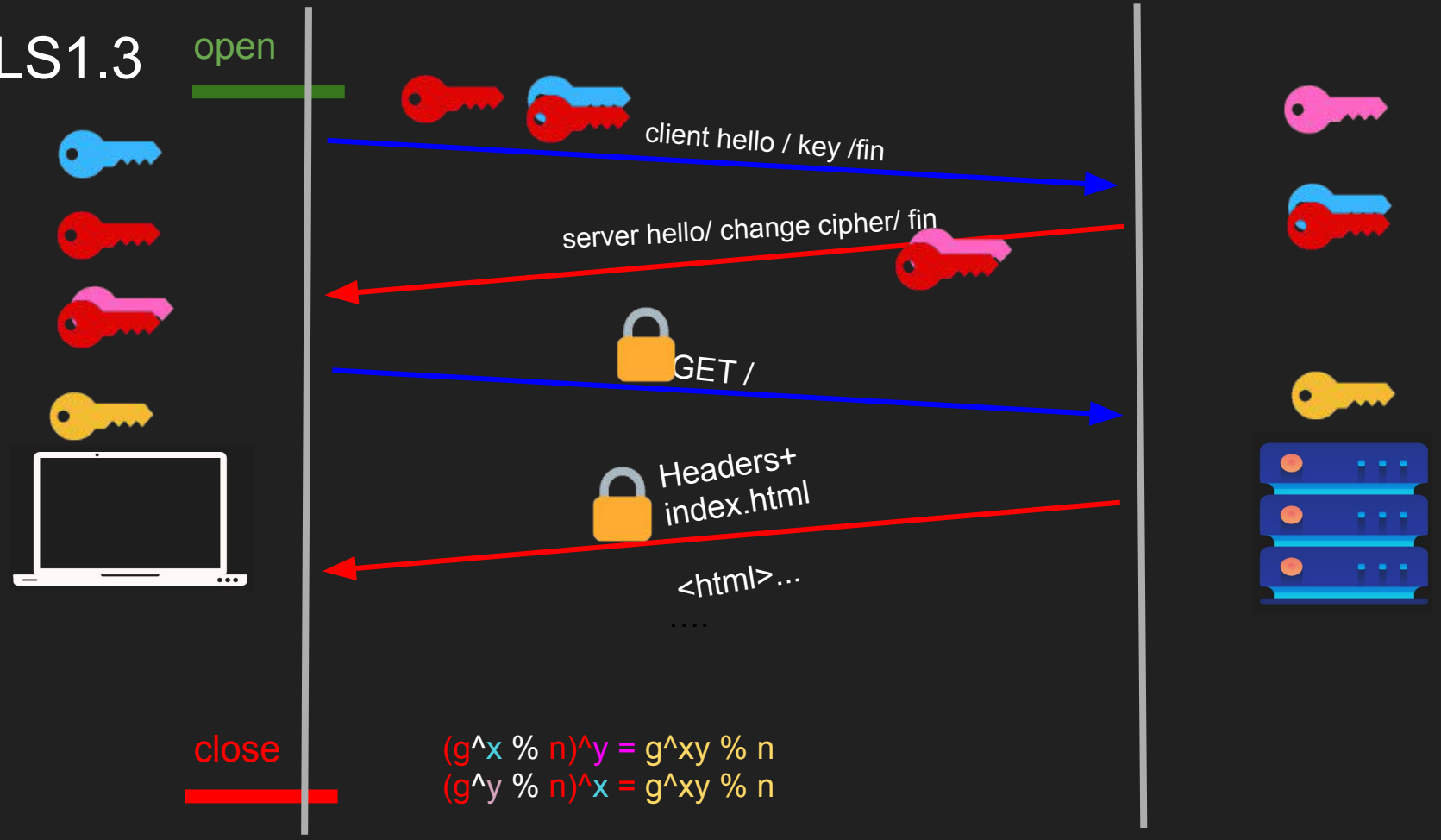


(g^x % n)^y = g^xy % n
(g^y % n)^x = g^xy % n

TLS Extensions

- SNI
  - Server Name Indication
- ALPN
  - Application Layer Protocol Negotiation
- Pre-shared key
  - 0RTT
- ECH
  - Encrypted Client Hello

TLS1.3


open
client hello / key /fin
server hello/ change cipher/ fin
GET /
Headers+
index.html
<html>...
....
close
(g^x % n)^y = g^xy % n
(g^y % n)^x = g^xy % n

TLS Summary

- Vanilla HTTP
- HTTPS
- TLS 1.2 Handshake (two round trips)
- Diffie Hellman
- TLS 1.3 Improvements (one round trip can be zero)

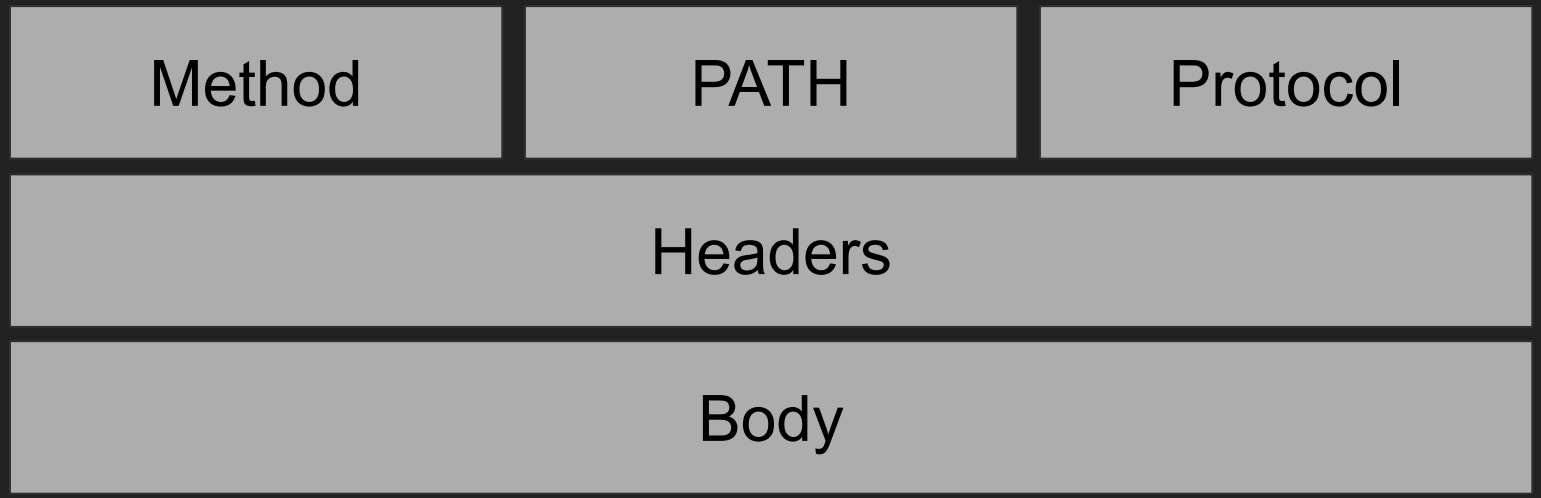
HTTP/1.1

Simple web protocol lasts decades

Client / Server

- (Client) Browser, python or javascript app, or any app that makes HTTP request
- (Server) HTTP Web Server, e.g. IIS, Apache Tomcat, NodeJS, Python Tornado

HTTP Request

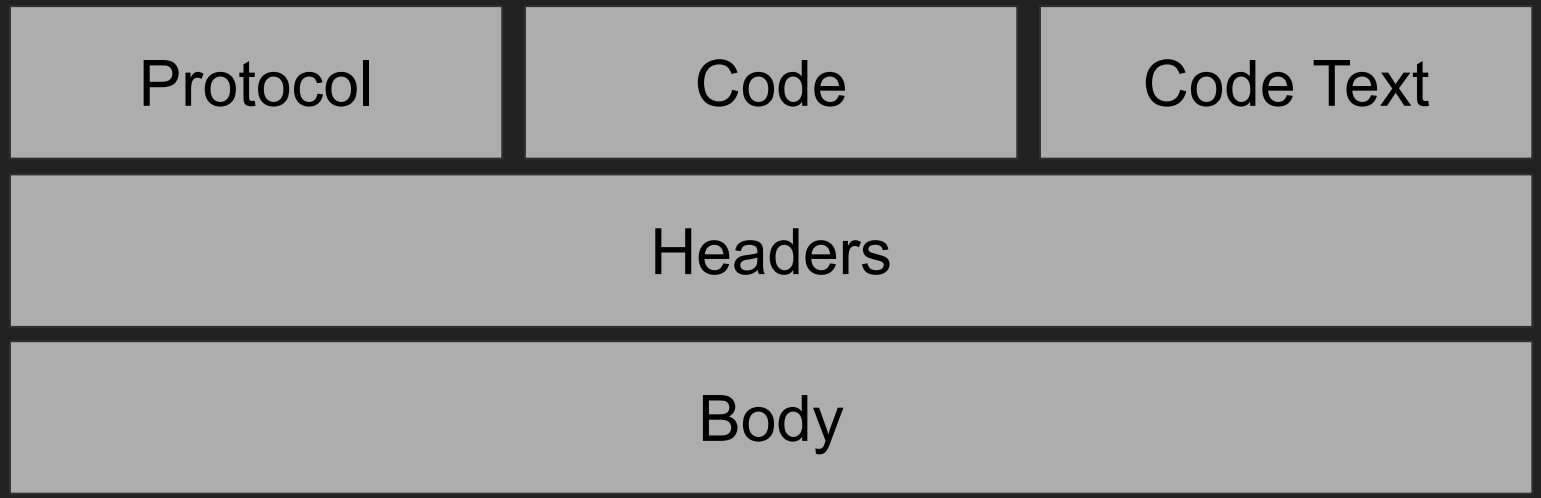


HTTP Request

```
curl -v http://husseinnasser.com/about
```

```
> GET /about HTTP/1.1
> Host: husseinnasser.com
> User-Agent: curl/7.79.1
> Accept: */*
```

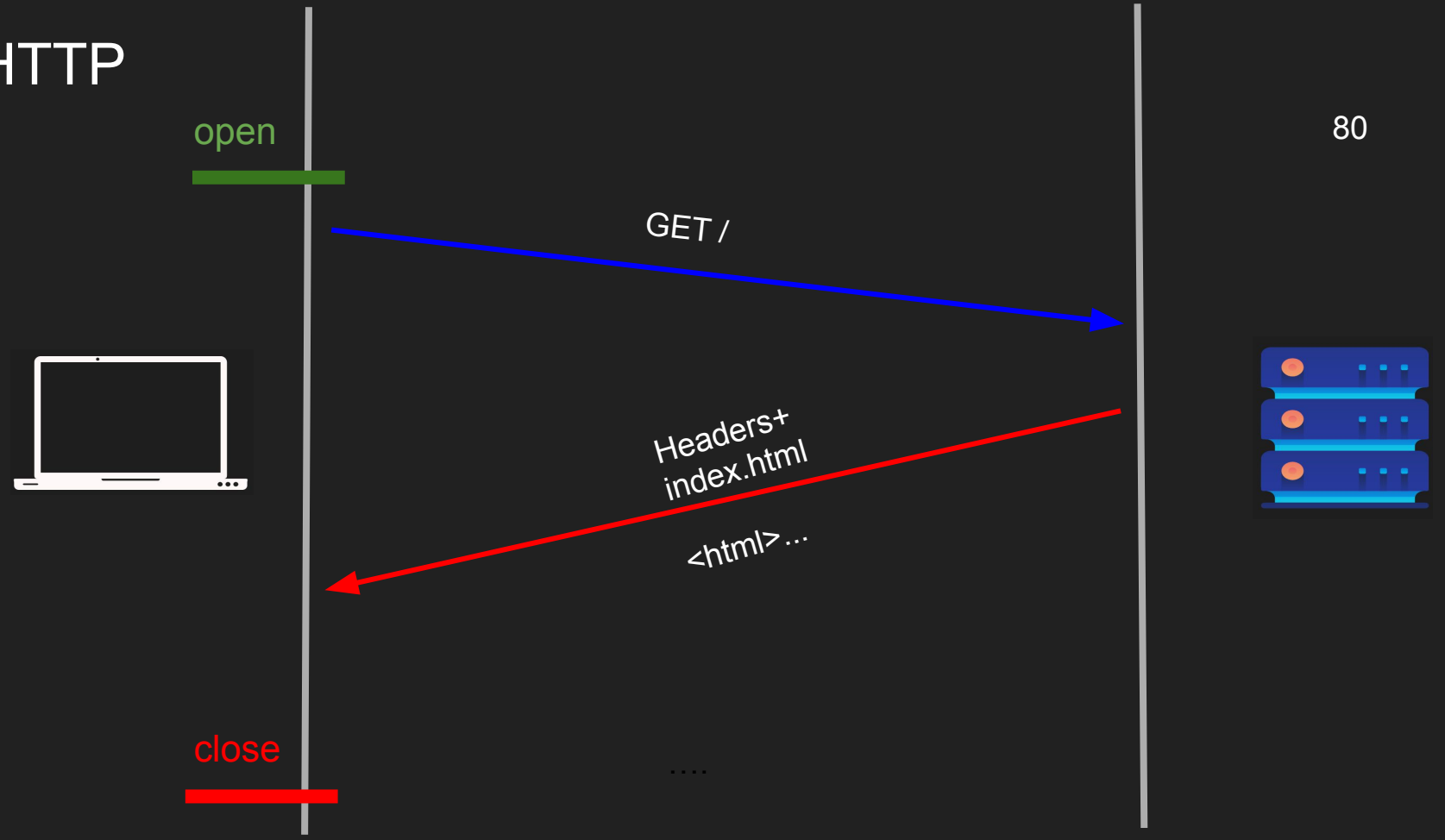
HTTP Response

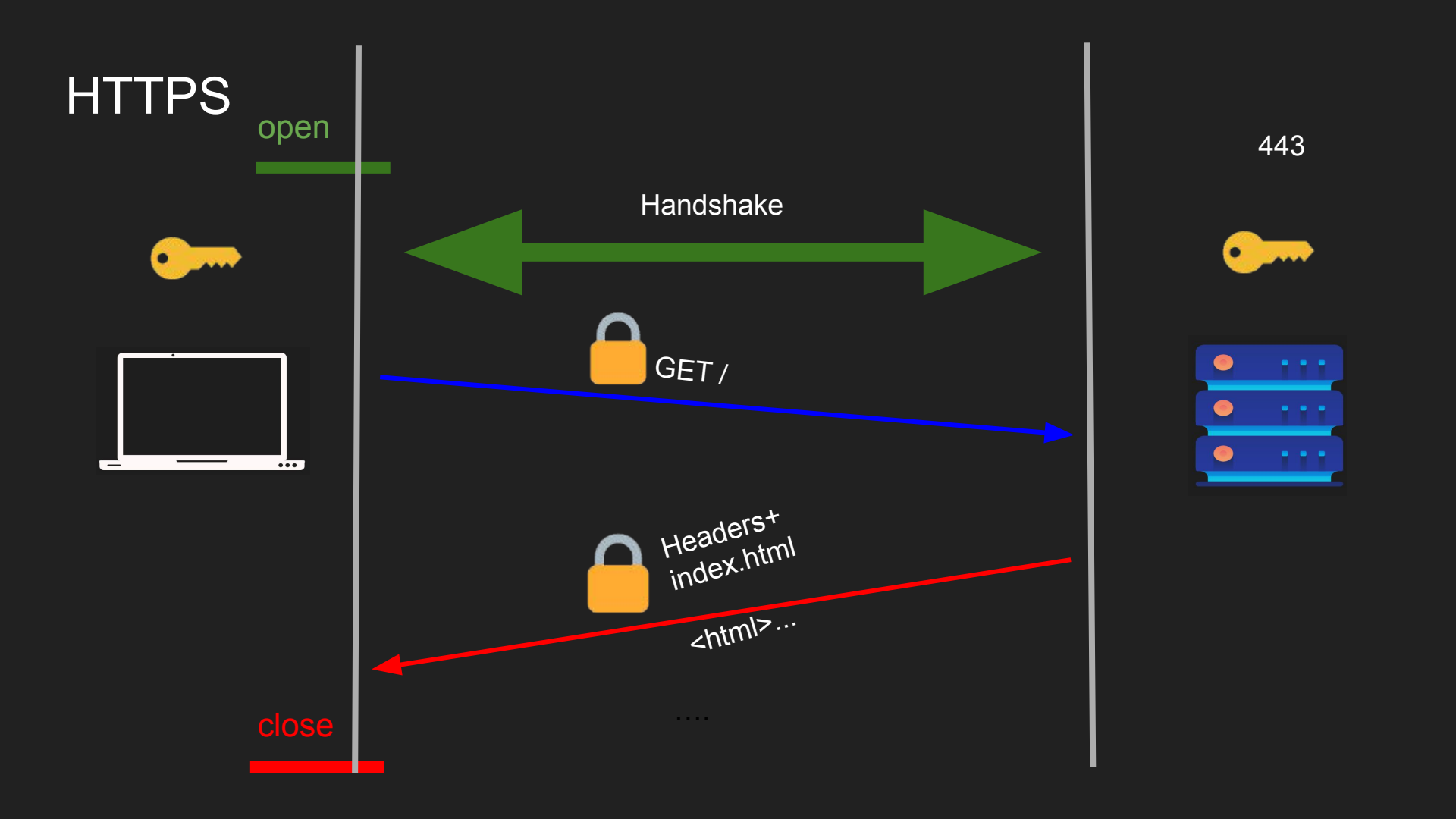


HTTP Response

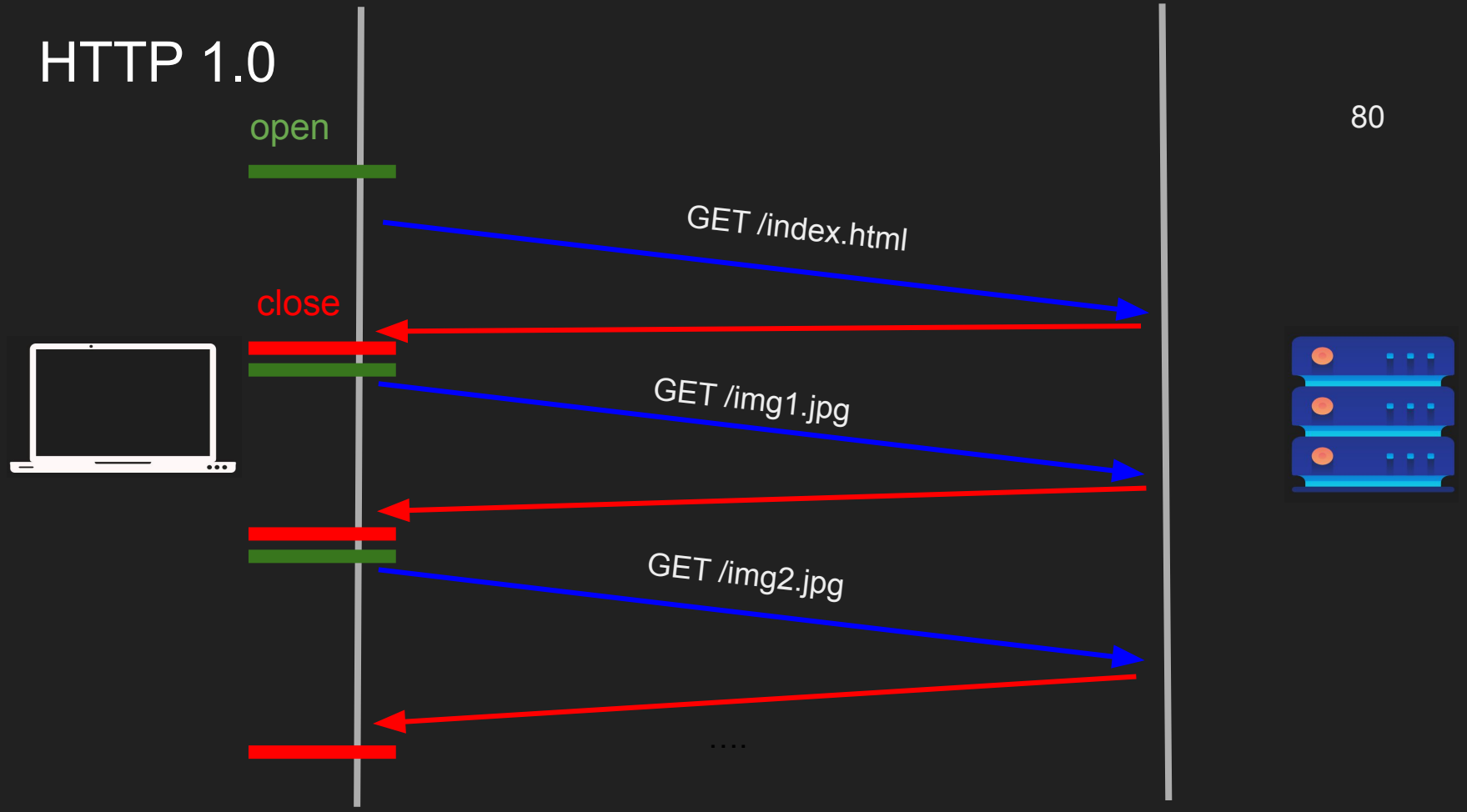
```
< HTTP/2 301
< location: https://www.husseinnasser.com/about
< date: Wed, 26 Oct 2022 17:10:59 GMT
< content-type: text/html; charset=UTF-8
< server: ghs
< content-length: 232
< x-xss-protection: 0
< x-frame-options: SAMEORIGIN
<
<HTML><HEAD><meta http-equiv="content-type" content
<TITLE>301 Moved</TITLE></HEAD><BODY>
```

HTTP




HTTPS
open
443
Handshake
GET /
Headers+
index.html
<html>...
....
close

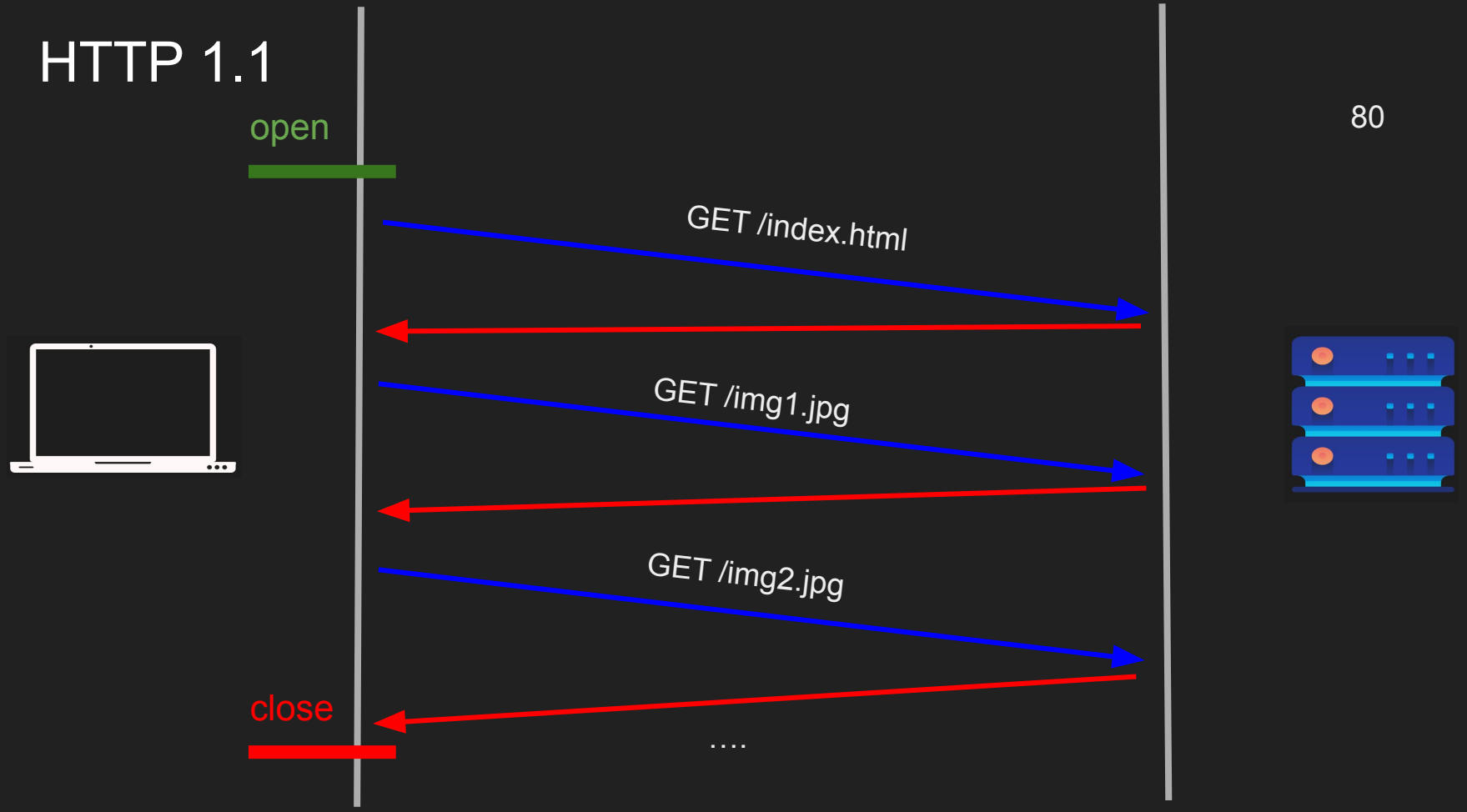
HTTP 1.0



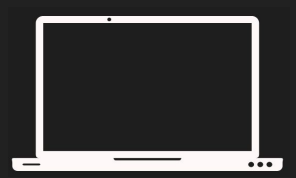
HTTP 1.0

- New TCP connection with each request
- Slow
- Buffering (transfer-encoding:chunked didn’t exist)
- No multi-homed websites (HOST header)

HTTP 1.1



HTTP 1.1 Pipelining



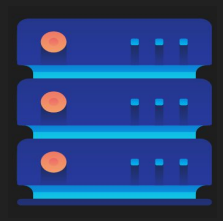
open

80

GET /index.html

GET /img1.jpg

GET /img2.jpg



close

....

HTTP 1.1

- Persisted TCP Connection
- Low Latency & Low CPU Usage
- Streaming with Chunked transfer
- Pipelining (disabled by default)
- Proxying & Multi-homed websites

HTTP/2

- SPDY
- Compression
- Multiplexing
- Server Push
- Secure by default
- Protocol Negotiation during TLS (NPN/ALPN)

HTTP over QUIC (HTTP/3)

- Replaces TCP with QUIC (UDP with Congestion control)
- All HTTP/2 features
- Without HOL

Summary

- HTTP Anatomy
- HTTP/1.0 over TCP
- HTTP/1.1 over TCP
- HTTP/2 over TCP
- HTTP over QUIC (HTTP/3)

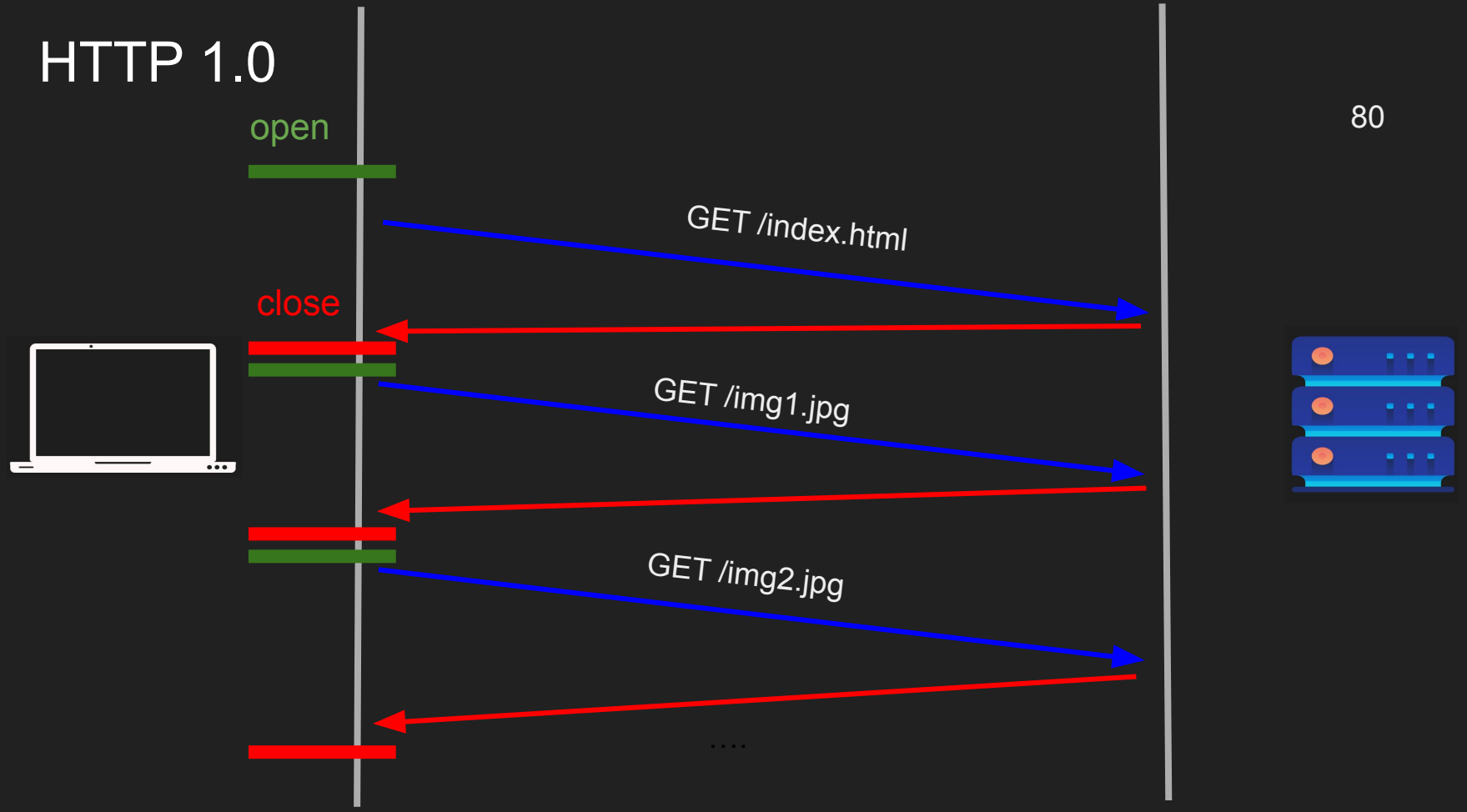
WebSockets

Bidirectional communications on the web

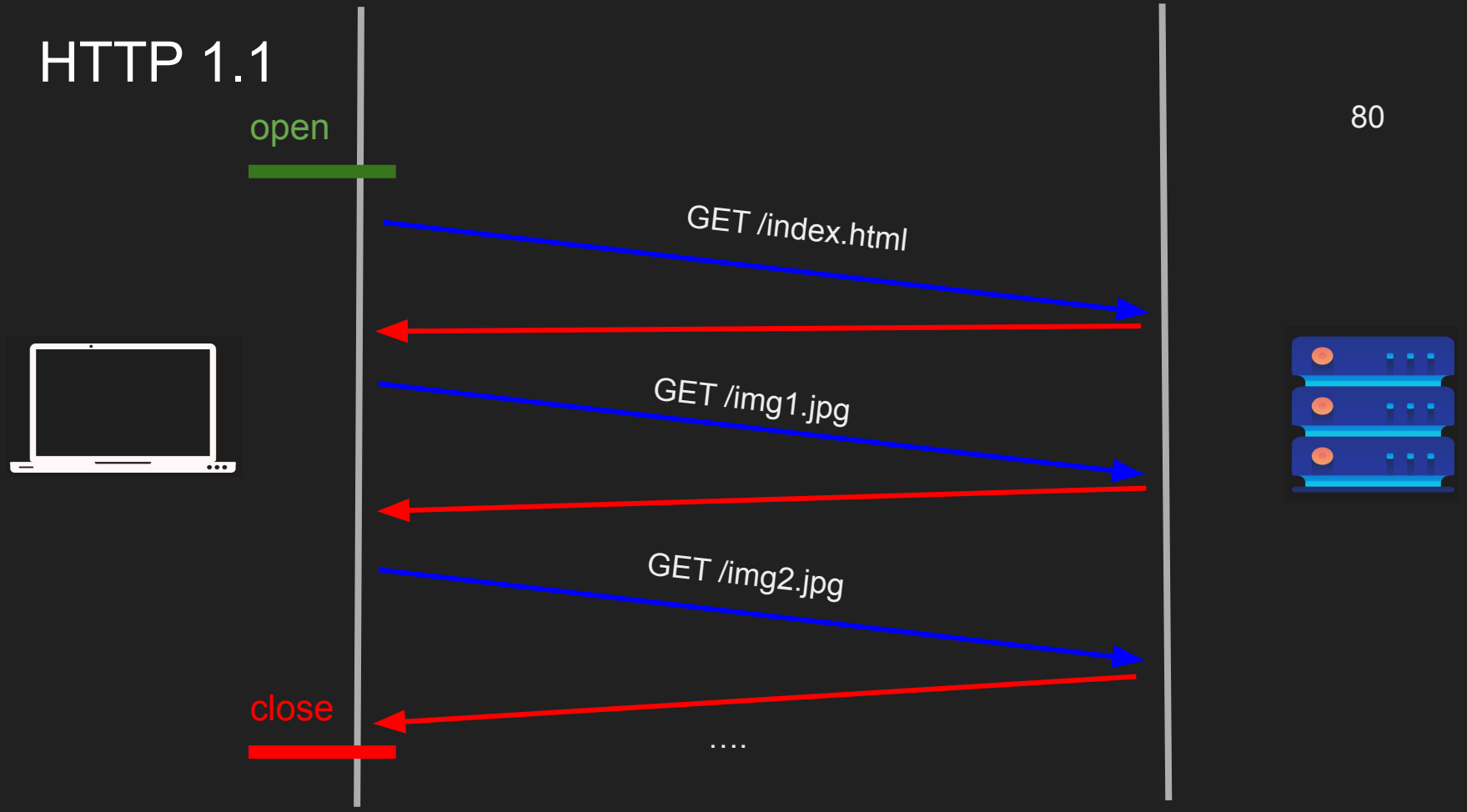
Agenda

- HTTP
- WebSockets
- WebSockets Handshake
- WebSockets use cases
- WebSockets Example
- WebSockets Pros and Cons

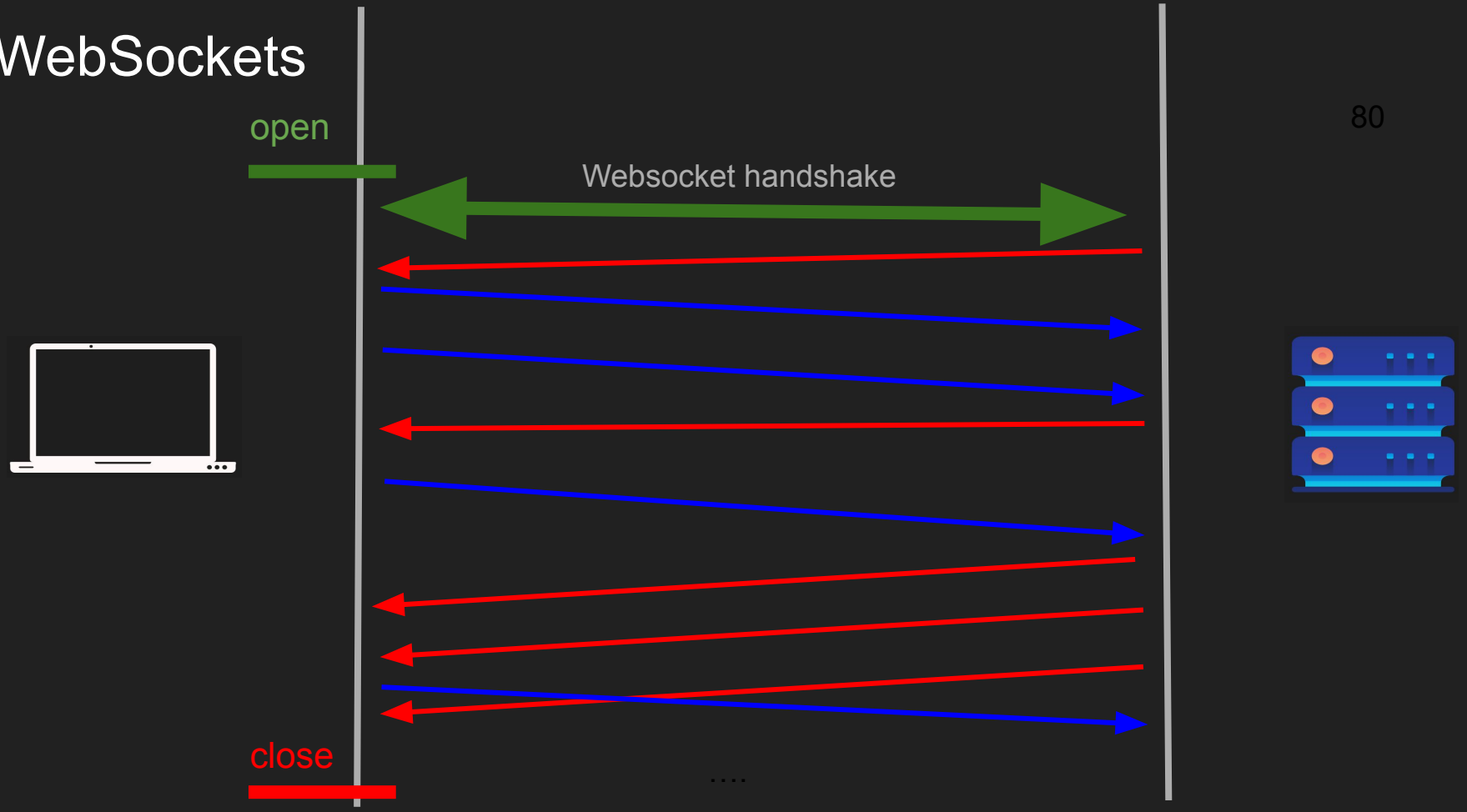
HTTP 1.0



HTTP 1.1



WebSockets



WebSockets Handshake ws:// or wss://

WebSocket Handshake

```
GET /chat HTTP/1.1
Host: server.example.com
Upgrade: websocket
Connection: Upgrade
Sec-WebSocket-Key: x3JJHMbDL1EzLkh9GBhXDw==
Sec-WebSocket-Protocol: chat, superchat
Sec-WebSocket-Version: 13
Origin: http://example.com
```

Client

```
HTTP/1.1 101 Switching Protocols
Upgrade: websocket
Connection: Upgrade
Sec-WebSocket-Accept: HSmrc0sMlYUkAGmm5OPpG2HaGWk=
Sec-WebSocket-Protocol: chat
```

Server

WebSockets use cases

- Chatting
- Live Feed
- Multiplayer gaming
- Showing client progress/logging

WebSockets Pros and Cons

Pros

- Full-duplex (no polling)
- HTTP compatible
- Firewall friendly (standard)

Cons

- Proxying is tricky
- L7 LB challenging (timeouts)
- Stateful, difficult to horizontally scale

Do you have to use WebSockets?

- No
- Rule of thumb - do you absolutely need bidirectional communication?
- Long polling
- Server Sent Events

Summary

- HTTP
- WebSockets
- WebSockets Handshake
- WebSockets use cases
- WebSockets Example
- WebSockets Pros and Cons

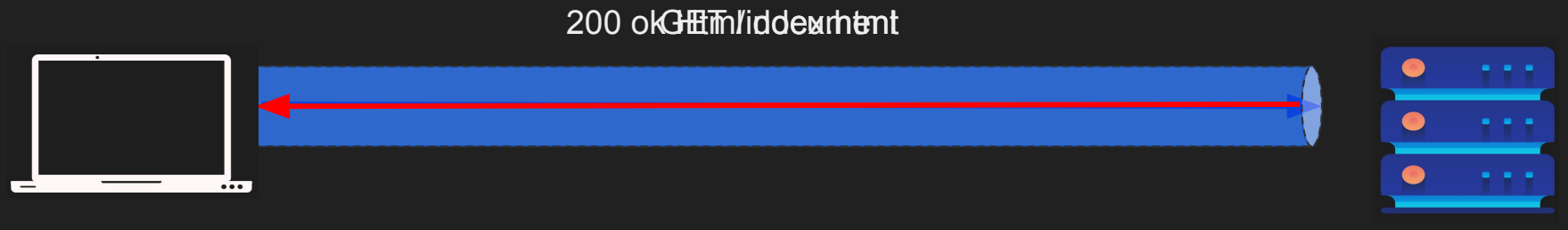
HTTP/2

Improving HTTP/1.1

Agenda

- How HTTP 1.1 Works?
- How HTTP/2 Works?
- HTTP/2 Pros & Cons
- H1 vs H2 performance test!

HTTP 1.1



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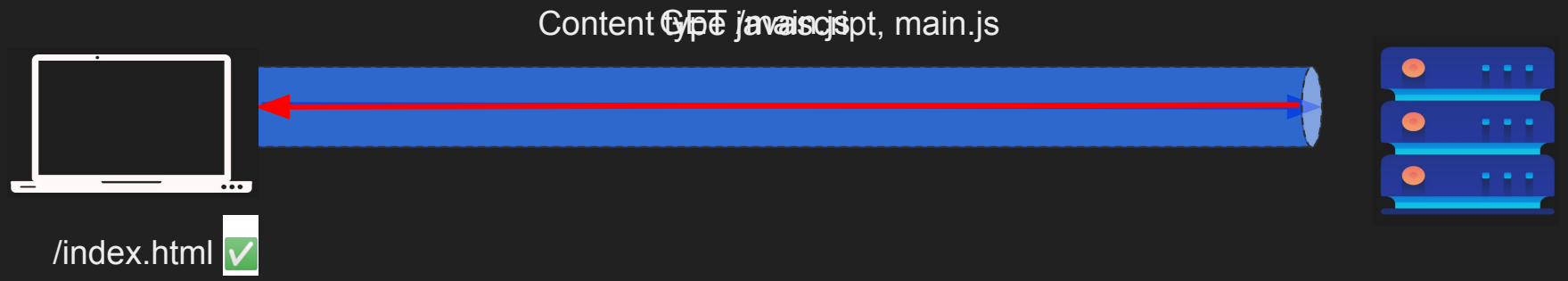
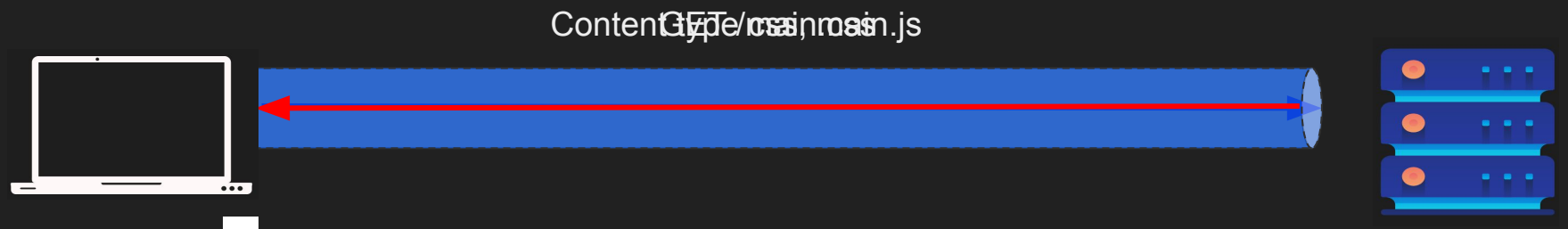
HTTP 1.1



/index.html ✅
/main.js
/main.css
/img1.jpg

....

HTTP 1.1



/index.html ✅
/main.js ✅
/main.css
/img1.jpg

....

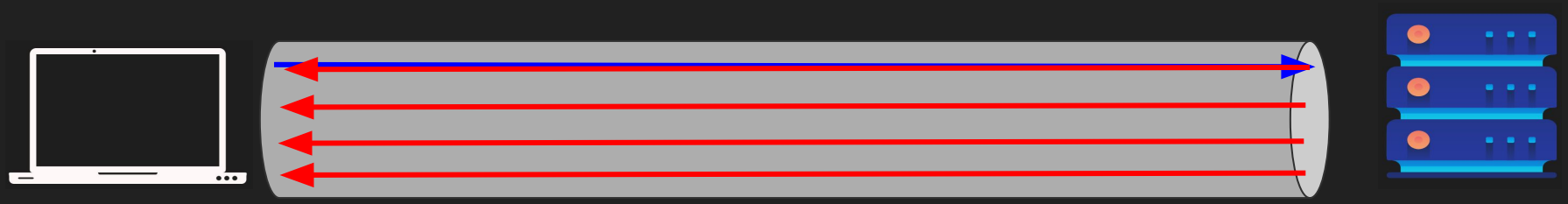
HTTP 1.1 (Browsers use 6 connections)

HTTP/2

GET /main.js (stream1)
GET /main.css (stream3)
GET /img1.jpg (stream5)
GET /img2.jpg (stream7)

HTTP/2 with Push

GET /index.html



Index.html
Main.css
Main.js
img.jpg

....

HTTP/2 Pros

- Multiplexing over Single Connection (save resources)
- Compression (Headers & Data)
- Server Push
- Secure by default
- Protocol Negotiation during TLS (ALPN)

HTTP/2 Cons

- TCP head of line blocking
- Server Push never picked up
- High CPU usage

Summary

- How HTTP 1.1 Works?
- How HTTP/2 Works?
- HTTP/2 Pros & Cons
- H1 vs H2 performance test

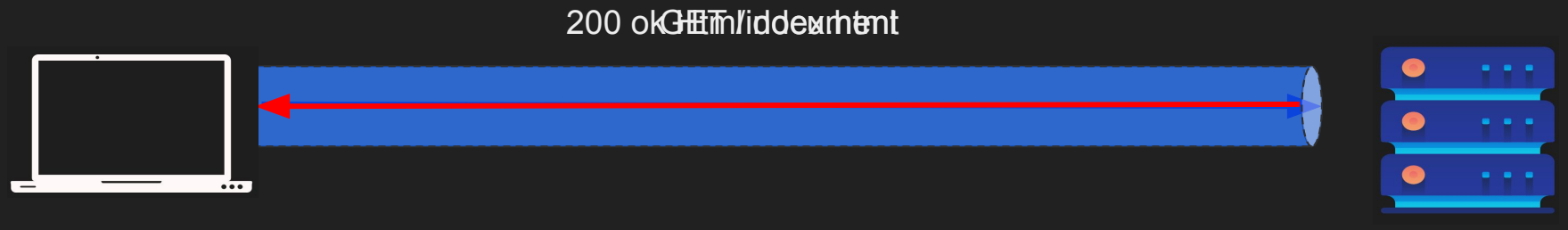
HTTP/3

HTTP over QUIC
Multiplexed streams

Agenda

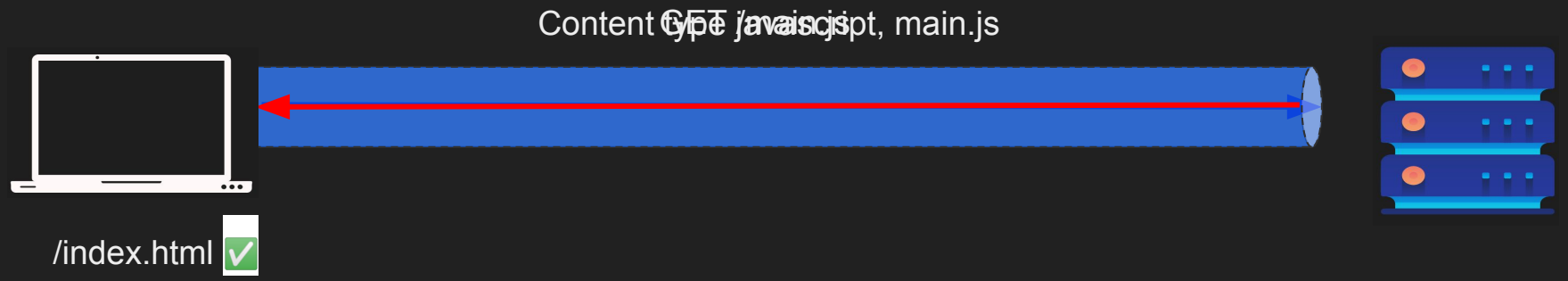
- How HTTP 1.1 Works?
- How HTTP/2 Works?
- HTTP/2 Cons
- How HTTP/3 & QUIC saves the day
- HTTP/3 & QUIC Pros & Cons

HTTP 1.1



....

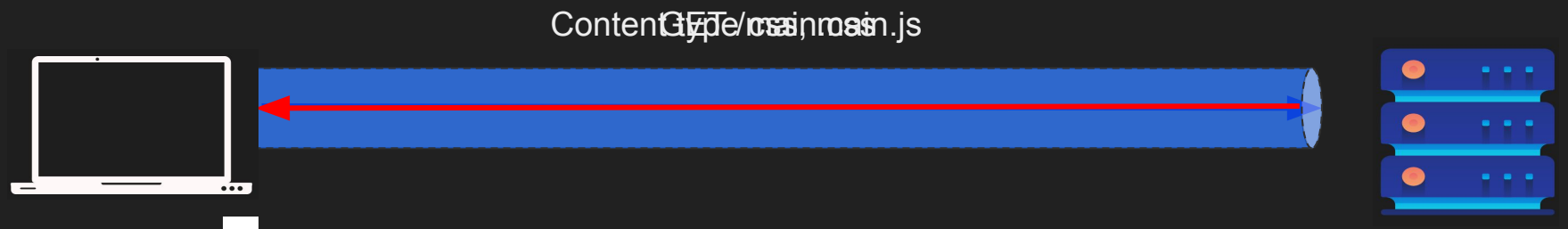
HTTP 1.1



/index.html ✅
/main.js
/main.css
/img1.jpg

....

HTTP 1.1



/index.html ✅
/main.js ✅
/main.css
/img1.jpg

....

HTTP 1.1 (Browsers use 6 connections)

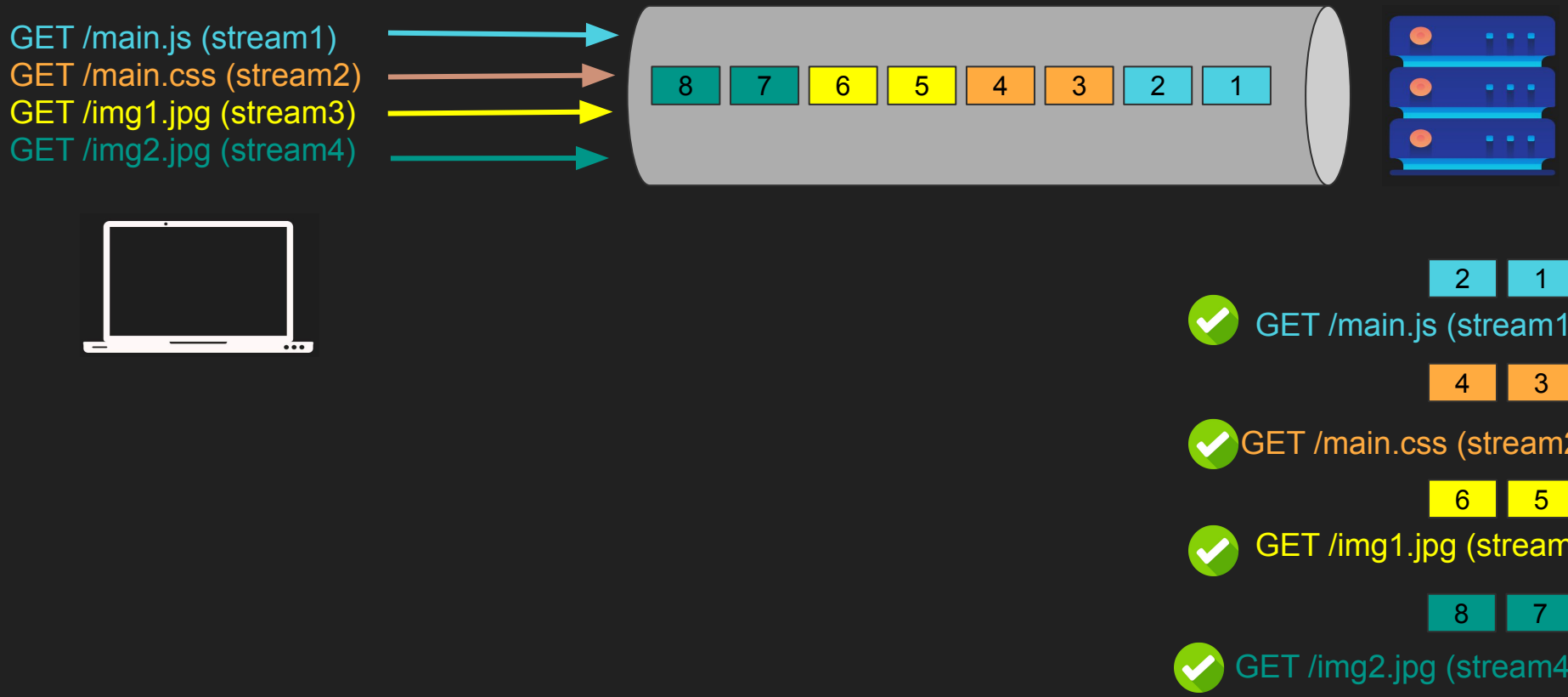
HTTP/2

GET /main.js (stream1)
GET /main.css (stream3)
GET /img1.jpg (stream5)
GET /img2.jpg (stream7)

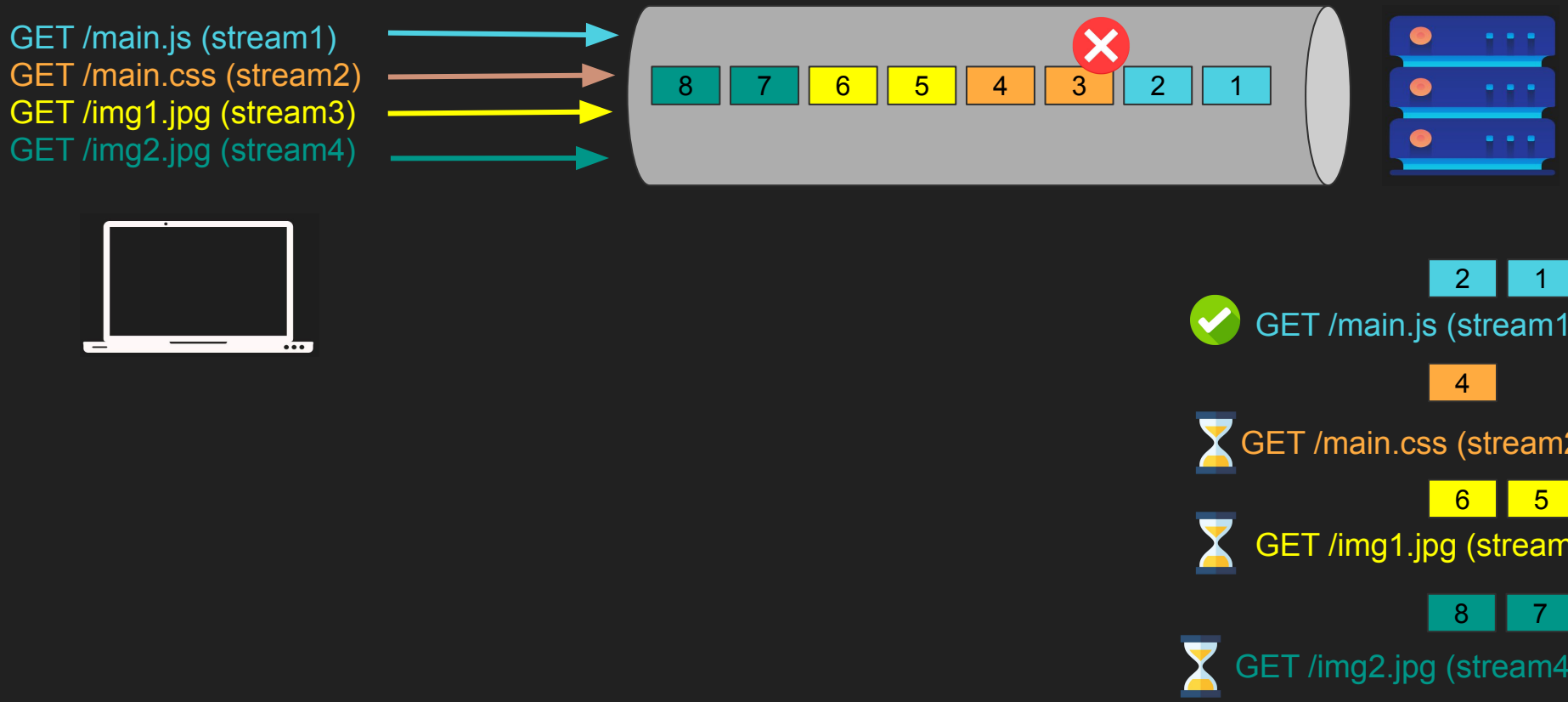
TCP head of line blocking

- TCP segments must be delivered in order
- But streams don’t have to
- Blocking requests

HTTP/2 TCP HOL



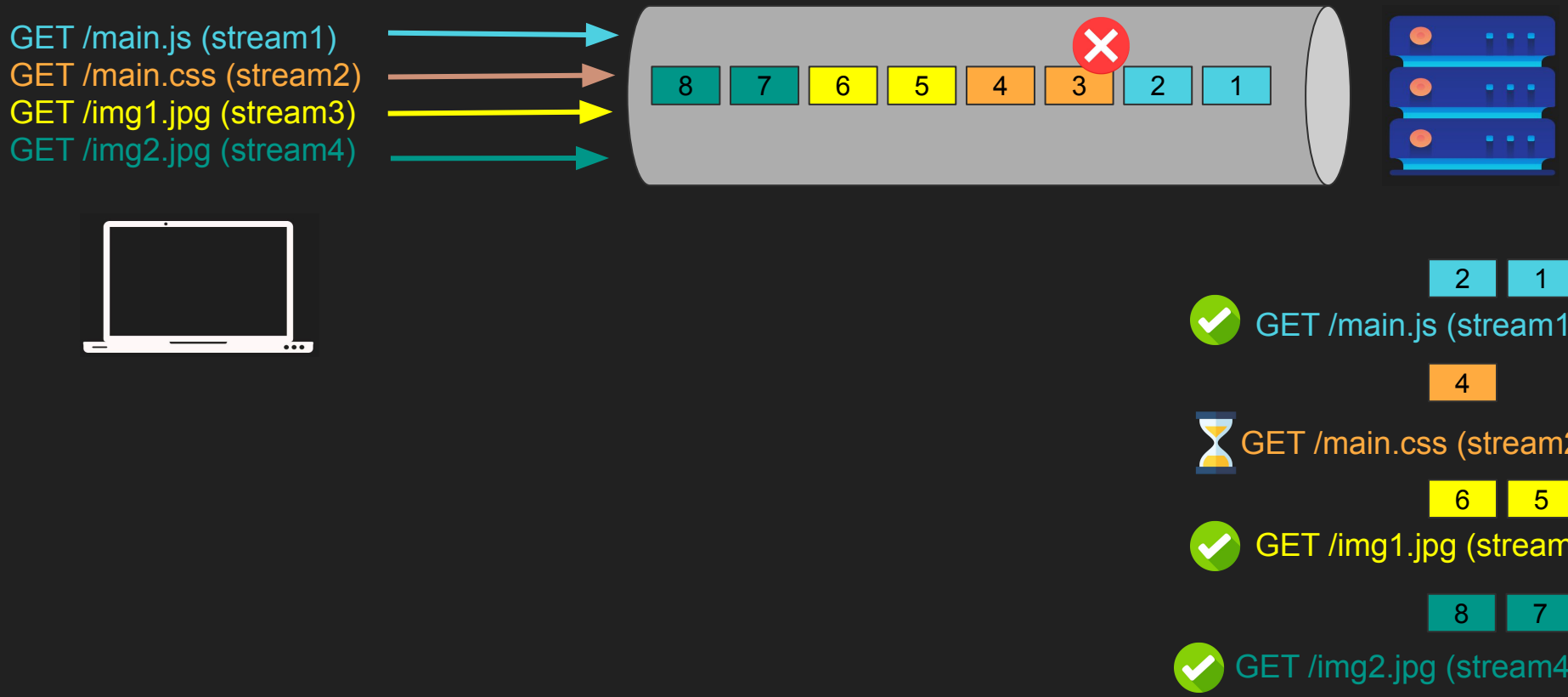
HTTP/2 TCP HOL



HTTP/3 & QUIC

- HTTP/3 uses QUIC
- Like HTTP/2, QUIC has streams
- But QUIC use UDP instead
- Application decides the boundary

HTTP/3 Streams



HTTP/3 & QUIC Pros

- QUIC has many other benefits
- Merges Connection setup + TLS in one handshake
- Has congestion control at stream level
- Connection migration (connectionID)
- Why not HTTP/2 over QUIC?
    - Header compression algorithm

HTTP/3 & QUIC Cons

- Takes a lot of CPU (parsing logic)
- UDP could be blocked
- IP Fragmentations is the enemy

Summary

- How HTTP 1.1 Works?
- How HTTP/2 Works?
- HTTP/2 Cons
- How HTTP/3 & QUIC saves the day
- HTTP/3 & QUIC Pros & Cons

gRPC

Taking HTTP/2 to the next level

Agenda

- Motivation
  - Client/Server Communication
  - Problem with Client Libraries
  - Why gRPC was invented?
- gRPC
  - Unary gRPC
  - Server Streaming
  - Client Streaming
  - Bidirectional

- Coding!
- gRPC Pros & Cons
- I’m tired of new protocols
- Summary

Client Server Communication

- SOAP, REST, GraphQL
- SSE, WebSockets
- Raw TCP

The Problem with Client Libraries

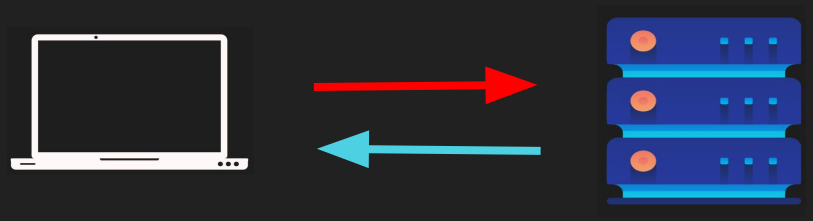
- Any communication protocol needs client library for the language of choice
  - SOAP Library
  - HTTP Client Library
- Hard to maintain and patch client libraries
  - HTTP/1.1 HTTP/2, new features, security etc.

Why gRPC was invented?

- Client Library: One library for popular languages
- Protocol: HTTP/2 (hidden implementation)
- Message Format: Protocol buffers as format

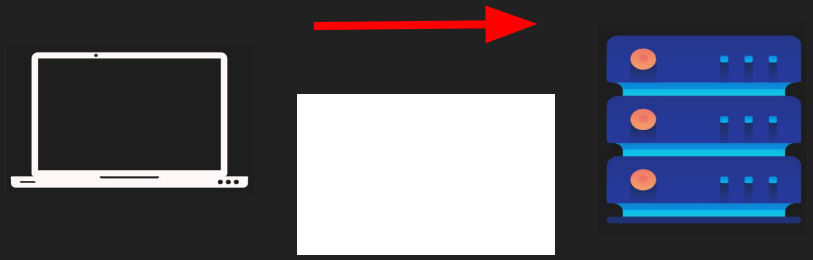
gRPC modes

- Unary RPC
- Server streaming RPC
- Client streaming RPC
- Bidirectional streaming RPC



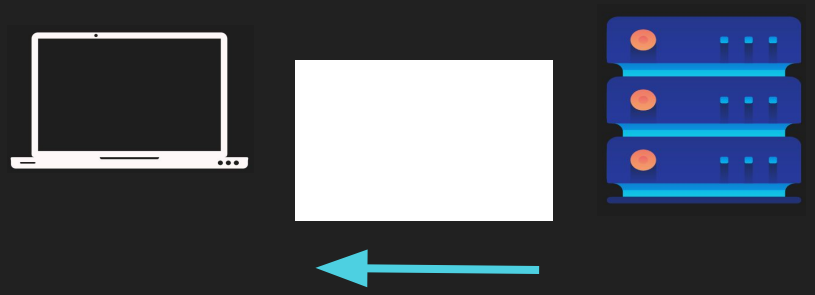
gRPC modes

- Unary RPC
- Server streaming RPC
- Client streaming RPC
- Bidirectional streaming RPC



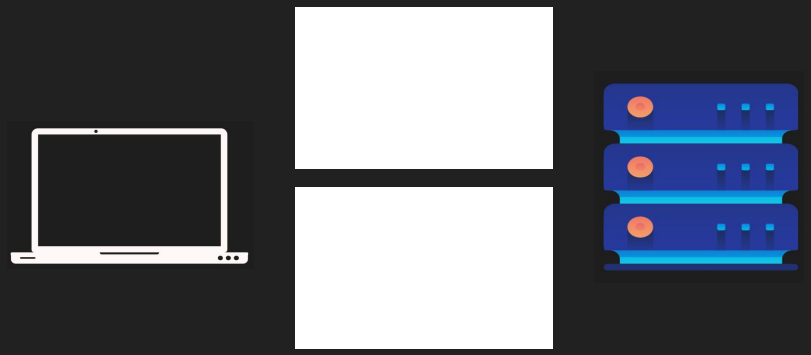
gRPC modes

- Unary RPC
- Server streaming RPC
- Client streaming RPC
- Bidirectional streaming RPC



gRPC modes

- Unary RPC
- Server streaming RPC
- Client streaming RPC
- Bidirectional streaming RPC



Coding time!

- Todo application (server, client) with gRPC
- createTodo ( )
- readTodos ( ) //synchronous
- readTodos ( ) //server stream

gRPC Pros & Cons

Pros

- Fast & Compact
- One Client Library
- Progress Feedback (upload)
- Cancel Request (H2)
- H2/Protobuf

Cons

- Schema
- Thick Client
- Proxies
- Error handling
- No native browser support
- Timeouts (pub/sub)

Can I write my own protocol too?

- Yes, you can, Spotify did (Hermes) but guess what
- Only you will be using it so…
- Spotify moved to gRPC not because of limitation of Hermes but because they are isolated.

Summary

- Motivation
  - Client/Server Communication
  - Problem with Client Libraries
  - Why gRPC was invented?
- gRPC
  - Unary gRPC
  - Server Streaming
  - Client Streaming
  - Bidirectional

- Coding!
- gRPC Pros & Cons
- I’m tired of new protocols
- Summary

WebRTC

Realtime communication on the web

Agenda

- WebRTC Overview
- WebRTC Demystified
  - NAT, STUN, TURN, ICE, SDP, Signaling the SDP
- Demo
- WebRTC Pros & Cons
- More WebRTC content beyond this content

WebRTC Overview

- Stands for Web Real-Time Communication
- Find a peer to peer path to exchange video and audio in an efficient and low latency manner
- Standardized API
- Enables rich communications browsers, mobile, IOT devices

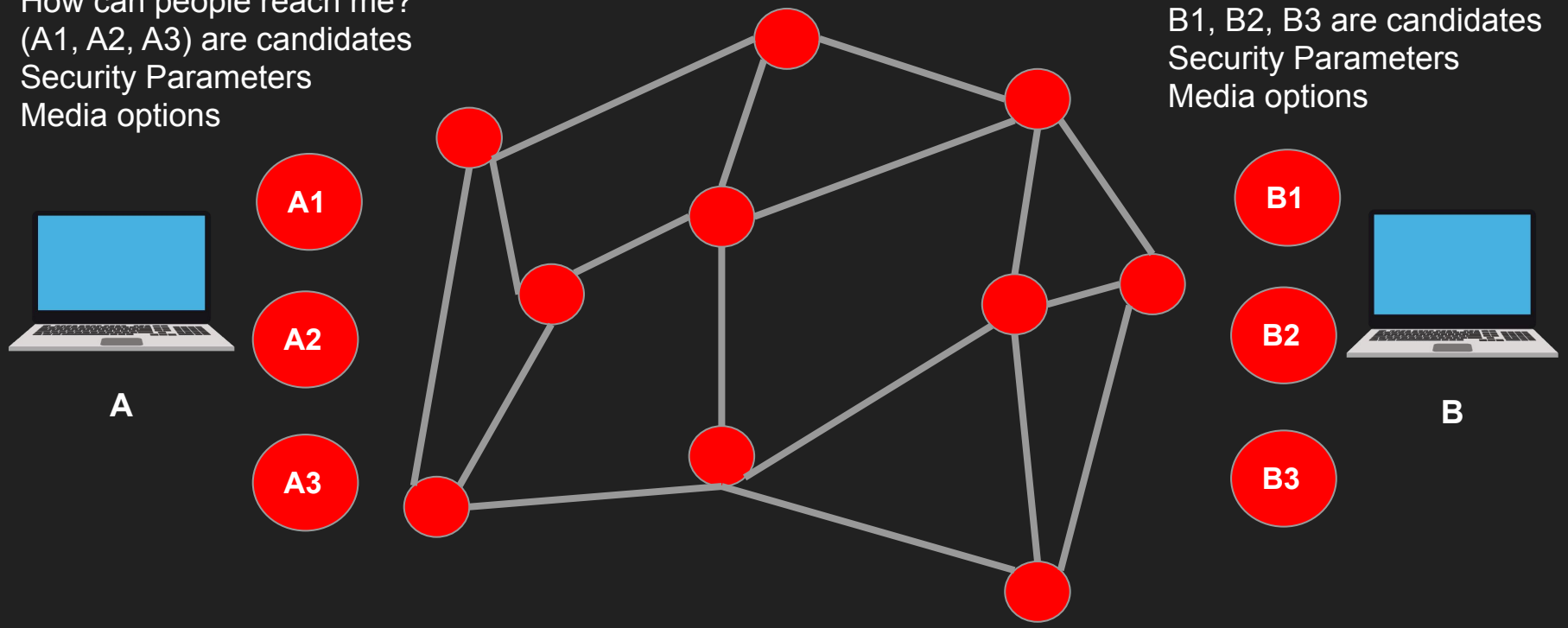
WebRTC Overview

- A wants to connect to B
- A finds out all possible ways the public can connect to it
- B finds out all possible ways the public can connect to it
- A and B signal this session information via other means
  - WhatsApp, QR, Tweet, WebSockets, HTTP Fetch..
- A connects to B via the most optimal path
- A & B also exchanges their supported media and security

WebRTC Overview

How can people reach me?
(A1, A2, A3) are candidates
Security Parameters
Media options

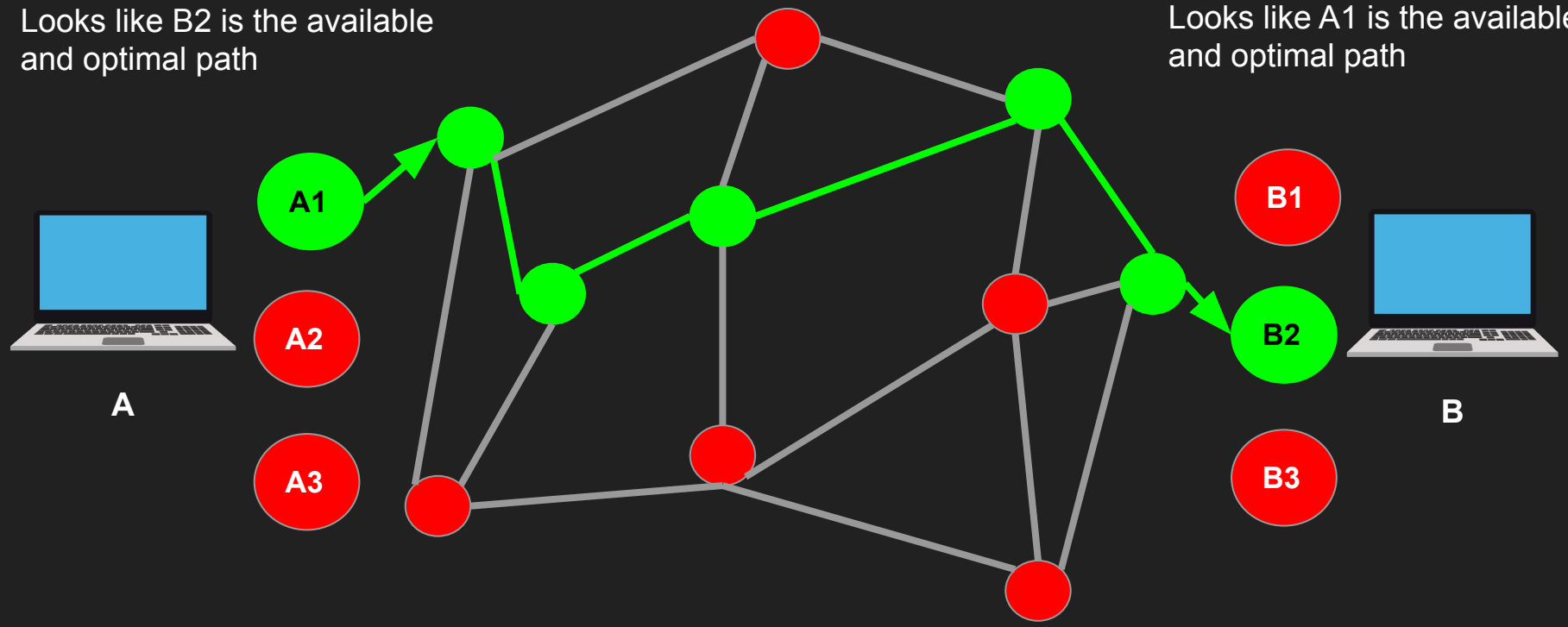
How can people reach me?
B1, B2, B3 are candidates
Security Parameters
Media options



WebRTC Overview

Looks like B2 is the available and optimal path

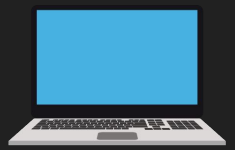
Looks like A1 is the available and optimal path



WebRTC Demystified

- NAT
- STUN, TURN
- ICE
- SDP
- Signaling the SDP

Network Address Translation

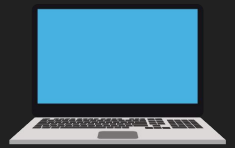


10.0.0.2

8992	10.0.0.2	GET/	4.4.4.4	80

10.0.0.1 5.5.5.5

8992	10.0.0.2	GET/	4.4.4.4	80
3333	5.5.5.5	GET/	4.4.4.4	80

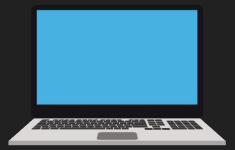


4.4.4.4:80

3333	5.5.5.5	GET/	4.4.4.4	80

Internal IP	Internal Port	External IP	Ext. Port	Dest IP	Dest Port
10.0.0.2	8992	5.5.5.5	3333	4.4.4.4	80

Network Address Translation

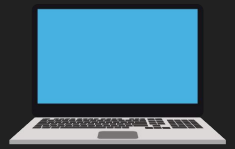


10.0.0.2

80	4.4.4.4	200 OK	10.0.0.2	8992

10.0.0.1 5.5.5.5

80	4.4.4.4	200 OK	5.5.5.5	3333
80	4.4.4.4	200 OK	10.0.0.2	8992



4.4.4.4:80

80	4.4.4.4	200 OK	5.5.5.5	3333

Internal IP	Internal Port	Ext IP	Ext Port	Dest IP	Dest Port
10.0.0.2	8992	5.5.5.5	3333	4.4.4.4	80

NAT Translations Method

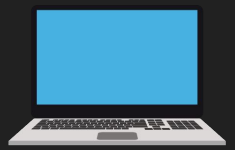
- One to One NAT (Full-cone NAT)
- Address restricted NAT
- Port restricted NAT
- Symmetric NAT

One to One NAT (Full cone NAT)

- Packets to external IP:port on the router always maps to internal IP:port without exceptions

Internal IP	Internal Port	Ext IP	Ext Port	Dest IP	Dest Port
10.0.0.2	8992	5.5.5.5	3333	4.4.4.4	80

One to One NAT



10.0.0.2

10.0.0.1 5.5.5.5

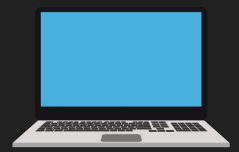
Internal IP	Internal Port	Ext IP	Ext Port	Dest IP	Dest Port
10.0.0.2	8992	5.5.5.5	3333	4.4.4.4	80
10.0.0.2	9999	5.5.5.5	4444	3.3.3.3	80

Address Restricted NAT

- Packets to external IP:port on the router always maps to internal IP:port as long as source address from packet matches the table (regardless of port)
- Allow if we communicated with this host before

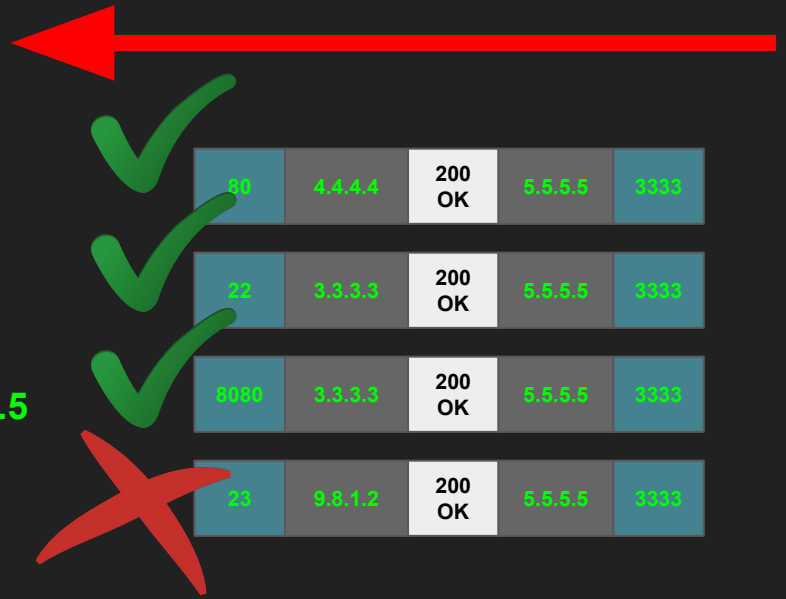
Internal IP	Internal Port	Ext IP	Ext Port	Dest IP	Dest Port
10.0.0.2	8992	5.5.5.5	3333	4.4.4.4	80

Address Restricted NAT



10.0.0.2

10.0.0.1 5.5.5.5



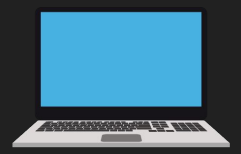
Internal IP	Internal Port	Ext IP	Ext Port	Dest IP	Dest Port
10.0.0.2	8992	5.5.5.5	3333	4.4.4.4	80
10.0.0.2	9999	5.5.5.5	4444	3.3.3.3	80

Port Restricted NAT

- Packets to external IP:port on the router always maps to internal IP:port as long as source address and port from packet matches the table
- Allow if we communicated with this host:port before

Internal IP	Internal Port	Ext IP	Ext Port	Dest IP	Dest Port
10.0.0.2	8992	5.5.5.5	3333	4.4.4.4	80

Port Restricted NAT



10.0.0.2

10.0.0.1 5.5.5.5

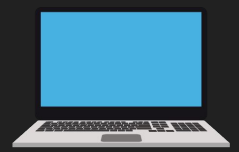
Internal IP	Internal Port	Ext IP	Ext Port	Dest IP	Dest Port
10.0.0.2	8992	5.5.5.5	3333	4.4.4.4	80
10.0.0.2	9999	5.5.5.5	4444	3.3.3.3	80
10.0.0.2	8888	5.5.5.5	2222	3.3.3.3	8080

Symmetric NAT

- Packets to external IP:port on the router always maps to internal IP:port as long as source address and port from packet matches the table
- Only Allow if the full pair match

Internal IP	Internal Port	Ext IP	Ext Port	Dest IP	Dest Port
10.0.0.2	8992	5.5.5.5	3333	4.4.4.4	80

Symmetric NAT



10.0.0.2

10.0.0.1

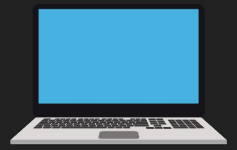
5.5.5.5

Internal IP	Internal Port	Ext IP	Ext Port	Dest IP	Dest Port
10.0.0.2	8992	5.5.5.5	3333	4.4.4.4	80
10.0.0.2	9999	5.5.5.5	4444	3.3.3.3	80
10.0.0.2	8888	5.5.5.5	2222	3.3.3.3	8080

STUN

- Session Traversal Utilities for NAT
- Tell me my public ip address/port through NAT
- Works for Full-cone, Port/Address restricted NAT
- Doesn’t work for symmetric NAT
- STUN server port 3478, 5349 for TLS
- Cheap to maintain

STUN Request



10.0.0.2

8992	10.0.0.2	STN	9.9.9.9	3478

10.0.0.1 5.5.5.5

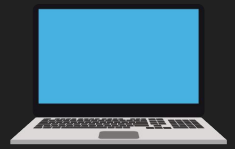
8992	10.0.0.2	STN	9.9.9.9	3478
3333	5.5.5.5	STN	9.9.9.9	3478

9.9.9.9:3478
(STUN Sever)

3333	5.5.5.5	STN	9.9.9.9	3478

Internal IP	Internal Port	Ext IP	Ext Port	Dest IP	Dest Port
10.0.0.2	8992	5.5.5.5	3333	9.9.9.9	3478

STUN Response



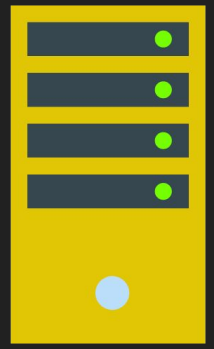
10.0.0.2

3478	9.9.9.9	RSP	10.0.0.2	8992

You are
5.5.5.5:3333

10.0.0.1 5.5.5.5

3478	9.9.9.9	RSP	5.5.5.5	3333
3478	9.9.9.9	RSP	10.0.0.2	8992



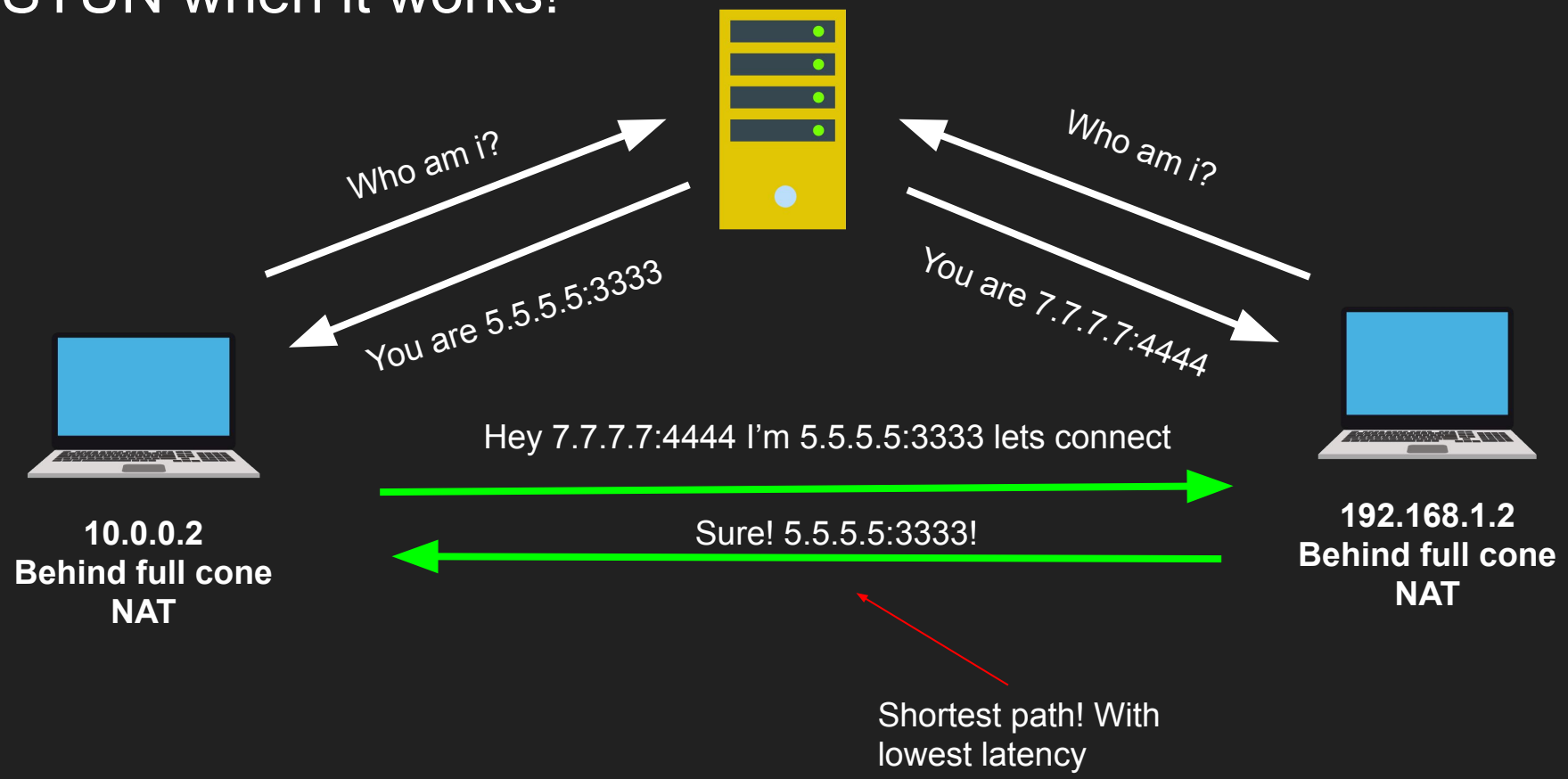
9.9.9.9:3478
(STUN Sever)

You are
5.5.5.5:3333

3478	9.9.9.9	RSP	5.5.5.5	3333

Internal IP	Internal Port	Ext IP	Ext Port	Dest IP	Dest Port
10.0.0.2	8992	5.5.5.5	3333	9.9.9.9	3478

STUN when it works!

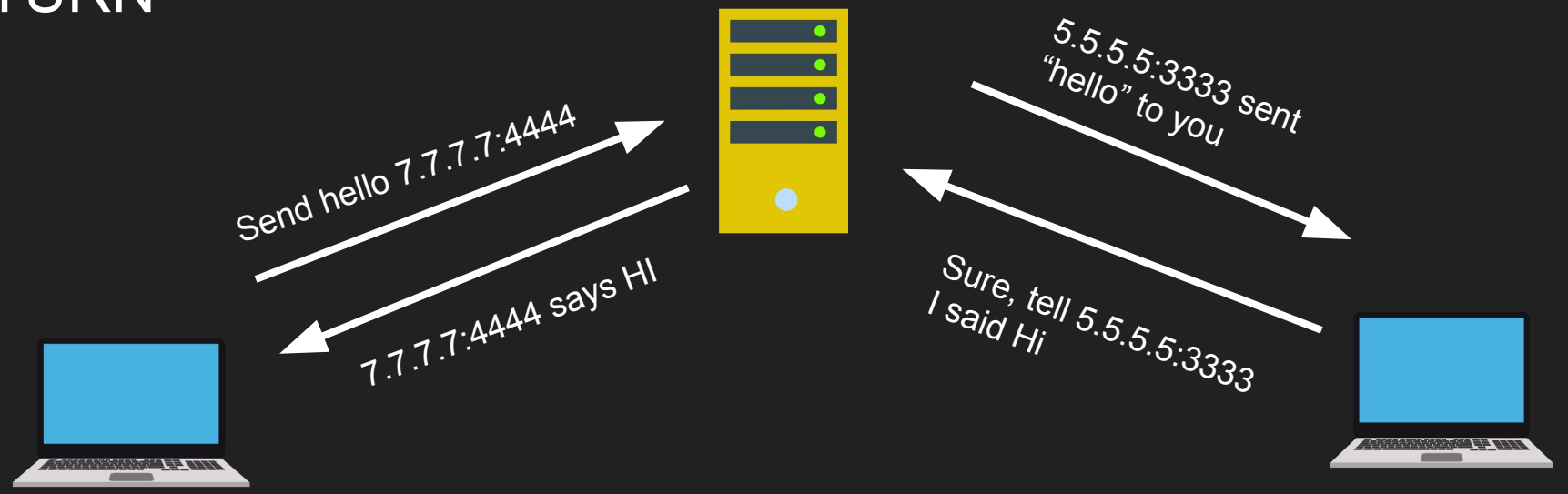


STUN when it doesn't!

TURN

- Traversal Using Relays around NAT
- In case of Symmetric NAT we use TURN
- It's just a server that relays packets
- TURN default server port 3478, 5349 for TLS
- Expensive to maintain and run

TURN



10.0.0.2
(5.5.5.5:3333)
Behind
symmetric NAT

192.168.1.2
(7.7.7.7:4444)
Behind
symmetric NAT

ICE

- Interactive Connectivity Establishment
- ICE collects all available candidates (local IP addresses, reflexive addresses – STUN ones and relayed addresses – TURN ones)
- Called ice candidates
- All the collected addresses are then sent to the remote peer via SDP

SDP

- Session Description Protocol
- A format that describes ice candidates, networking options, media options, security options and other stuff
- Not really a protocol its a format
- Most important concept in WebRTC
- The goal is to take the SDP generated by a user and send it “somehow” to the other party

SDP Example

```
v=0
o=- 9148204791819634656 3 IN IP4 127.0.0.1
s=-
t=0 0
a=group:BUNDLE audio video data
a=msid-semantic: WMS kyaiqbOs7S2h3EoSHabQ3JlBqZ67cFqZmWFN
m=audio 50853 RTP/SAVPF 111 103 104 0 8 107 106 105 13 126
c=IN IP4 192.168.1.64
a=rtcp:50853 IN IP4 192.168.1.64
a=candidate:3460887983 1 udp 2113937151 192.168.1.64 50853 typ host generation 0
a=candidate:3460887983 2 udp 2113937151 192.168.1.64 50853 typ host generation 0
...
```

Signaling

- SDP Signaling
- Send the SDP that we just generated somehow to the other party we wish to communicate with
- Signaling can be done via a tweet, QR code, Whatsapp, WebSockets, HTTP request DOESN’T MATTER! Just get that large string to the other party

WebRTC Demystified

1. A wants to connect to B
2. A creates an “offer”, it finds all ICE candidates, security options, audio/video options and generates SDP, the offer is basically the SDP
3. A signals the offer somehow to B (whatsapp)
4. B creates the “answer” after setting A’s offer
5. B signals the “answer” to A
6. Connection is created
7.

WebRTC Demo

- We will connect two browsers (Browser A & Browser B)
- A will create an offer (sdp) and set it as local description
- B will get the offer and set it as remote description
- B creates an answer sets it as its local description and signal the answer (sdp) to A
- A sets the answer as its remote description
- Connection established, exchange data channel

WebRTC Pros & Cons

- Pros
  - P2p is great ! low latency for high bandwidth content
  - Standardized API I don’t have to build my own
- Cons
  - Maintaining STUN & TURN servers
  - Peer 2 Peer falls apart in case of multiple participants (discord case)

More WebRTC stuff!

So more to discuss beyond this content

Media API

- getUserMedia to access microphone, video camera
- RTCPConnection.addTrack(stream)
- https://www.html5rocks.com/en/tutorials/webrtc/basics/

onIceCandidate and addIceCandidate

- To maintain the connection as new candidates come and go
- onIceCandidate tells user there is a new candidate after the SDP has already been created
- The candidate is signaled and sent to the other party
- The other party uses addiceCandidate to add it to its SDP

Set custom TURN and STUN Servers

```
const iceConfiguration = {
   iceServers: [{
            urls: 'turn:turnserver.company.com:3478',
            username: 'optional-username',
            credentials: 'auth-token'},
           { urls: "stun:stun.services.mozilla.com",
             username: "test@mozilla.com",
             credential: "webrtcdemo"}
         ]
}
const pc = new RTCPeerConnection(configuration);
```

Create your own STUN & TURN server

- COTURN open source project
- https://github.com/coturn/coturn

Public STUN servers

- `stun1.l.google.com:19302`
- `stun2.l.google.com:19302`
- `stun3.l.google.com:19302`
- `stun4.l.google.com:19302`
- `stun.stunprotocol.org:3478`

Many ways to HTTPS

And how each affect latency

HTTPS Communication Basics

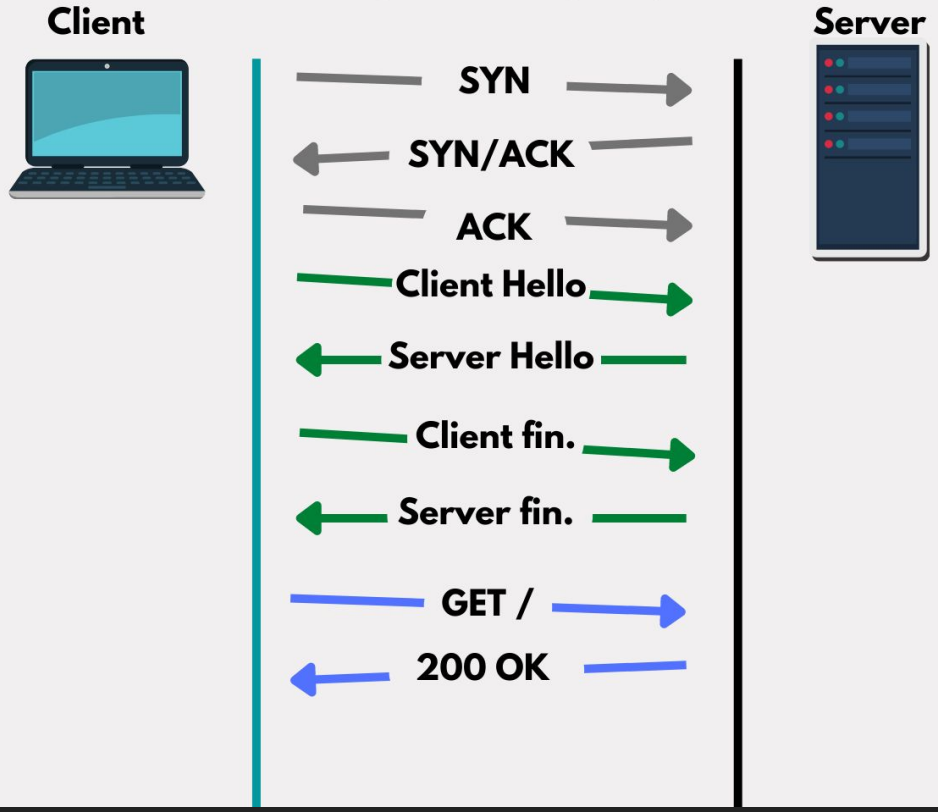
- Establish Connection
- Establish Encryption
- Send Data
- Close Connection (when absolutely done)

HTTPs communications

- HTTPS over TCP with TLS 1.2
- HTTPS over TCP with TLS 1.3
- HTTPS over QUIC
- HTTPS over TCP fast Open
- HTTPS over TCP with TLS 1.3 0RTT
- HTTPS over QUIC with 0RTT

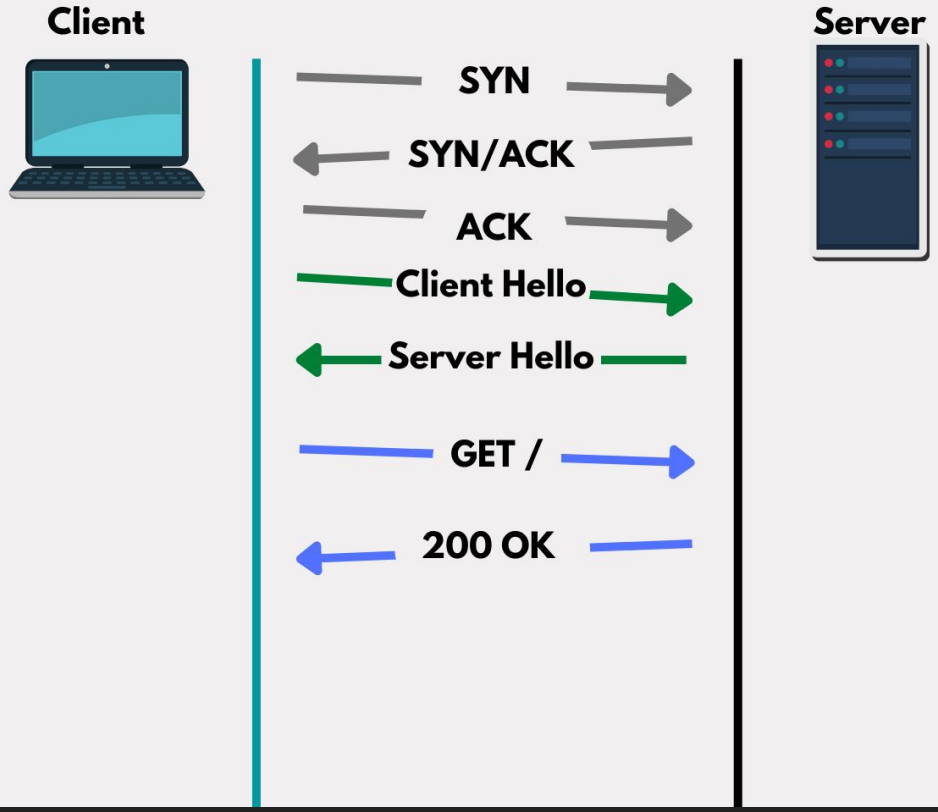
HTTPS over TCP With TLS 1.2

HTTPS over TCP with TLS 1.2



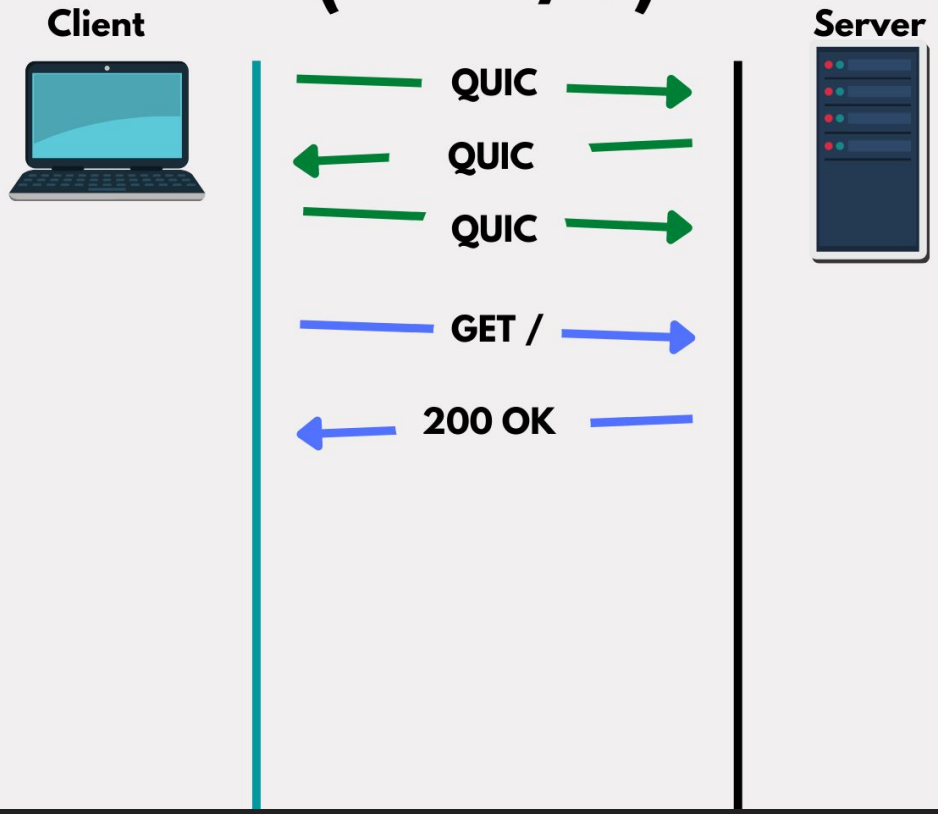
HTTPS over TCP With TLS 1.3

HTTPS over TCP with TLS 1.3



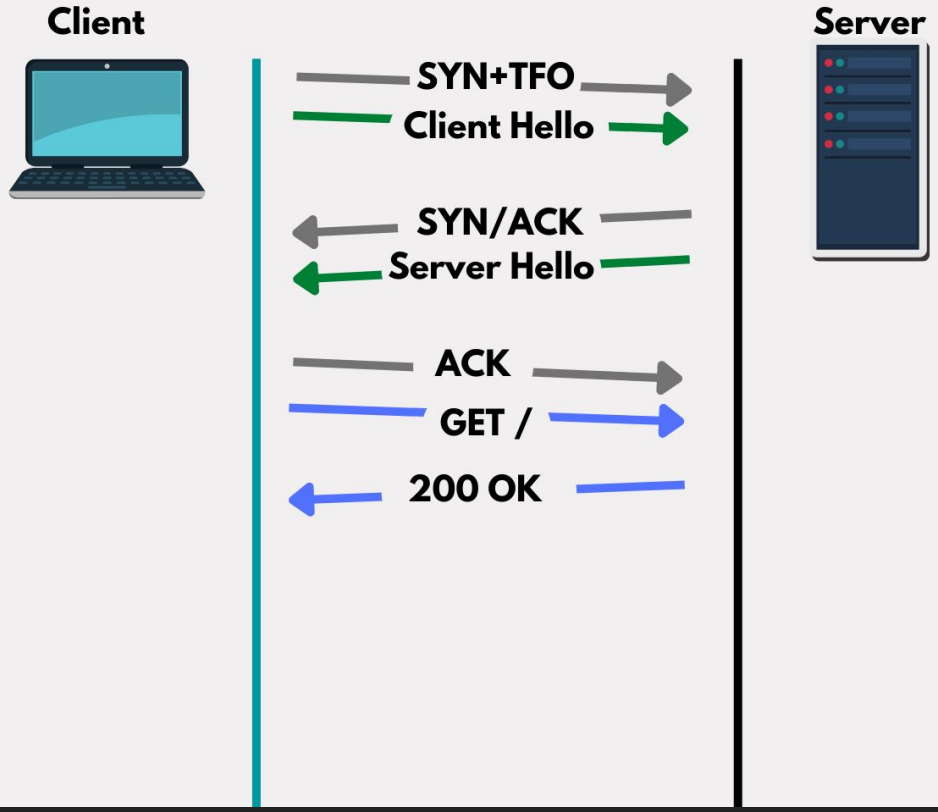
HTTPS over QUIC (HTTP/3)

HTTPS over QUIC (HTTP/3)



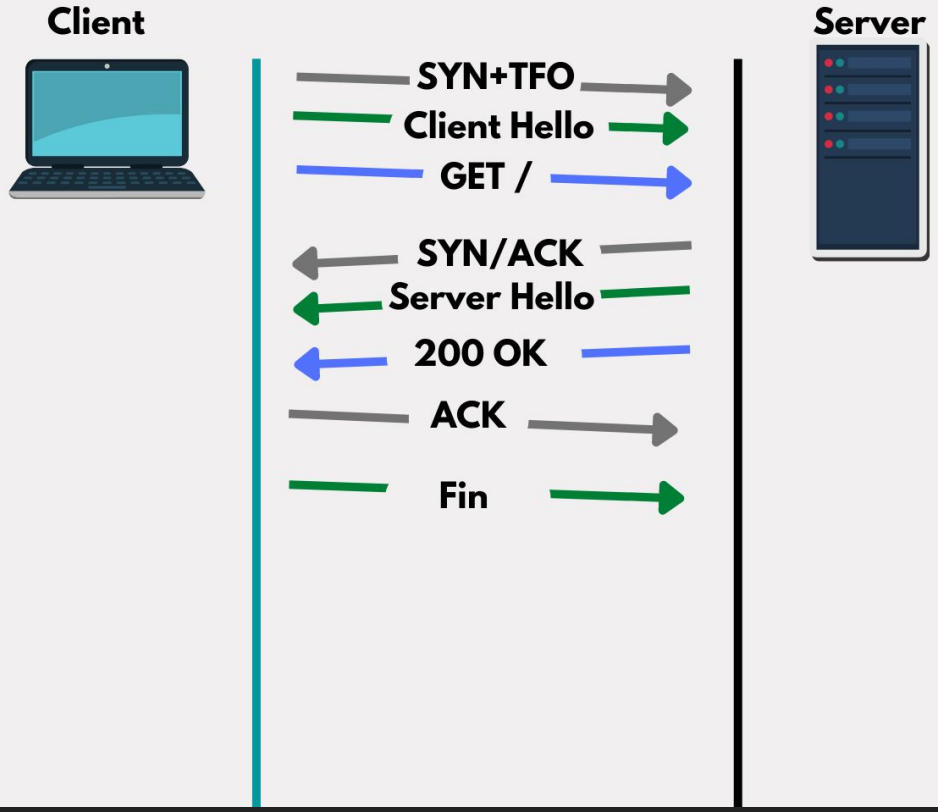
HTTPS over TFO with TLS 1.3

HTTPS over TCP Fast Open with TLS 1.3 (theory)



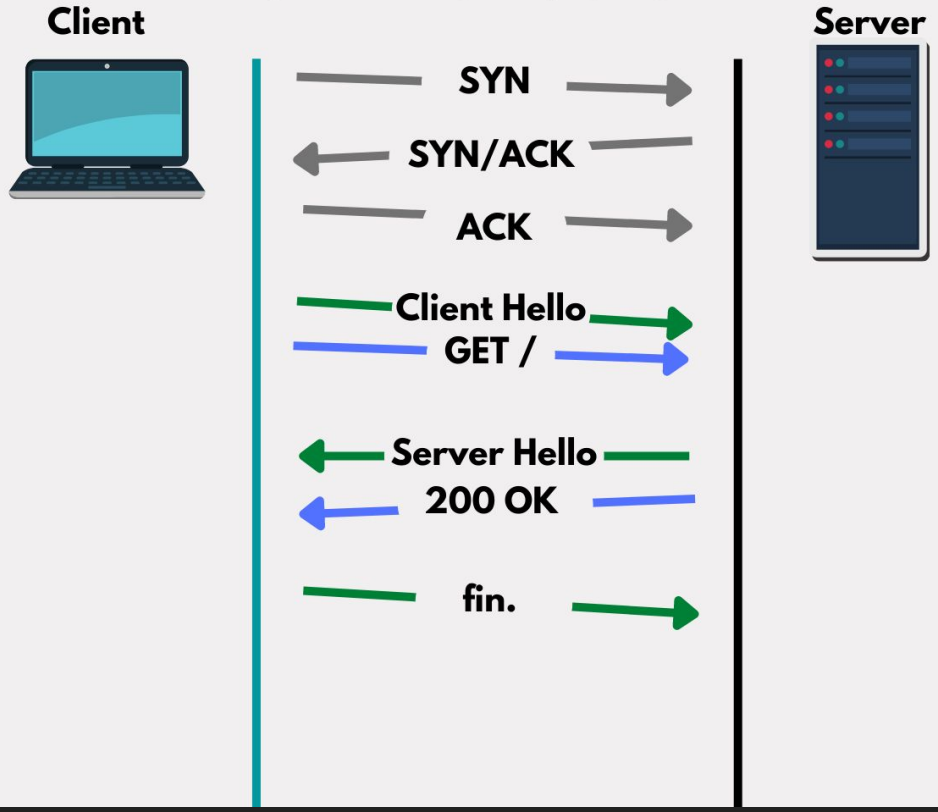
HTTPS over TFO with TLS 1.3 0RTT

HTTPS over TCP Fast Open with TLS 1.3 0RTT (theory)



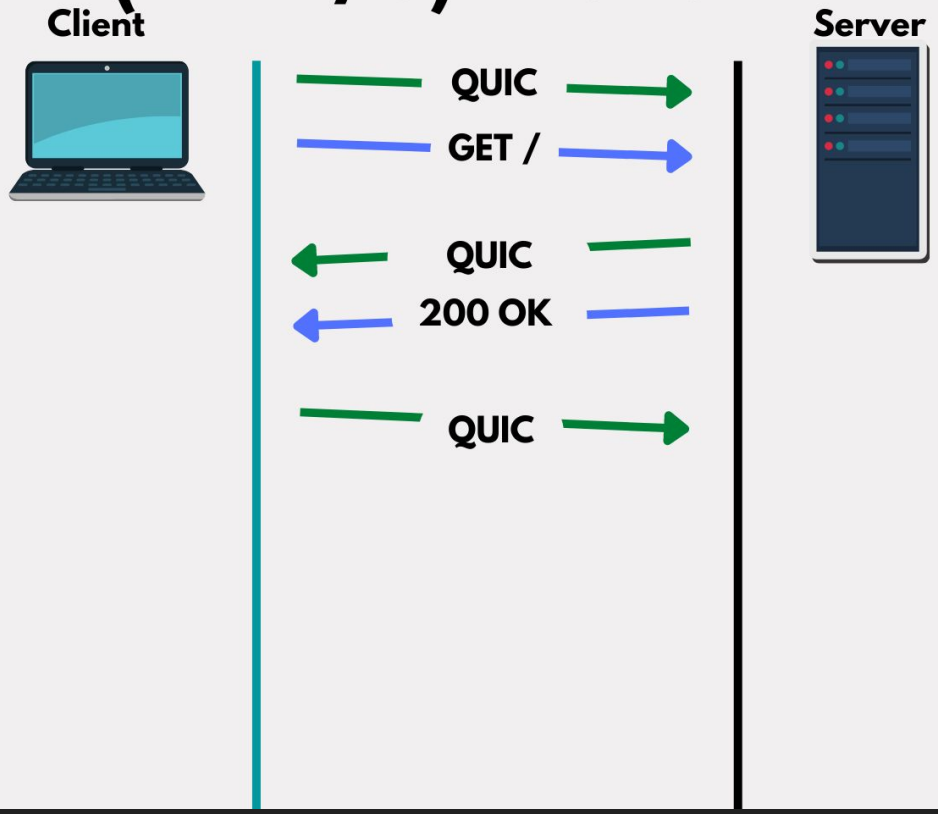
HTTPS over TCP with TLS1.3 0RTT

HTTPS over TCP with TLS 1.3 0RTT



HTTPS over QUIC 0RTT

HTTPS over QUIC (HTTP/3) with 0RTT



Backend Execution Patterns

How Backends accept, dispatch and execute requests

Process vs Thread

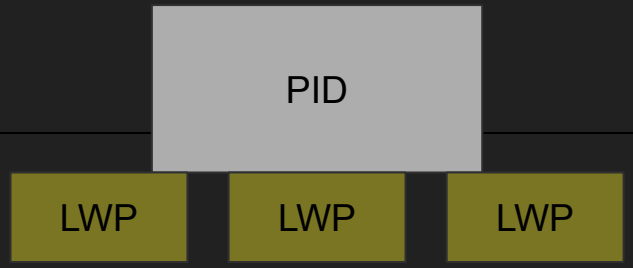
What is the difference?

What is a Process?

PID

- A set of instructions
- Has an isolated memory
- Has a PID
- Scheduled in the CPU

What is a Thread?



- Light weight Process (LWP)
- A set of instructions
- Shares memory with parent process
- Has a ID
- Scheduled in the CPU

Single Threaded Process

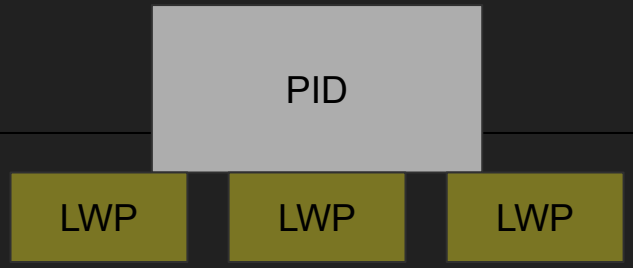
PID

- One Process with a single thread
- Simple
- Examples NodeJS

Multi-Processes

- App has multiple processes
- Each has its own Memory
- Examples NGINX/Postgres
- Take advantage of multi-cores
- More memory but isolated
- Redis backup routine (COW)

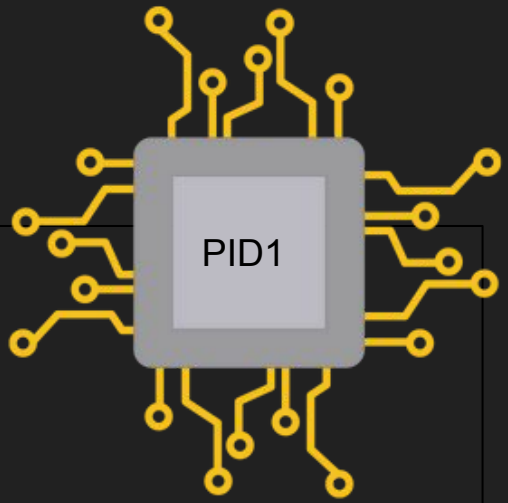
Multi-Threaded



- One Process, multiple threads
- Shared Memory (compete)
- Take advantage of multi-cores
- Require less memory
- Race conditions
- Locks and Latches (SQL Server)
- Examples Apache, Envoy

How many is too many?

- Too many processes/threads
- CPU context switch
- Multiple Cores help
- Rule of thumb -> # Cores = # Processes



How Connections are Established

SYN/Accept Queues

Connection Establishment

- TCP Three way handshake
- SYN/SYN-ACK/ACK
- But what happens on the backend?

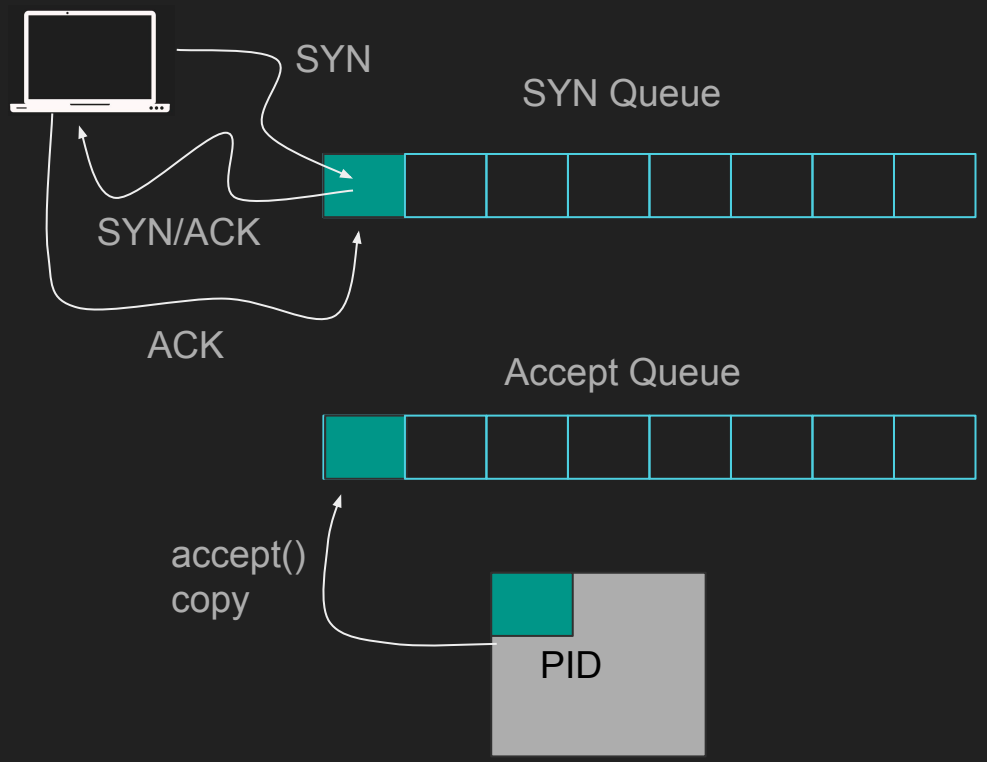
Connection Establishment

- Server Listens on an address:port
- Client connects
- Kernel does the handshake creating a connection
- Backend process “Accepts” the connection

Connection Establishment

- Kernel creates a socket & two queues SYN and Accept
- Client sends a SYN
- Kernels adds to SYN queue, replies with SYN/ACK
- Client replies with ACK
- Kernel finish the connection
- Kernel removes SYN from SYN queue
- Kernel adds full connection to Accept queue
- Backend accepts a connection, removed from accept queue
- A file descriptor is created for the connection

Connection Establishment



Problems with accepting connections

- Backend doesn’t accept fast enough
- Clients who don’t ACK
- Small backlog

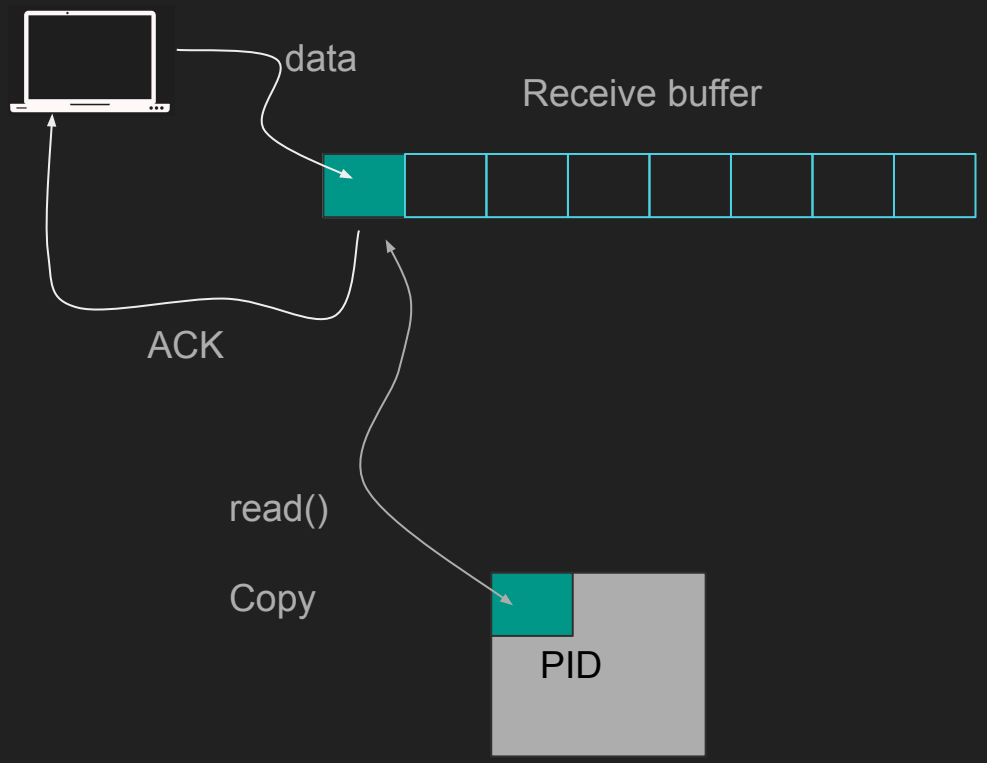
Reading and Sending Data

Receive vs Send buffers

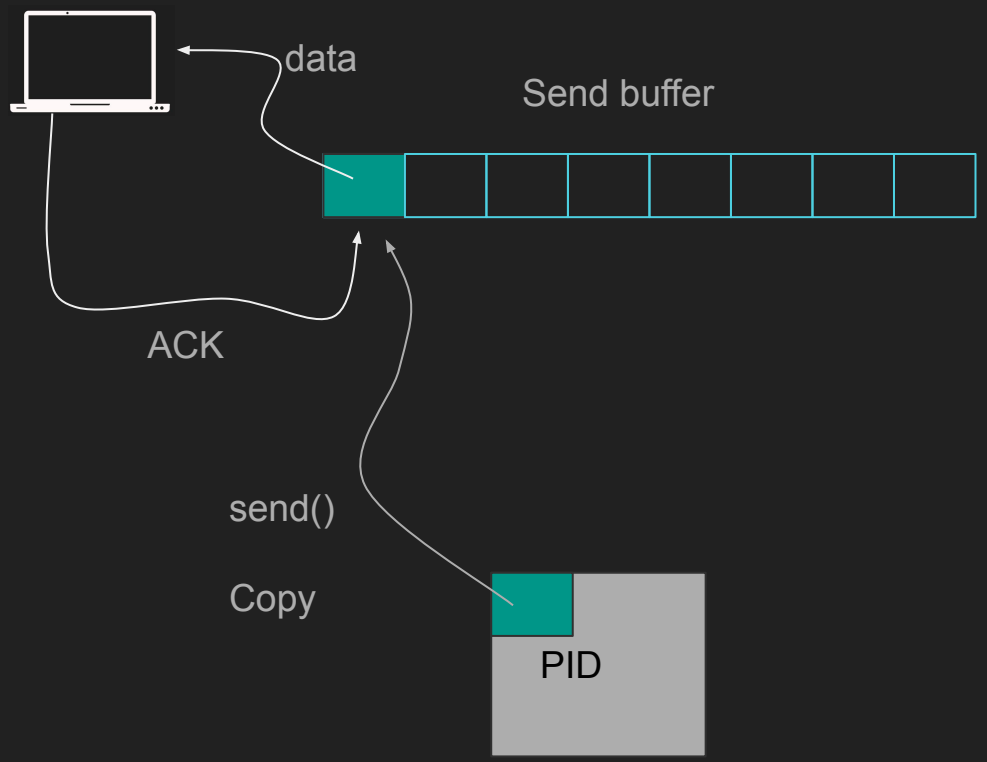
Send and receive buffers

- Client sends data on a connection
- Kernel puts data in receive queue
- Kernel ACKs (may delay) and update window
- App calls read to copy data

Receive buffers



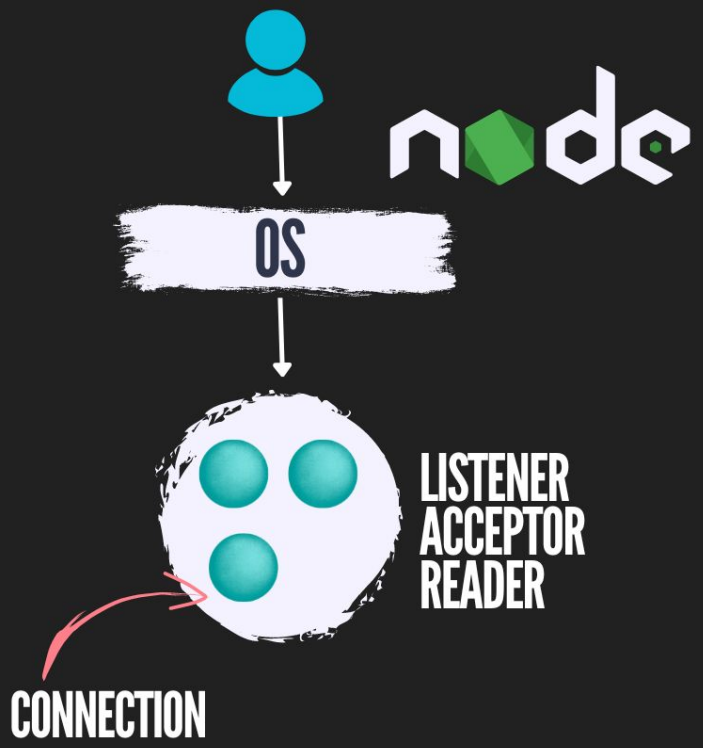
Send buffers



Problems with reading and sending

- Backend doesn’t read fast enough
- Receive queue is full
- Client slows down

Single Listener/Single Worker Thread



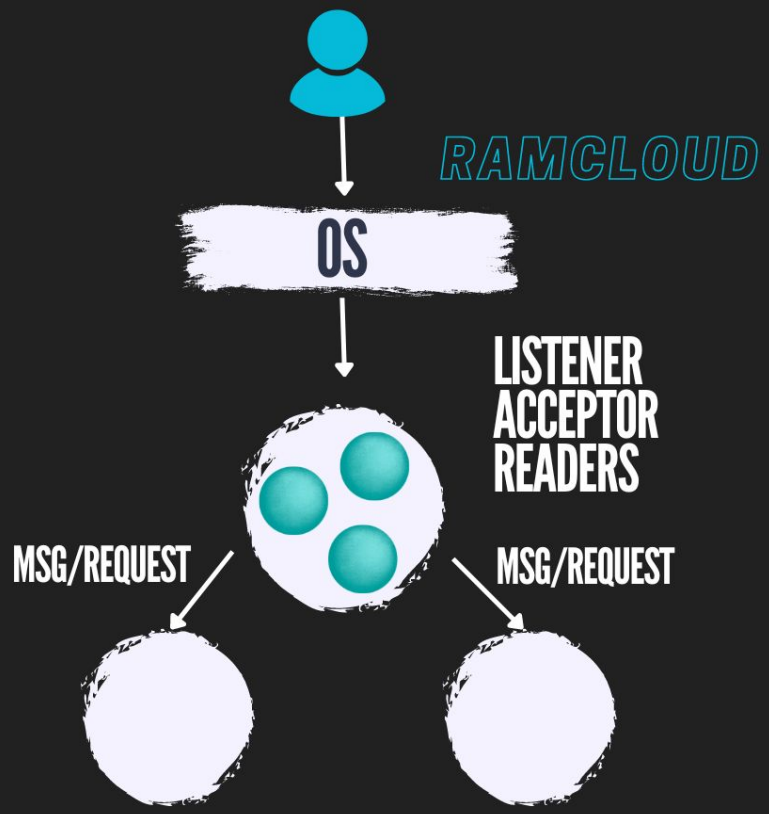
SINGLE THREAD

Single Listener/Multiple Worker threads

MULTIPLE THREADS

W/ SINGLE ACCEPTOR

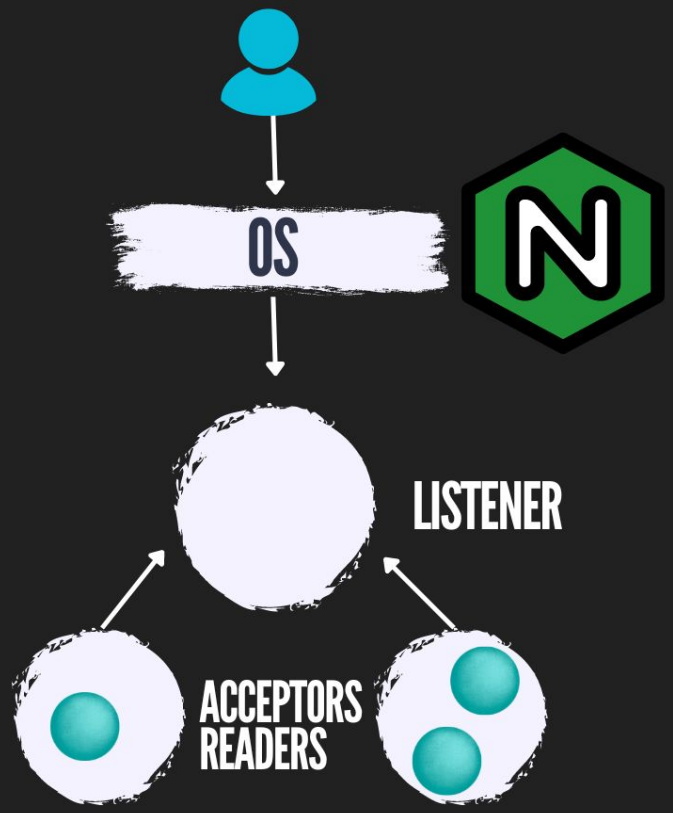
Single Listener/Multiple Worker threads with load balancing



MULTIPLE THREADS

W/MESSAGE LOAD BALANCING

Multiple Threads single Socket

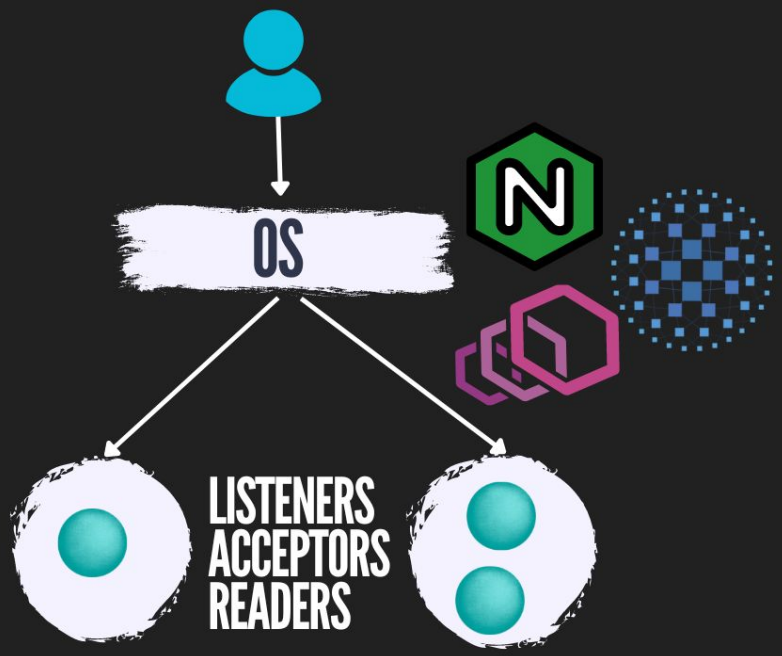


MULTIPLE THREADS

W/ MULTIPLE ACCEPTORS

Multiple Listeners on the same port

SO_REUSEPORT



MULTIPLE THREADS

W/ SOCKET SHARDING (SO_REUSEPORT)

Idempotency

Resending the Request without affecting backend

What is idempotency?

- API /postcomment
- Takes comment and appends it to table
- What if the user/proxy retries it?
- You will have duplicate comments
- Very bad for financial systems

What is idempotency?

- Idempotent request can be retried without affecting backend
- Easy implementation send a requestId
- If requestId has been processed return
- Also known as idempotency token

In HTTP

- GET is idempotent
- POST isn’t, but we can make it
- Browsers and proxies treat GET as idempotent
- Make sure your GETs are

Proxying & Load Balancing

Core of backend engineering

Proxy vs Reverse Proxy

A fundamental component of backend networking

Layer 4 vs Layer 7 Load balancers

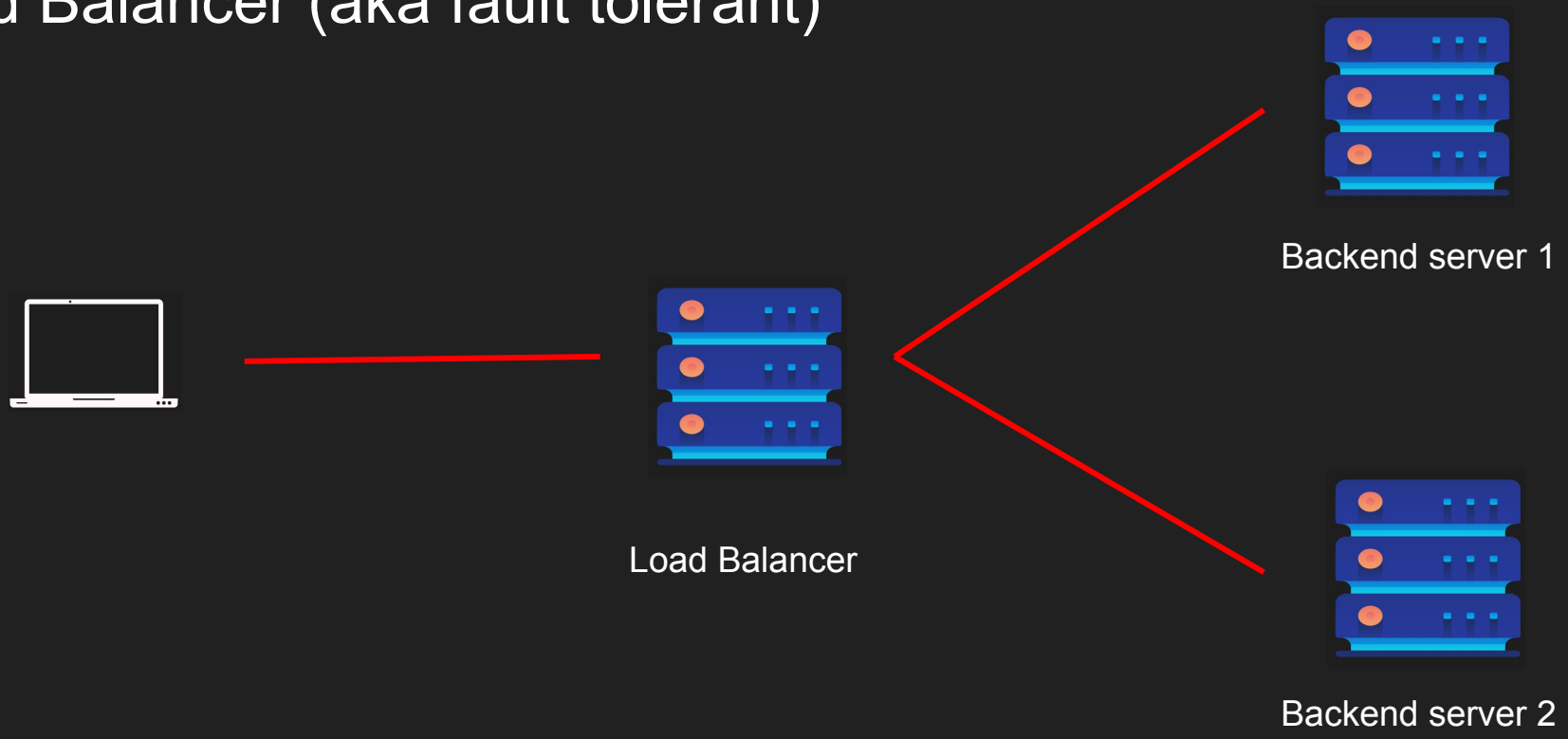
A fundamental component of backend networking

Agenda

- Layer 4 vs Layer 7
- Load Balancer
- Layer 4 Load Balancer (pros and cons)
- Layer 7 Load Balancer (pros and cons)

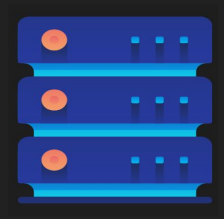
Layer 7 Application	Application
Layer 6 Presentation	Presentation
Layer 5 Session	Session
Layer 4 Transport	Transport
Layer 3 Network	Network
Layer 2 Data Link	Data Link
Layer 1 Physical	Physical

Load Balancer (aka fault tolerant)



Layer 4 Load Balancer

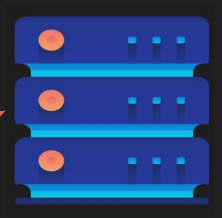
44.1.1.1



L4 Load Balancer
44.1.1.2

TCP Connection

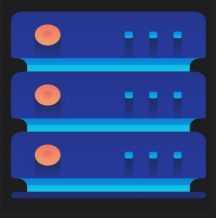
SYN
SYN/ACK
ACK



Backend server 1
44.1.1.3

TCP Connection

SYN
SYN/ACK
ACK



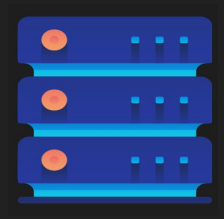
Backend server 2
44.1.1.4

Layer 4 Load Balancer

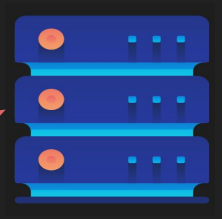
44.1.1.1

TCP Connection

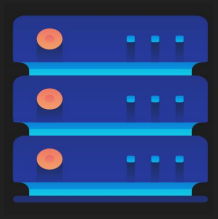
SYN
SYN/ACK
ACK



L4 Load Balancer
44.1.1.2



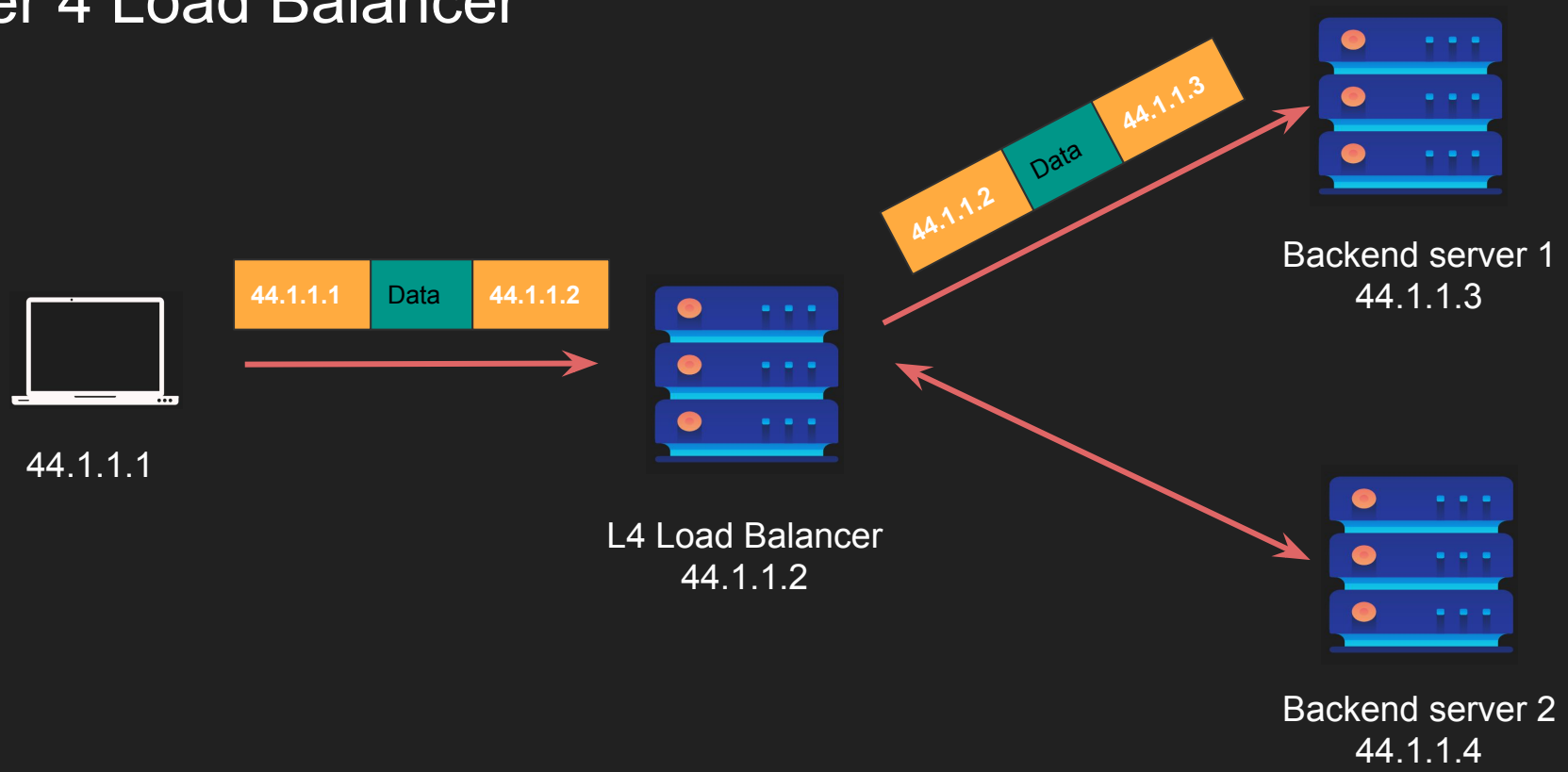
Backend server 1
44.1.1.3



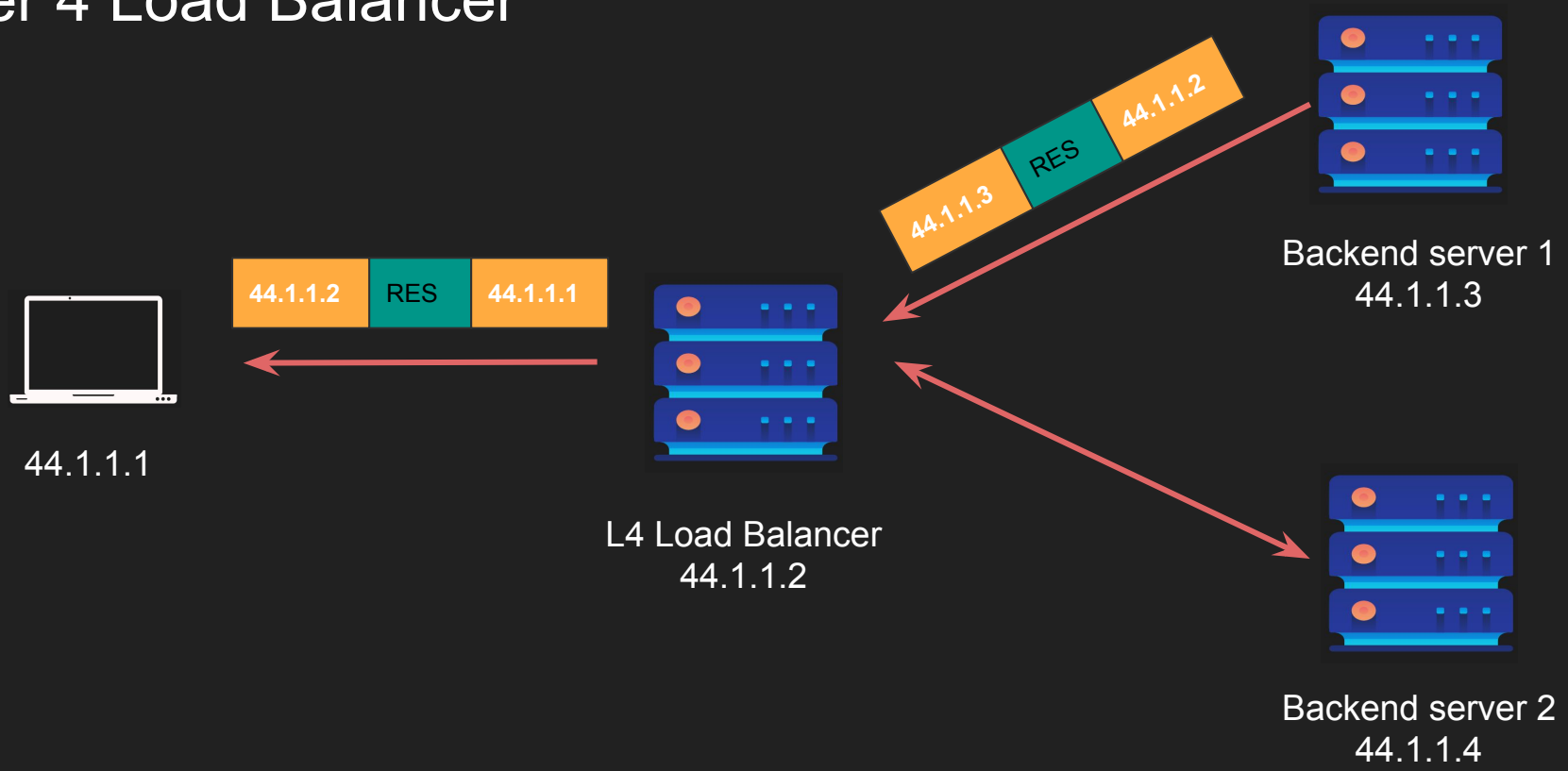
Backend server 2
44.1.1.4

When a client connects to the L4 load balancer, the LB chooses one server and all segments for that connections go to that server

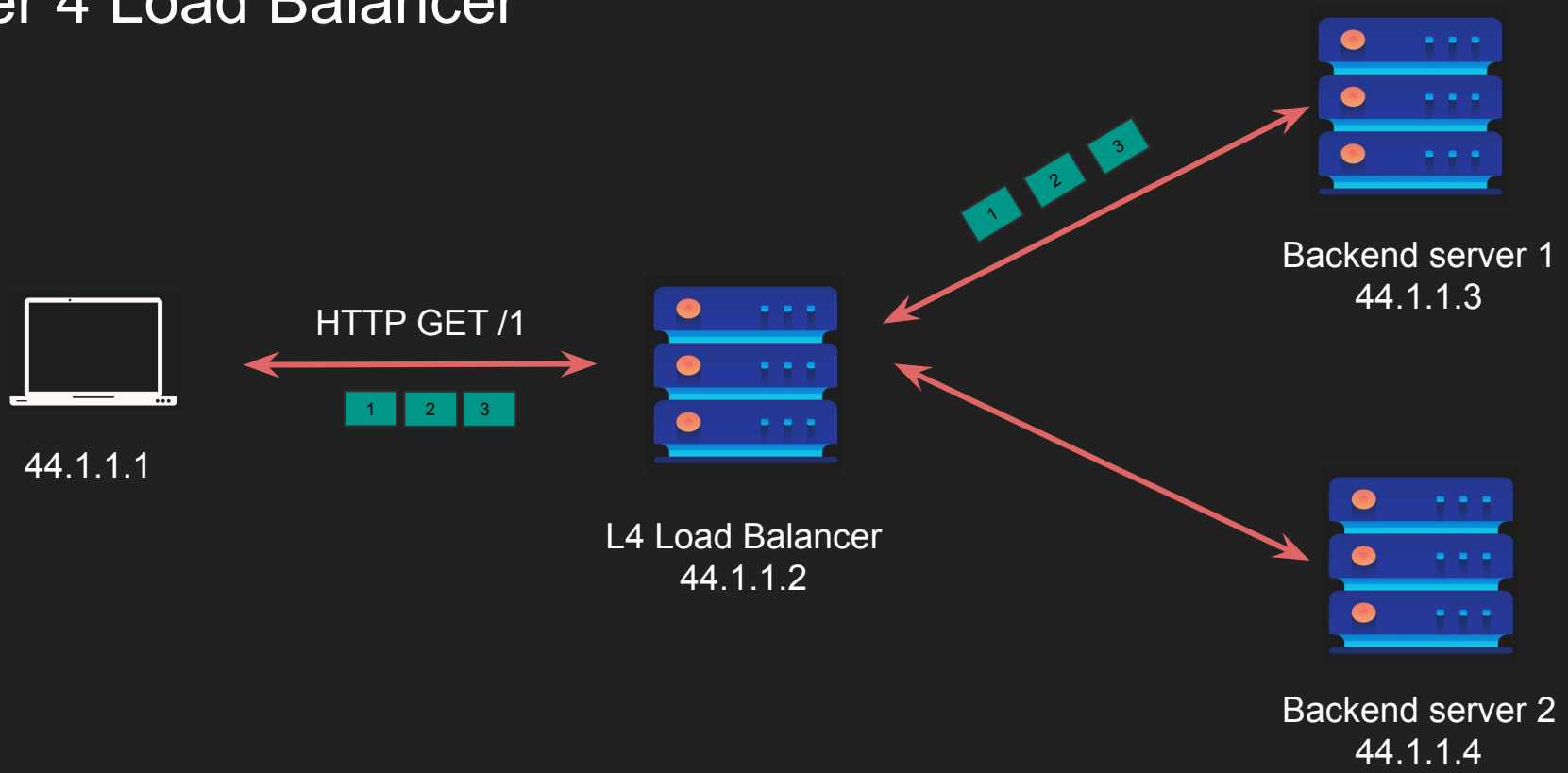
Layer 4 Load Balancer



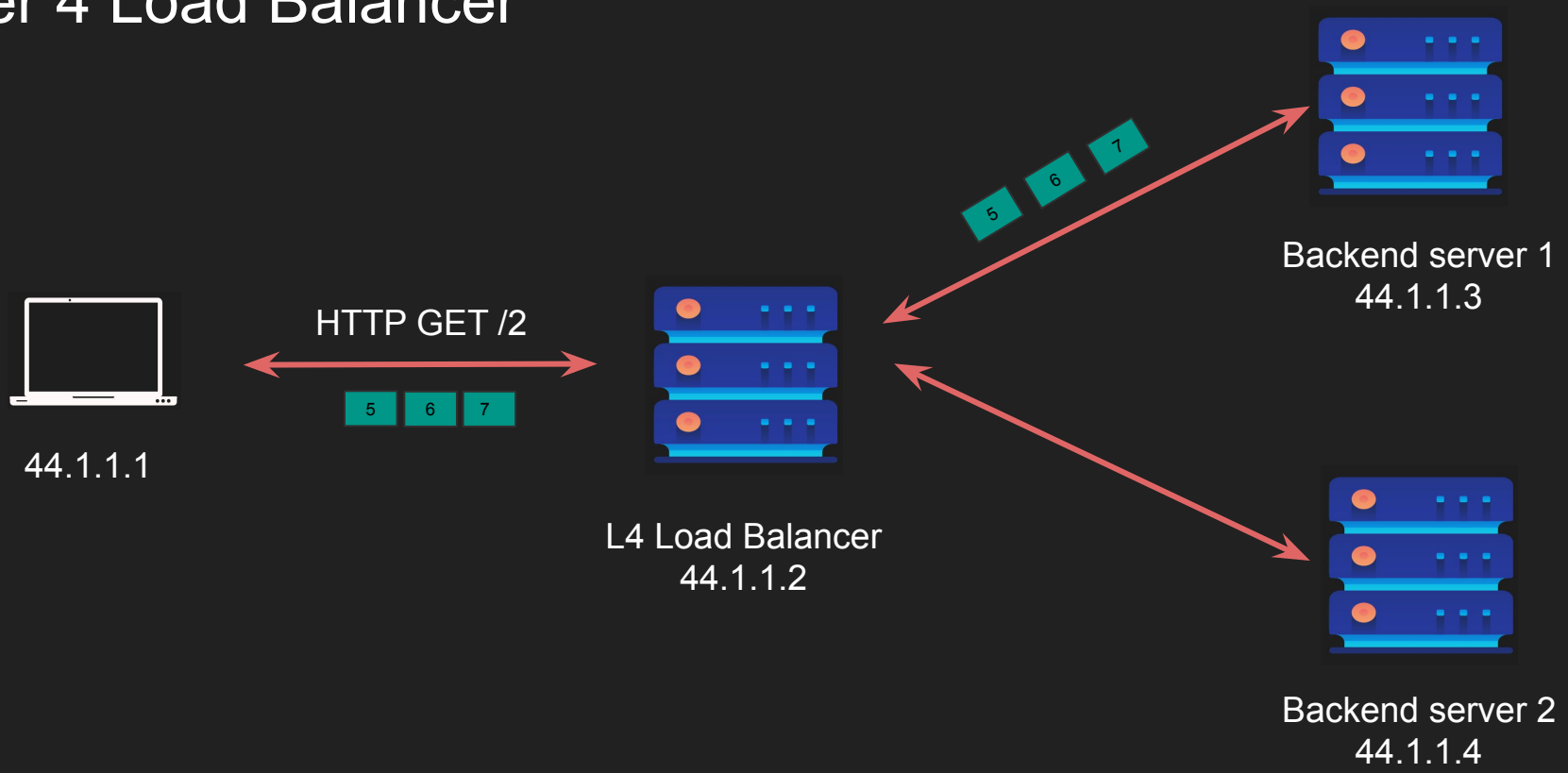
Layer 4 Load Balancer



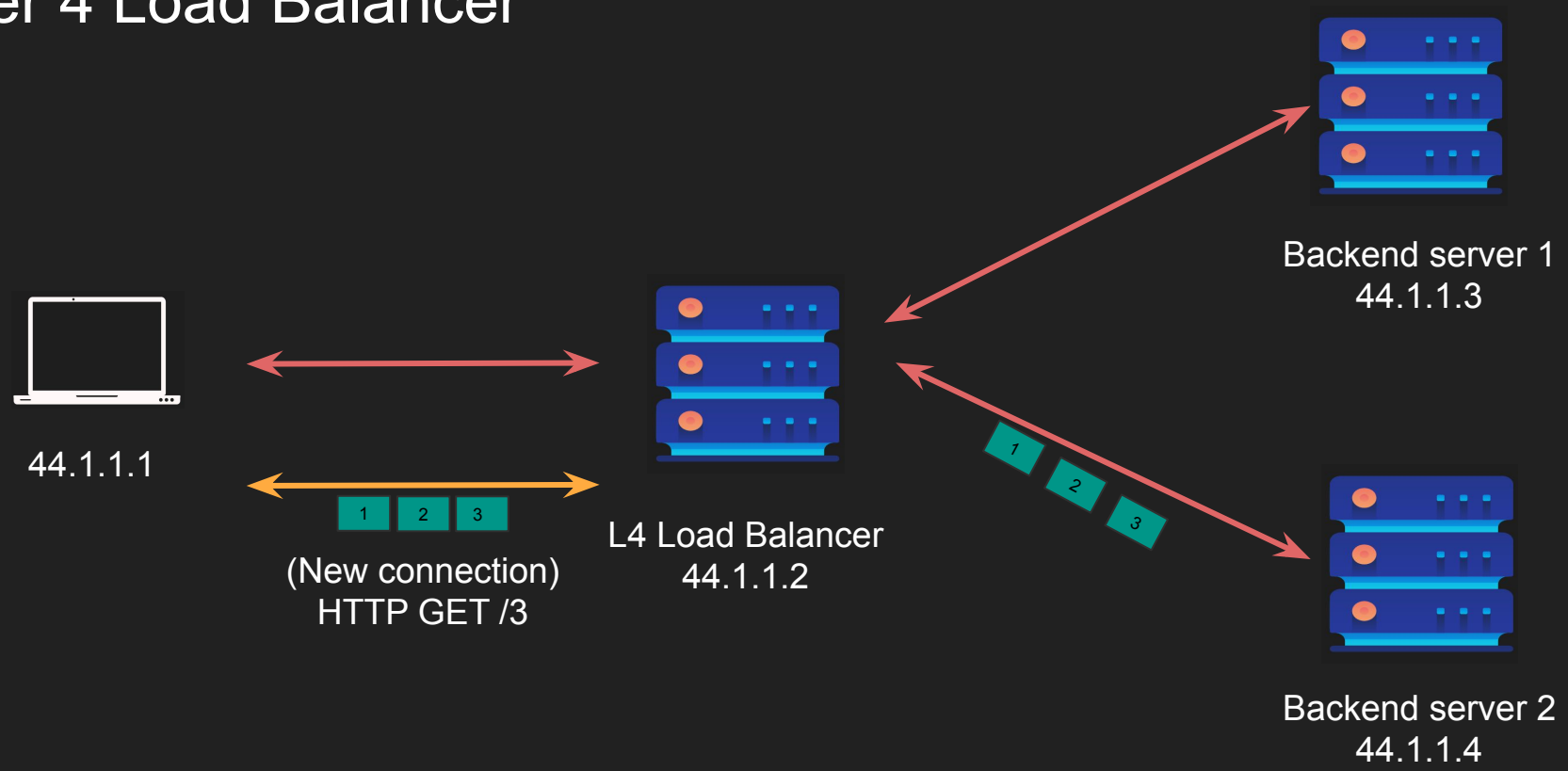
Layer 4 Load Balancer



Layer 4 Load Balancer



Layer 4 Load Balancer



Layer 4 Load Balancer (Pros and Cons)

Pros

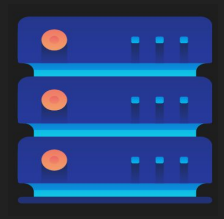
- Simpler load balancing
- Efficient (no data lookup)
- More secure
- Works with any protocol
- One TCP connection (NAT)

Cons

- No smart load balancing
- NA microservices
- Sticky per connection
- No caching
- Protocol unaware (can be dangerous) bypass rules

Layer 7 Load Balancer

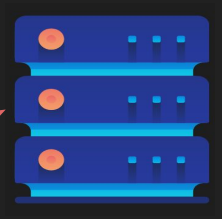
44.1.1.1



L7 Load Balancer
44.1.1.2

TCP Connection

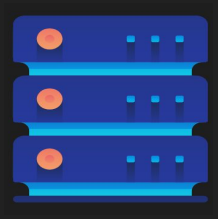
SYN
SYN/ACK
ACK



Backend server 1
44.1.1.3

TCP Connection

SYN
SYN/ACK
ACK



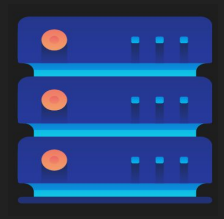
Backend server 2
44.1.1.4

Layer 7 Load Balancer

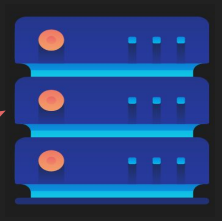
44.1.1.1

TCP Connection

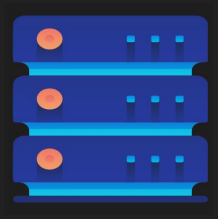
SYN
SYN/ACK
ACK



L7 Load Balancer
44.1.1.2



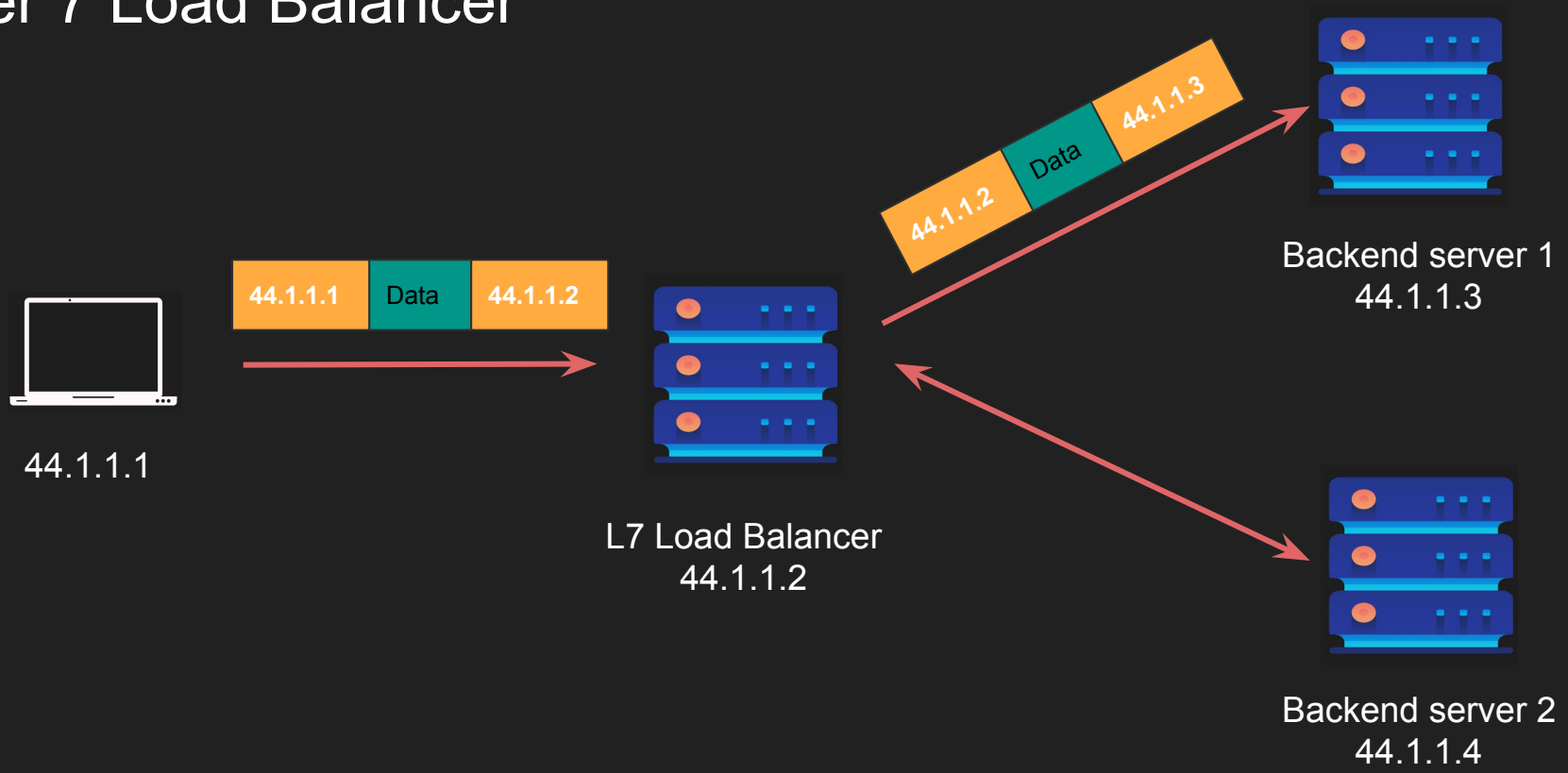
Backend server 1
44.1.1.3



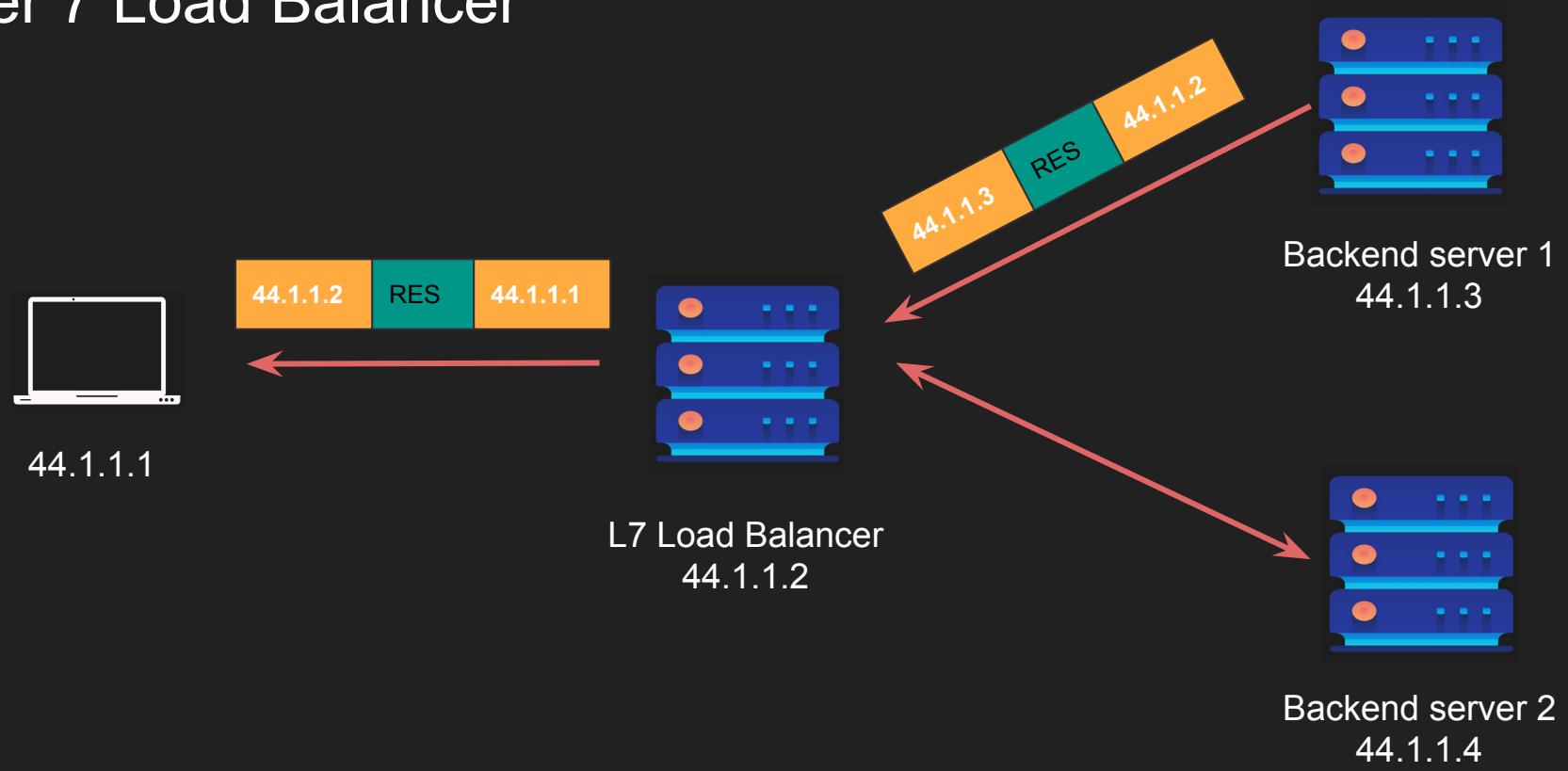
Backend server 2
44.1.1.4

When a client connects to the L7 load balancer, it becomes protocol specific. Any logical “request” will be forwarded to a new backend server. This could be one or more segments

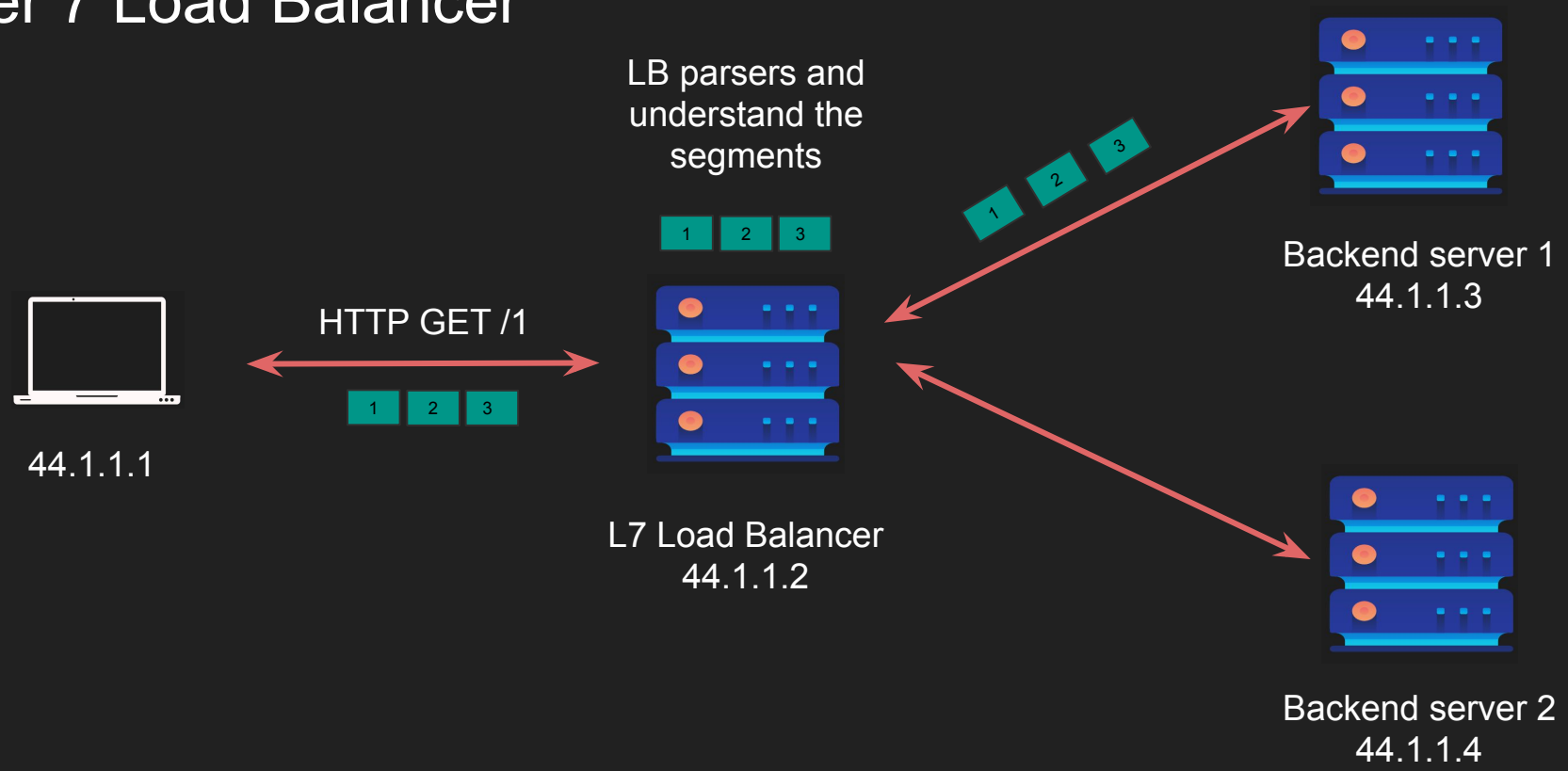
Layer 7 Load Balancer



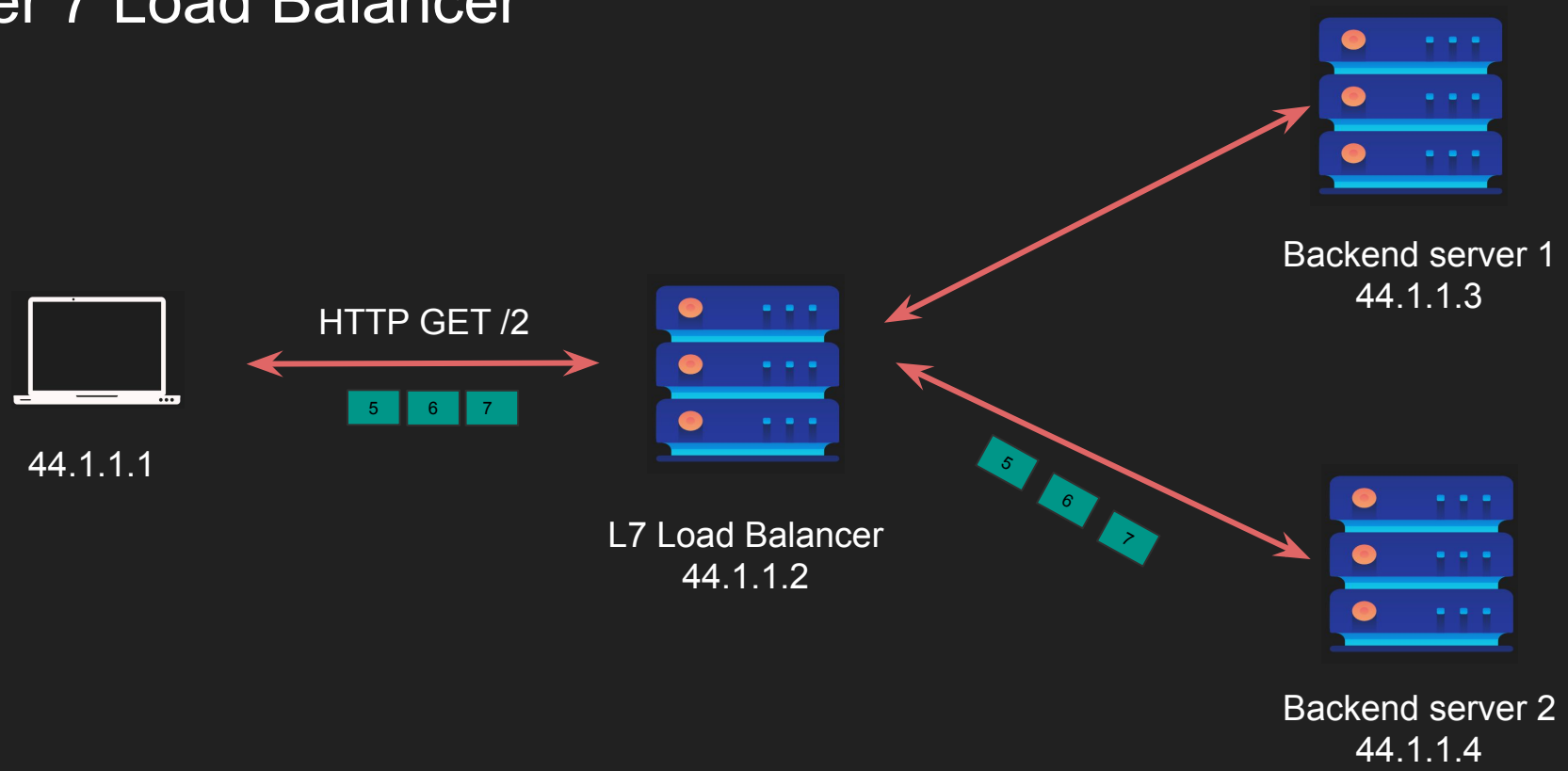
Layer 7 Load Balancer



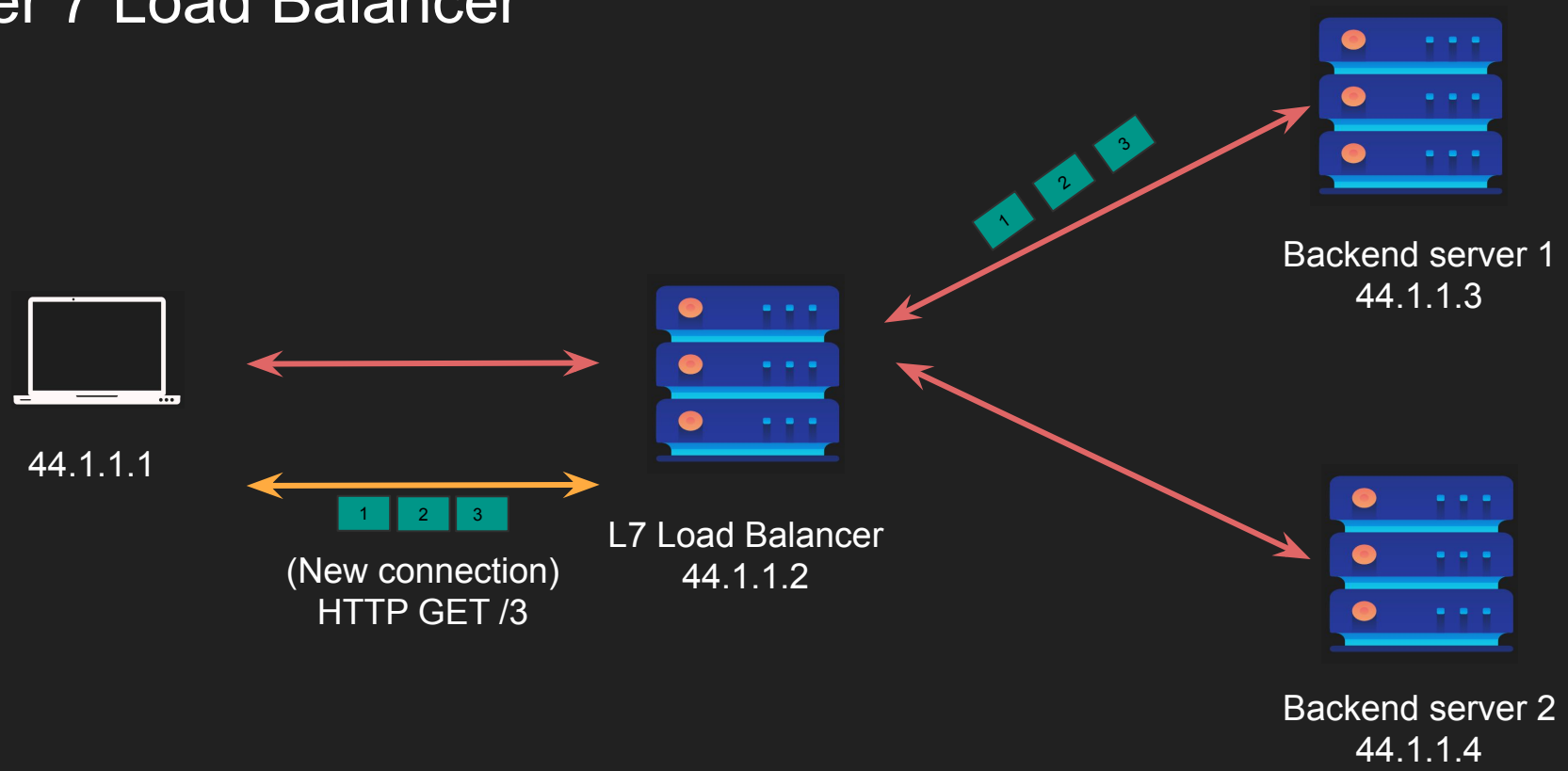
Layer 7 Load Balancer



Layer 7 Load Balancer



Layer 7 Load Balancer



Layer 7 Load Balancer (Pros and Cons)

Pros

- Smart load balancing
- Caching
- Great for microservices
- API Gateway logic
- Authentication

Cons

- Expensive (looks at data)
- Decrypts (terminates TLS)
- Two TCP Connections
- Must share TLS certificate
- Needs to buffer
- Needs to understand protocol

Summary

- Layer 4 vs Layer 7
- Load Balancer
- Layer 4 Load Balancer (pros and cons)
- Layer 7 Load Balancer (pros and cons)